

# Physiology

World of Revision

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## Physiology

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Male reproductive system • Female reproductive system • Exercise physiology •  
Regulation of body temperature

# GENERAL & CELLULAR PHYSIOLOGY : PART 1

----- Active space -----

## Homeostasis

00:01:51

- Concept of constancy.
- Walter Cannon Termed → **Homeostasis**.
- Claude Bernard Termed → **milieu interior** (Internal environment).

### Maintenance :

**Dynamic** process.

Control systems  :

|           | Negative feedback                                                                                                                                        | Positive feedback                                                                                                                                                                                                                                        | Feed forward control        |
|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|
| mechanism | ↑ f/b ↓                                                                                                                                                  | <ul style="list-style-type: none"> <li>• ↑ f/b ↑</li> <li>• Amplification (Vicious cycle)</li> </ul>                                                                                                                                                     | <b>Anticipatory</b> changes |
| Examples  | <ul style="list-style-type: none"> <li>• Insulin regulation of glucose</li> <li>• Baroreceptor reflex</li> <li>• Endocrine hormone regulation</li> </ul> | <b>'CLAPS' :</b> <ul style="list-style-type: none"> <li>• <b>C</b>lotting</li> <li>• <b>Ca<sup>2+</sup></b> entry into SR</li> <li>• <b>LH</b> surge</li> <li>• <b>A</b>ction potential</li> <li>• <b>P</b>arturition</li> <li>• <b>S</b>hock</li> </ul> | Body temperature regulation |

Gain  :

- Assessment of effectiveness of negative feedback.

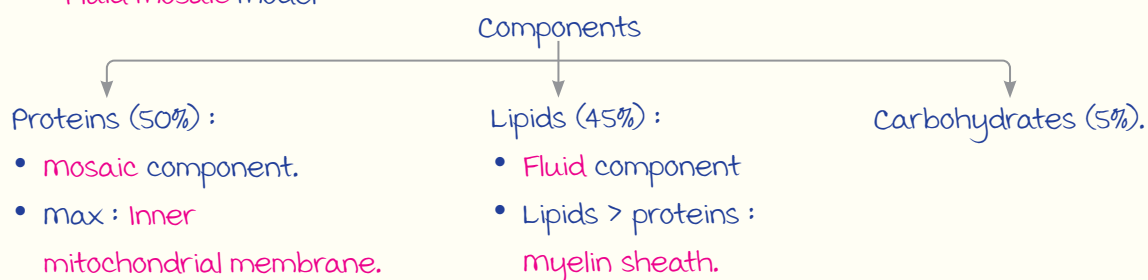
$$\text{Gain} = \frac{\text{Correction}}{\text{Error remaining}}$$

- If Error = 0, **Gain** =  $\infty$  (Role of kidneys in BP regulation).

## Cell Membrane

00:15:24

- **Fluid mosaic** model :



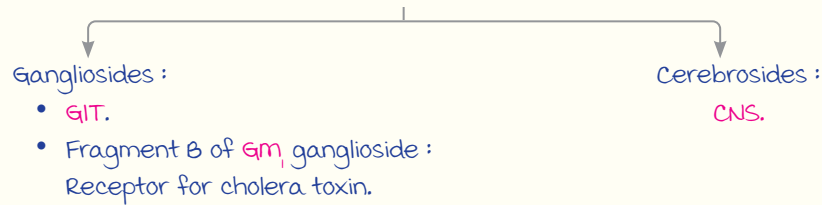
----- Active space -----

membrane Lipids :

## 1. Phospholipids :

|                      | Site                                                                  | Features                                                                                                                        |
|----------------------|-----------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| Phosphatidylcholine  | Lung                                                                  | <ul style="list-style-type: none"> <li>AKA <b>lecithin</b></li> <li>major surfactant</li> </ul>                                 |
| Sphingomyelin        | <ul style="list-style-type: none"> <li>Nerve</li> <li>Lung</li> </ul> | <ul style="list-style-type: none"> <li>minor surfactant</li> <li>L/S ratio <math>\geq 2</math> : Fetal lung maturity</li> </ul> |
| Phosphatidylserine   | Cell membrane :<br>Inner surface                                      | If exposed extracellularly :<br>Signal for apoptosis ( <b>Eat me signal</b> )                                                   |
| Phosphatidylinositol | Endocrine                                                             | Second messenger                                                                                                                |
| Cardiolipin          | Heart : mitochondria                                                  | -                                                                                                                               |

## 2. Glycolipids : Lipids + carbohydrates.



## 3. Sterols :

- Cholesterol → Vitamin D synthesis.
- Cholesterol → **Fluidity buffer.**

## • membrane fluidity :

|                             | Increased fluidity                                                                                                                           | Decreased fluidity                                                                    |
|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| Effect on membrane function | Good                                                                                                                                         | Bad                                                                                   |
| Type of fatty acids         | <b>unsaturated</b> (essential)                                                                                                               | <b>Saturated</b>                                                                      |
| Examples                    | <ul style="list-style-type: none"> <li>Linoleic acid</li> <li>Linolenic acid</li> <li>Arachidonic acid</li> <li>Omega-3-FA (Fish)</li> </ul> | <ul style="list-style-type: none"> <li>Stearic acid</li> <li>Palmitic acid</li> </ul> |

membrane Proteins :

|                                                | Location | Features                                                                                                                                                                                               |
|------------------------------------------------|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Transmembrane/<br>integral membrane<br>protein | -        | <ul style="list-style-type: none"> <li>Receptor : GPCR</li> <li>Pump : <math>\text{Na}^+ - \text{K}^+</math> ATPase</li> <li>Channel (<math>\text{Cl}^-</math>) : <b>CFTR</b> mutation → CF</li> </ul> |
| Peripheral protein                             | RBC      | mutation of : <ul style="list-style-type: none"> <li>Spectrin : Hereditary elliptocytosis</li> <li>Ankyrin : Hereditary spherocytosis</li> </ul>                                                       |
| Lipid/<br>GPI anchored protein                 |          | <ul style="list-style-type: none"> <li>CD-55 } mutation → Paroxysmal nocturnal hemoglobinuria</li> <li>CD-59 }</li> </ul>                                                                              |

## Cell Organelles

00:35:34

----- Active space -----

### Endoplasmic Reticulum :

|           | RER (Granular)                                                                                                                                                                                                                            | SER (Agranular)                                                                                                    |
|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Ribosomes | ⊕                                                                                                                                                                                                                                         | ⊖                                                                                                                                                                                                     |
| Functions | <ul style="list-style-type: none"> <li>Protein biosynthesis</li> <li>Protein folding with chaperones</li> <li>ER associated degradation :<br/>Destruction of misfolded proteins<br/>↓<br/>Accumulation<br/>↓<br/>Prion disease</li> </ul> | <ul style="list-style-type: none"> <li>Drug metabolism (Liver)</li> <li>Calcium storage :<br/>Sarcoplasmic reticulum (Skeletal muscle)</li> <li>Steroid hormone synthesis (Adrenal cortex)</li> </ul> |

### Golgi Apparatus :

Located close to RER.

Functions :

- Cis end → Post translational modification → Trans end  
(Receives protein) (E.g. : Glycosylation) (Releases vesicles).
- Protein tagging, sorting & delivery :  
I - cell disease : mannose-6-phosphate tagging ⊖.

### Lysosomes :

- AKA suicidal bags/residual bodies.
- Acid mediated destruction :
  - Acid lipase.
  - Acid hydrolase.
- Autophagy : Starvation → mitochondrial destruction → Energy release.

### Peroxisomes :

- AKA microbodies.
- Abundant in liver & kidney.

Functions :

- VLCFA : Oxidation.
- $H_2O_2$  : Generation and degradation (Catalase).
- Synthesis of plasmalogens (myelin sheath).

----- Active space -----

Peroxisomal storage disorders :

- Zellweger syndrome : VLCFA accumulation.
- Refsum's disease : Phytanic acid accumulation.

mitochondria :

Derived from ovum : maternal inheritance.

mitochondrial DNA :

- Double stranded.
- 16,500 base pairs.
- mutation rate : 10 x nuclear DNA.

mitochondrial disorders :

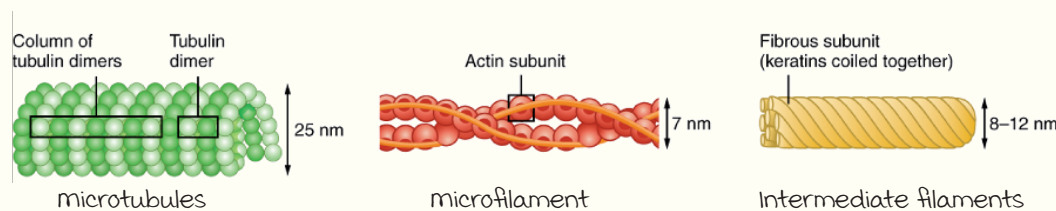
- MELAS : mitochondrial encephalomyopathy, lactic acidosis & stroke-like episodes.
- Affects organs with high metabolic rate : muscle, brain.

Nucleus :

- Contains chromosomes (DNA) : Blueprint.
- Nucleosome : DNA (String) + histone proteins (Beads).
- Chromatin : whole complex of DNA + proteins.
- Nuclear pore complex (NPC) :
  - Exportin.
  - Importin.
- Units of heredity - genes.

## Cytoskeletal Filaments

00:56:24

microtubules :

Types : Kinesin, dynein, tubulin.

Function :

- Axonal transport
    - Anterograde → Kinesin.
    - Retrograde (Rabies) → Dynein.
  - Ciliary motility : Defect → Kartagener syndrome
  - Chromosome separation → Cell division.
- Cancer Rx : Vincristine (microtubule inhibitor drug).

microfilaments :

----- Active space -----

Actin polymerization → Cell movement (*Listeria monocytogenes*).  
 → maintains cell structure.

Intermediate Filaments  :

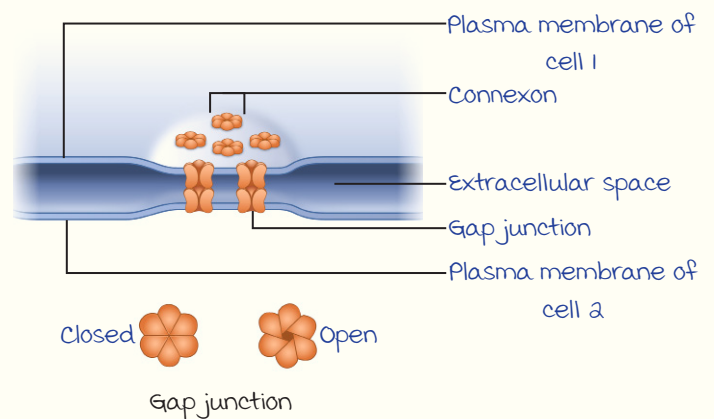
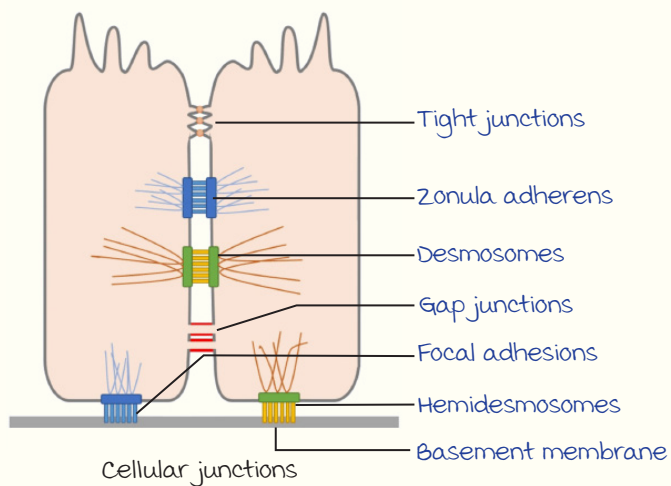
Functions :

- Resist external pressure
- Cellular/tumor markers :

| Filament                               | Tissue                          | marker for                    |
|----------------------------------------|---------------------------------|-------------------------------|
| Keratin                                | Epithelial cells                | Carcinoma                     |
|                                        | Liver ( <i>mallory bodies</i> ) | Alcoholic hepatitis           |
| Desmin                                 | muscle cells                    | Rhabdomyosarcoma              |
| Vimentin                               | Connective tissue (Fibroblast)  | Connective tissue tumors      |
| Glial fibrillary acidic protein (GFAP) | Astrocytes                      | Astrocytoma                   |
| Lamin                                  | Nucleus                         | Progeria<br>(Premature aging) |

Cellular Junctions

01:11:17



----- Active space -----

| Cell junctions                                                                                          | Proteins                                                                            | Features                                                                                                                                                                                                                                                                                          |
|---------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Cell-cell junctions  : |                                                                                     |                                                                                                                                                                                                                                                                                                   |
| Zonula adherens                                                                                         | Cadherins                                                                           | <ul style="list-style-type: none"> <li>Calcium dependent linkages</li> <li>Linked to <b>actin</b></li> </ul>                                                                                                                                                                                      |
| Desmosomes                                                                                              | <ul style="list-style-type: none"> <li>Desmogleins</li> <li>Desmocollins</li> </ul> | <ul style="list-style-type: none"> <li>Areas of mechanical stress :<br/>Skin, uterine cervix</li> <li>Antibodies against desmoglein<br/>↓<br/>Pemphigus vulgaris</li> </ul>                                                                                                                       |
| Zonula occludens/<br>Tight junctions                                                                    | <ul style="list-style-type: none"> <li>Occludin</li> <li>Claudin</li> </ul>         | <ul style="list-style-type: none"> <li>BBB, kidney, GIT</li> <li><b>Apicolateral</b></li> <li>Regulates <b>paracellular</b> transport</li> <li><b>Zonula occludens toxin :</b> <ul style="list-style-type: none"> <li>Produced in cholera</li> <li>Loosens tight junctions</li> </ul> </li> </ul> |
| Gap junctions                                                                                           | Connexon<br>(1 connexon = 6 connexins)                                              | Heart<br>(Functional syncytium arrangement)                                                                                                                                                                                                                                                       |
| Cell-basement membrane junctions :                                                                      |                                                                                     |                                                                                                                                                                                                                                                                                                   |
| Hemidesmosomes                                                                                          | -                                                                                   | A/w intermediate filaments                                                                                                                                                                                                                                                                        |
| Focal adhesions                                                                                         | -                                                                                   | A/w actin                                                                                                                                                                                                                                                                                         |

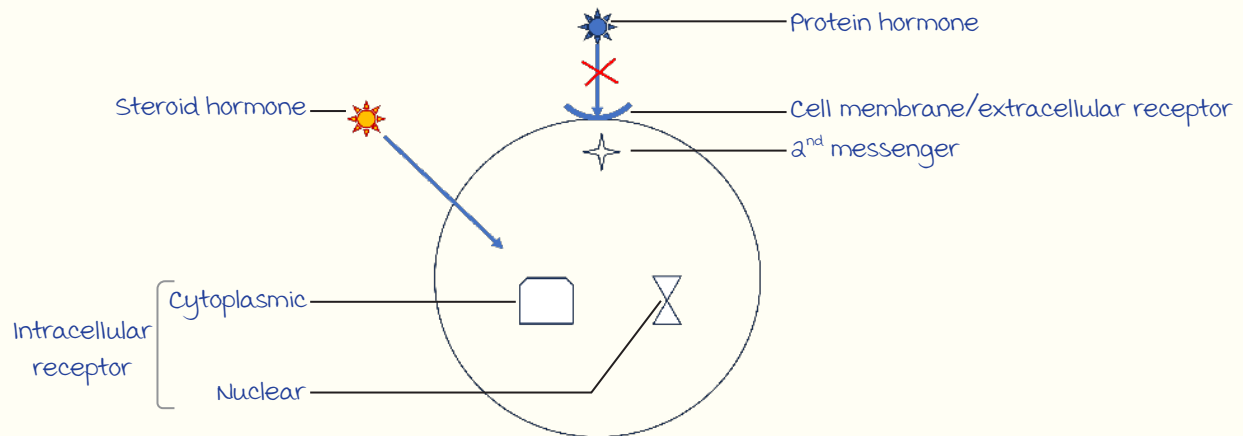
# GENERAL AND CELLULAR PHYSIOLOGY : PART 2

----- Active space -----

## Cellular Messengers and Receptors

00:00:06

### Hormones :



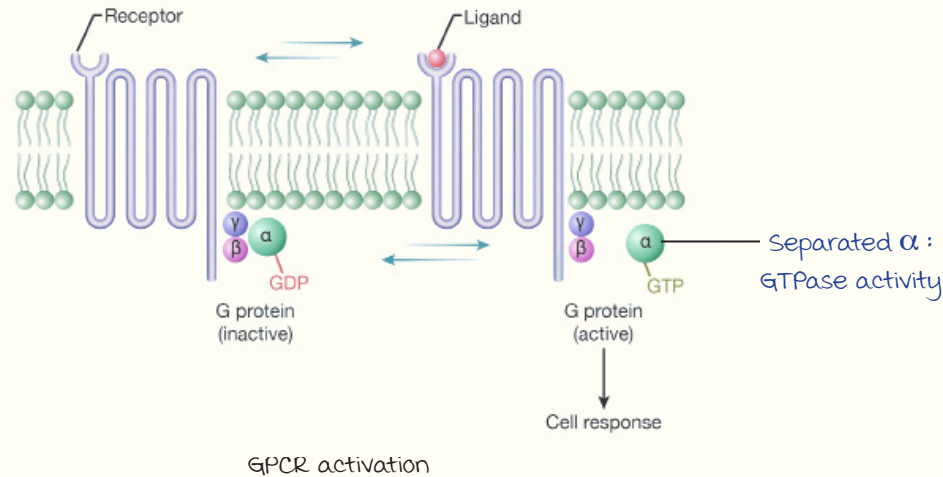
### Classification :

|                                              | Examples                                       |
|----------------------------------------------|------------------------------------------------|
| Amino acid derivatives                       | Tyrosine → • $T_3$ , $T_4$<br>• Catecholamines |
| Protein hormone                              | • Insulin : 51 aa<br>• PTH : 84 aa             |
| Steroid hormones/<br>Cholesterol derivatives | Aldosterone, cortisol, estrogen, testosterone  |
| Vitamin hormones                             | Vit D, Vit A                                   |

### Cell membrane Receptors :

|                                                                                                              | Features of hormones                                                                                     |
|--------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| G protein-coupled receptor (GPCR)                                                                            | • 7 transmembrane receptor<br>• Serpentine<br>• 3 subunits : $\alpha$ , $\beta$ , $\gamma$               |
| Tyrosine kinase receptor  | Insulin                                                                                                  |
| Janus Kinase receptor                                                                                        | • JAK-STAT pathway<br>• Cytokine receptor super family<br>• GEL : Growth hormone, Erythropoietin, Leptin |
| Serine Kinase receptor                                                                                       | Activin, AMH                                                                                             |

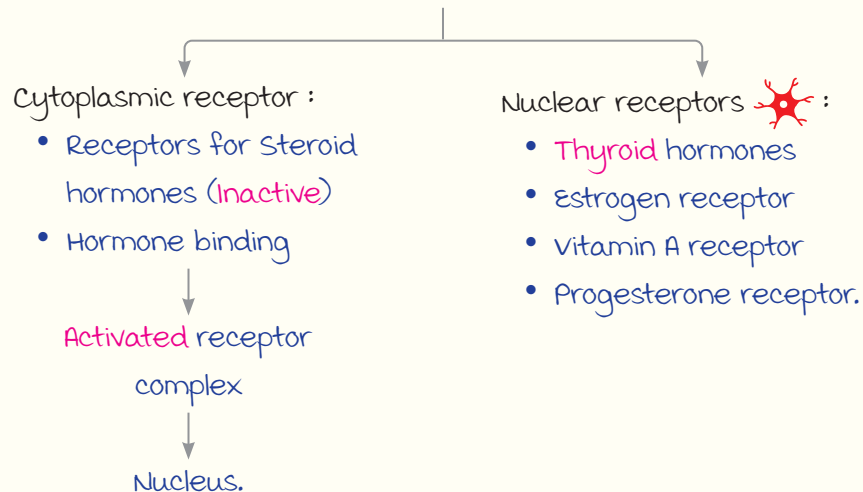
----- Active space -----



### Second messengers 🦋:

|         | Function                            | Hormone               |
|---------|-------------------------------------|-----------------------|
| cAMP    | Water reabsorption                  | Vasopressin ( $V_2$ ) |
|         | metabolism (Glycogenolysis)         | Epinephrine           |
|         | Electrolyte Secretion ( $HCO_3^-$ ) | Secretin              |
| Calcium | Vasoconstriction                    | Vasopressin ( $V_1$ ) |
|         | uterine contraction                 | Oxytocin              |
|         | Exocytosis of TSH                   | TRH                   |
| cGMP    | Vasodilation                        | Nitric oxide          |
|         | mesangial cell relaxation           | Natriuretic peptides  |

### Intracellular Receptors :



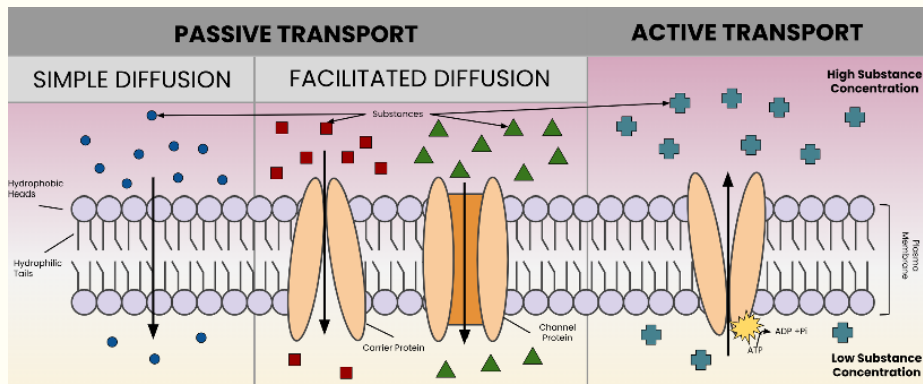
#### • Examples :

- Glucocorticoid receptor.
- Androgen receptor.
- Mineralocorticoid receptor.
- Vitamin D receptors.

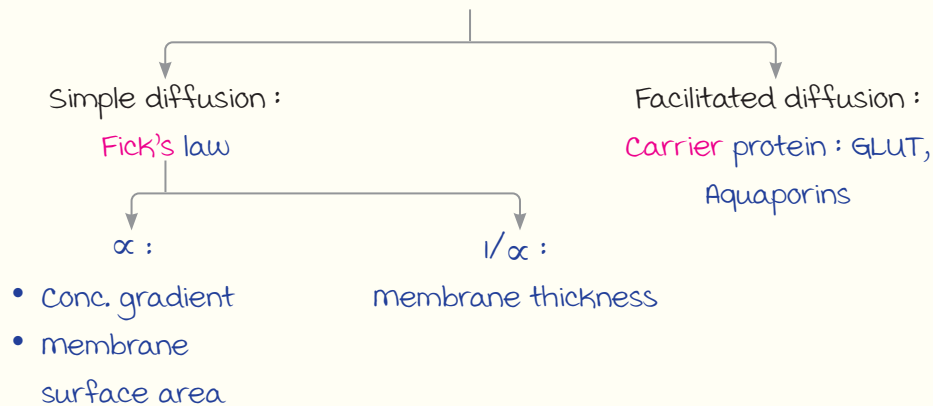
## Membrane Transport

00:16:43

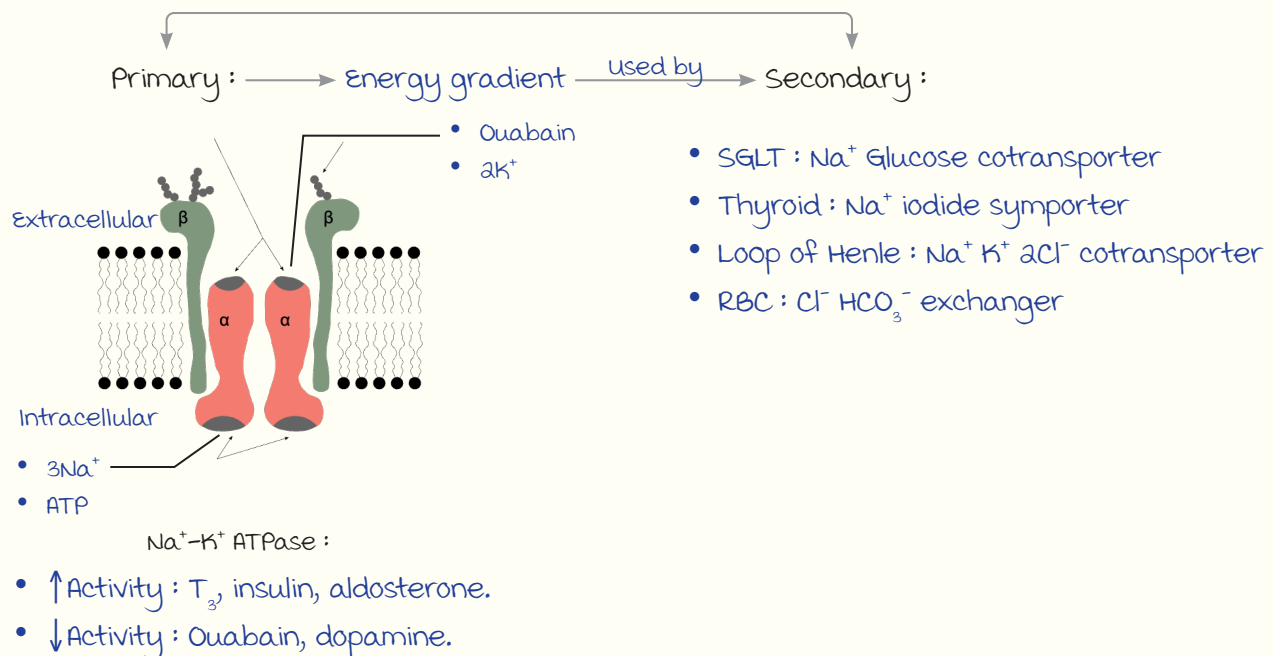
----- Active space -----



### Passive Transport (No ATP) :



### Active Transport (ATP) :



----- Active space -----

Vesicular transport :

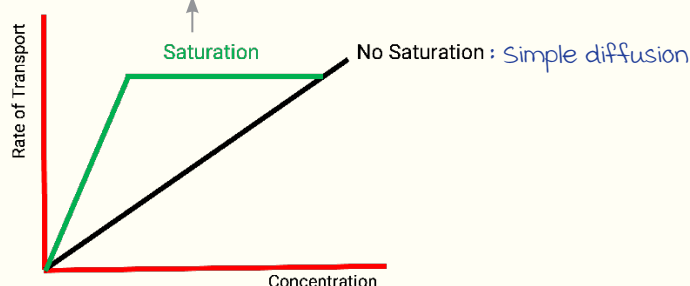
- Exocytosis :
  - $\text{Ca}^{2+}$  ions.
  - SNARE proteins.

• Endocytosis :

- Pinocytosis (Fluid).
- Phagocytosis (Bacteria).
- Receptor mediated (Clathrin) :  
LDL cholesterol.

Saturation kinetics :

- Facilitated diffusion (Carriers).
- Active transport (Channels).



## Membrane Potentials

00:34:01

Resting Membrane Potential (RMP) : $\text{K}^+$  moves out  $\rightarrow$  Increased negativity inside the cell.

Values :

| Stability                     | Cell  | RMP                                               |
|-------------------------------|------------------------------------------------------------------------------------------|---------------------------------------------------|
| Stable                        | Neuron                                                                                   | -70 mV                                            |
|                               | Skeletal muscle                                                                          | -90 mV                                            |
|                               | Cardiac ventricles                                                                       |                                                   |
| Unstable<br>(Pacemaker cells) | SA Node                                                                                  | -60 mV to -40 mV<br>(Restless membrane potential) |
|                               | Cajal cells                                                                              |                                                   |
|                               | Pre Botzinger complex (RS)                                                               |                                                   |

Clinical Significance  :

- Hyperkalemia :  $\downarrow \text{K}^+$  diffusion  $\rightarrow$  Depolarised cell  $\rightarrow$   $\uparrow$  excitable.
- Hypokalemia :  $\uparrow \text{K}^+$  diffusion  $\rightarrow$  Hyperpolarised cell  $\rightarrow$   $\downarrow$  excitable.

Equilibrium Potential :

No ion movement.

Nernst Equation :

$$E = \pm 61 \log \frac{C_o}{C_i}$$

$\rightarrow$  Outside conc.  
 $\rightarrow$  Inside conc.

Values :

| Ion              | Equilibrium potential   |
|------------------|-------------------------|
| $\text{Na}^+$    | +60 mV                  |
| $\text{K}^+$     | -90 mV                  |
| $\text{Cl}^-$    | -70 mV (RMP of neurons) |
| $\text{Ca}^{2+}$ | +130 mV                 |

----- Active space -----

Gibbs Donnan Effect  :

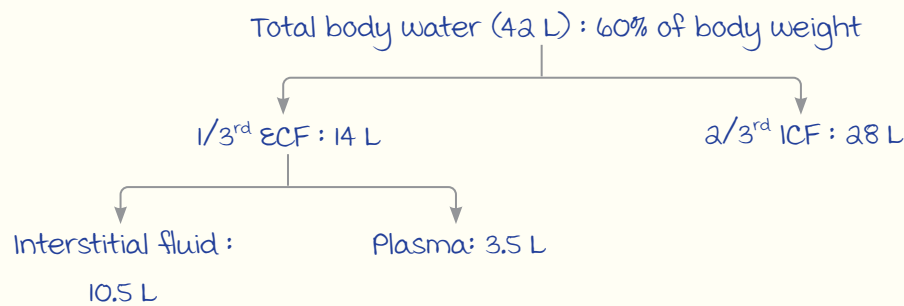
|                |                           |                           |
|----------------|---------------------------|---------------------------|
|                | Intracellular             | Extracellular             |
| Diffusible     | $\text{K}^+, \text{Cl}^-$ | $\text{K}^+, \text{Cl}^-$ |
| Non diffusible | Protein anions            | -                         |

Affects distribution 

## Cellular Fluids

00:43:26

Classification :



Measurement  :

Indicator dilution method :

| Volume                                               | Indicators                            |
|------------------------------------------------------|---------------------------------------|
| Direct measurement :                                 |                                       |
| Total body water (TBW)                               | Deuterium (m/c)                       |
| ECF                                                  | Inulin, Sucrose, mannitol             |
| Plasma volume (PV)                                   | Radiolabelled albumin, Evans blue dye |
| Blood volume = $\text{PV} / (1 - \text{Hematocrit})$ | Chromium tagged RBCs                  |
| Indirect measurement :                               |                                       |
| Interstitial fluid = $\text{ECF} - \text{Plasma}$    | -                                     |
| Intracellular fluid = $\text{TBW} - \text{ECF}$      | -                                     |

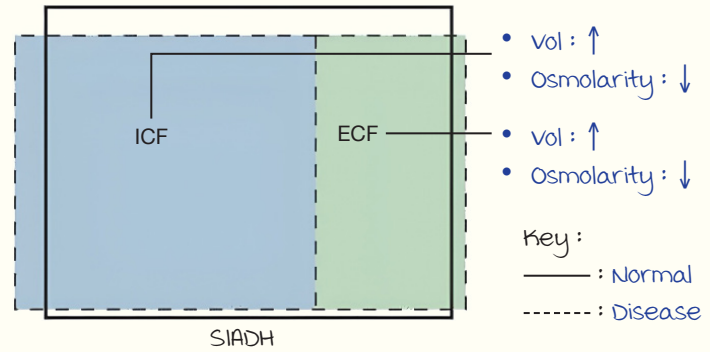
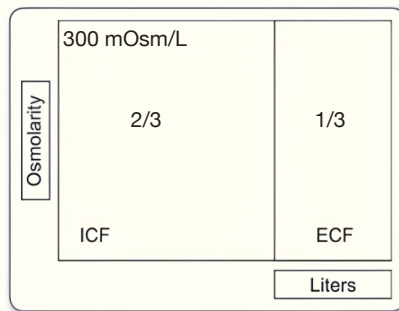
Percentage of body weight :

- male : 60%.
- Female : 50%.
- Children : 75%.

----- Active space -----

Dominant Ions :

| Dominant Ions | ECF           | ICF                                        |
|---------------|---------------|--------------------------------------------|
| Cations       | $\text{Na}^+$ | $\text{K}^+ > \text{mg}^{2+}$              |
| Anions        | $\text{Cl}^-$ | $\text{PO}_4^{3-} > \text{Protein anions}$ |

Darrow-Yannet diagram  :

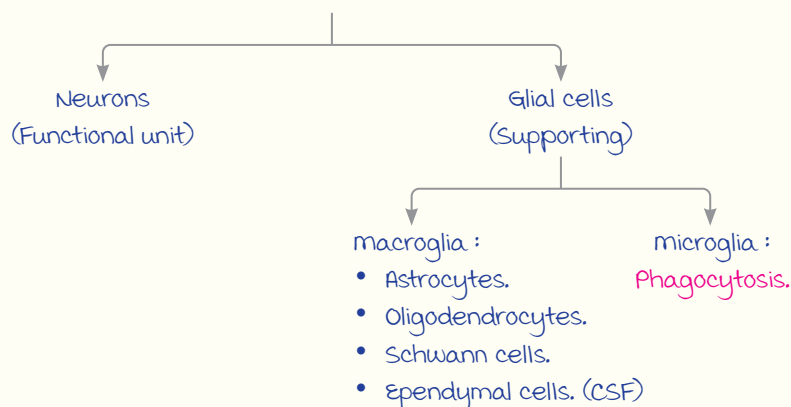
# NERVE MUSCLE PHYSIOLOGY

----- Active space -----

## Nerves

00:00:27

Cells in CNS  :



## Astrocytes :

Types :

- Protoplasmic : Gray matter.
- Fibrous : white matter.

Intermediate filament : Glial fibrillary acidic protein (GFAP).

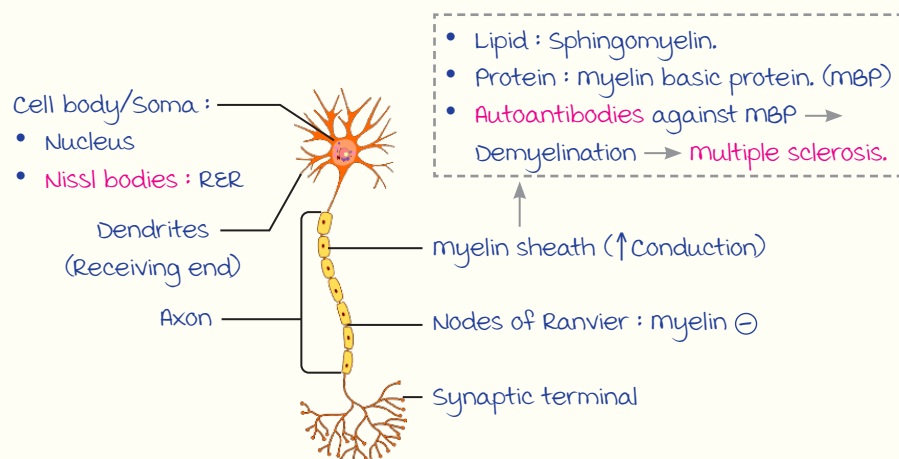
Functions :

- BBB.
- Produces nerve growth factor (NGF).
- Synaptic clearing of  $K^+$  & neurotransmitters.

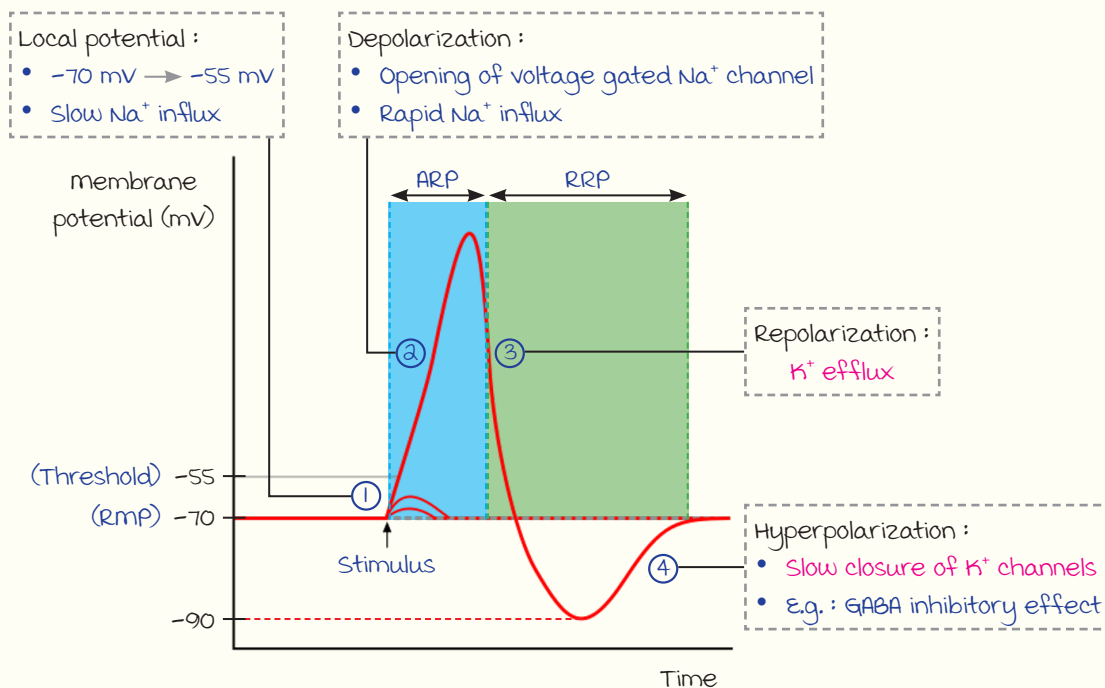
## Oligodendrocytes vs. Schwann Cells :

|             | Oligodendrocytes | Schwann cells |
|-------------|------------------|---------------|
| myelination | CNS              | PNS           |
| Ratio       | One to many      | 1 : 1         |

----- Active space -----

Structure of Neuron :Nerve Action Potential 

00:10:02

Phases :Refractory Periods 

|                                      | Absolute refractory period (ARP)                         | Relative refractory period (RRP)                               |
|--------------------------------------|----------------------------------------------------------|----------------------------------------------------------------|
| Definition                           | Threshold to initial 1/3 <sup>rd</sup> of repolarization | 1/3 <sup>rd</sup> of repolarization to end of action potential |
| Response to 2 <sup>nd</sup> stimulus | $\ominus$                                                | $\oplus$                                                       |
| $\text{Na}^+$ channel                | Inactivated                                              | Begin to open                                                  |

## Classification of Nerve Fibres & Nerve Injury

00:18:02

----- Active space -----

### Classification of Nerve Fibres :

Erlanger Gasser classification  :

| Nerve fibre | myelin     | Diameter                 | Conduction velocity     | Functions                             |
|-------------|------------|--------------------------|-------------------------|---------------------------------------|
| A           | A $\alpha$ | Largest<br>↓<br>Smallest | Fastest<br>↓<br>Slowest | Proprioception, $\alpha$ motor neuron |
|             | A $\beta$  |                          |                         | Touch, pressure                       |
|             | A $\gamma$ |                          |                         | motor to muscle spindle               |
|             | A $\delta$ |                          |                         | Fast pain                             |
| B           | +          |                          |                         | Preganglionic autonomic               |
| C           | -          |                          |                         | Slow pain, postganglionic sympathetic |

Susceptibility to compression : A $\alpha$  > B > C.

Lloyd Hunt classification (Only for sensory fibres):

| Nerve fibre | Function           | Fibre type |
|-------------|--------------------|------------|
| Ia          | muscle spindle     | A $\alpha$ |
| Ib          | Golgi tendon organ |            |
| II          | Touch              | A $\beta$  |
| III         | Fast pain          | A $\delta$ |
| IV          | Slow pain          | C          |

### Nerve Injury :

Distal/Wallerian degeneration  :

- < 24 hrs.
- Axonal degeneration → myelin degeneration → Cleaning of debris.

Proximal degeneration :

- 36 hrs after injury.
- Nucleus pushed to periphery, Nissl bodies → Chromatolysis.

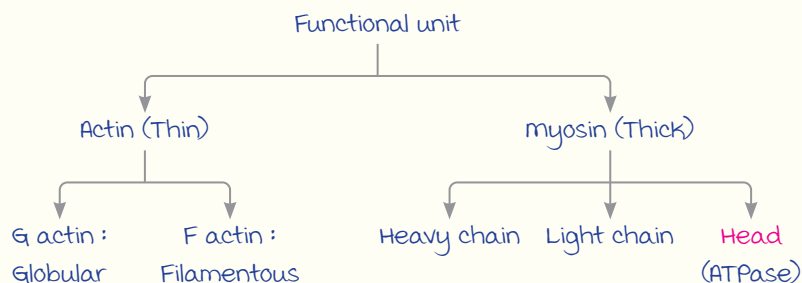
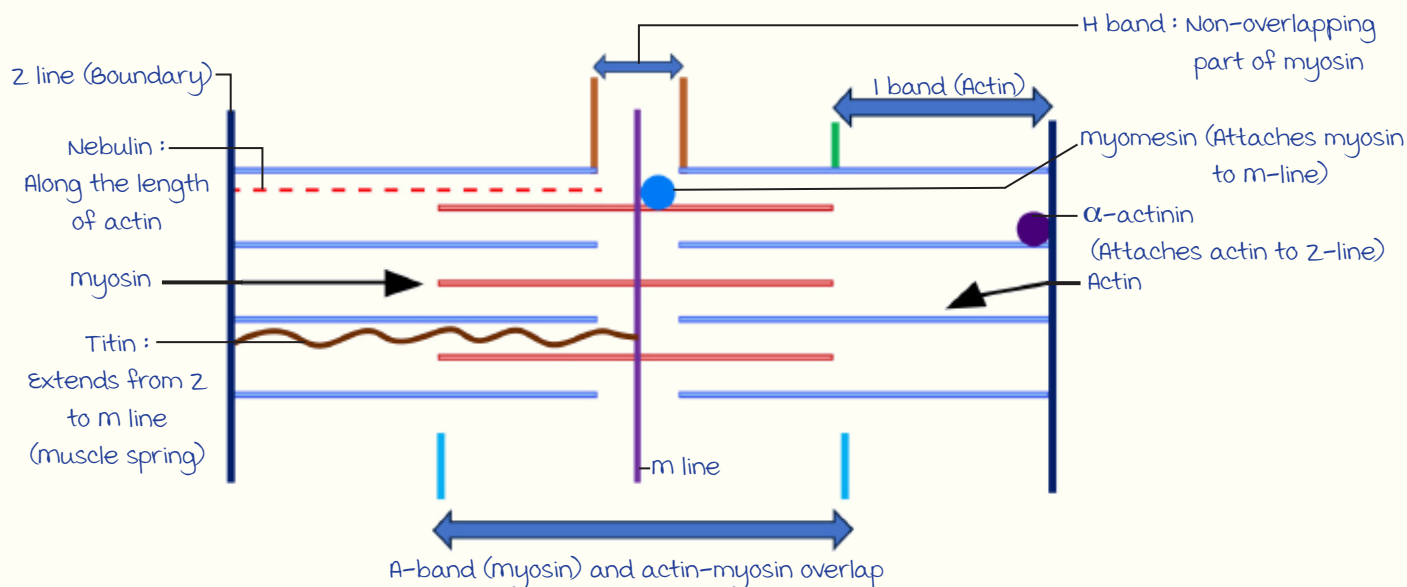
Regeneration :

- 90 hrs after injury.
- 1 - 3 mm/day.
- Depends on presence of NGF : Neurotrophins.
- Schwann cell : Reinnervation + myelination.
- Tinel sign : Tingling (+) on tapping area of injury → Regeneration (+).

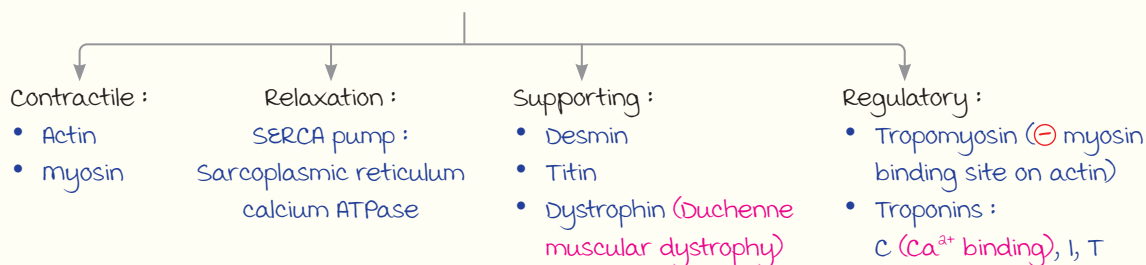
----- Active space -----

**Skeletal Muscle**

00:31:30

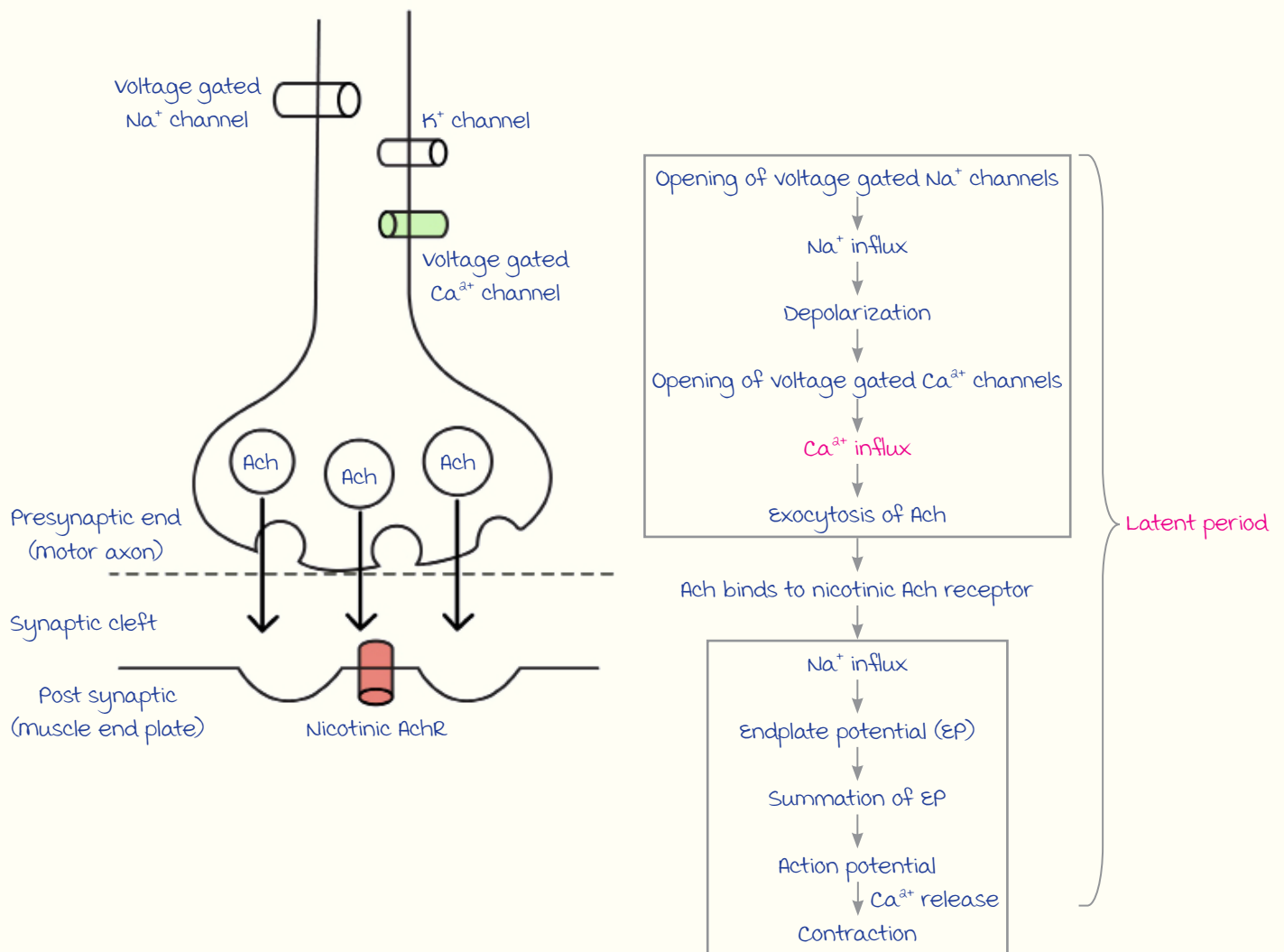
**Sarcomere :****Structure :****muscle contraction  :**

- Z lines move closer.
- I band shortens.
- H zone shortens.
- A band : **No change.**

**Skeletal muscle Proteins :**

## Neuromuscular Transmission :

----- Active space -----



Spontaneous release of ACh  $\rightarrow$  miniature endplate potential.

Latent period :

- Events before muscle contraction.
- Nerve impulse, transmission, excitation - contraction coupling,  $\text{Ca}^{2+}$  release.

Diseases :

Autoantibodies against :

- voltage-gated  $\text{Ca}^{2+}$  channels  $\rightarrow$  Lambert Eaton myasthenic syndrome (LEMS).
- Nicotinic ACh receptors  $\rightarrow$  myasthenia gravis.

----- Active space -----

Skeletal muscle Calcium Channels :

Dihydropyridine receptor (DHPR) :

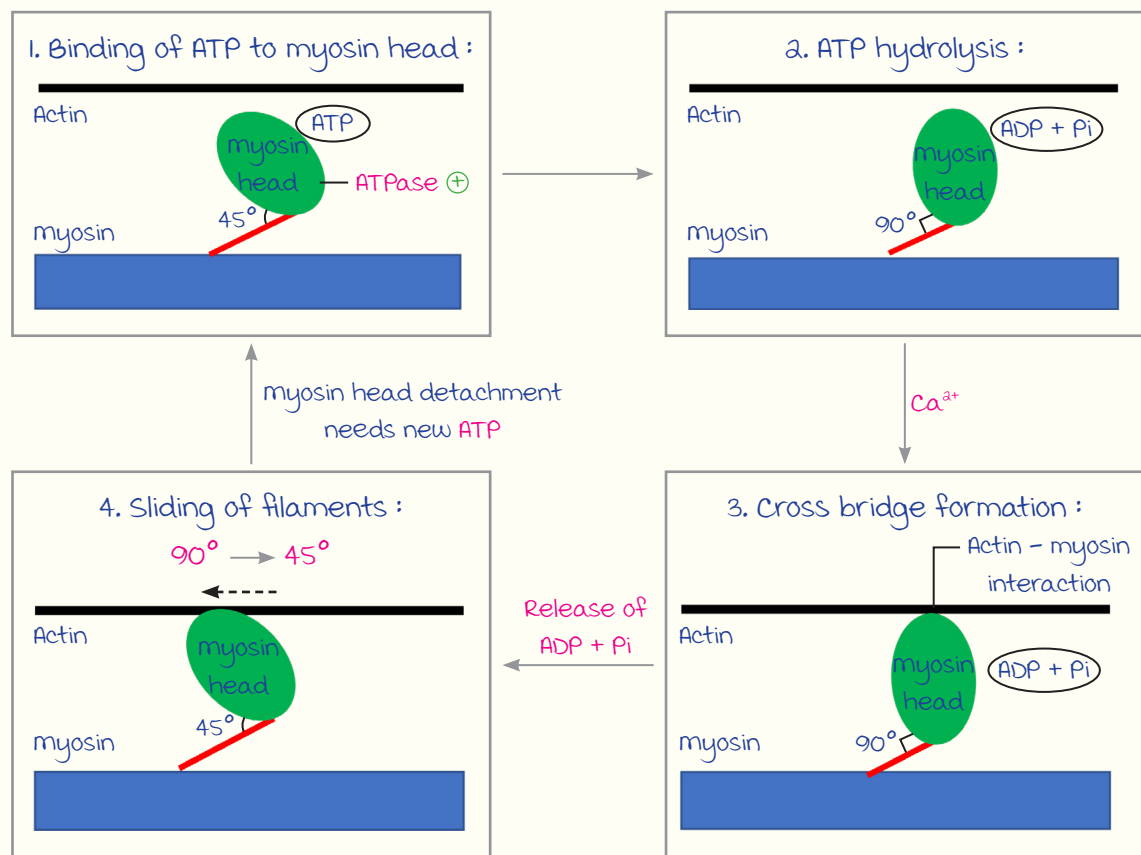
- Sarcolemma.
- Opens RyR (mechanical interaction).

Ryanodine receptor (RyR) :

- Sarcoplasmic reticulum.
- $\text{Ca}^{2+}$  release  $\rightarrow$  Contraction.
- mutation  $\rightarrow$  malignant hyperthermia (Rx : Dantrolene sodium).

Lock & key model  :

Depolarization  $\rightarrow$  DHPR (key) opens RyR (Lock)  $\rightarrow$   $\text{Ca}^{2+}$  release  $\rightarrow$  muscle contraction.

SERCA pump :  $\text{Ca}^{2+}$  uptake  $\rightarrow$  muscle relaxation.molecular mechanism of muscle Contraction :Sliding filament theory  :

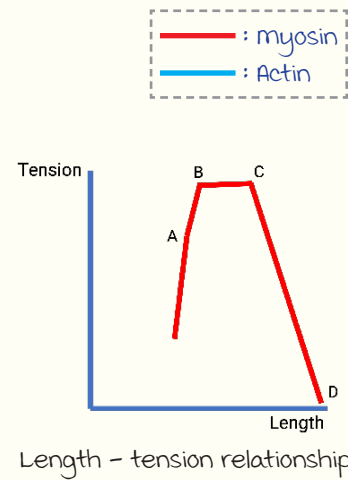
After death  $\rightarrow$  No ATP  $\rightarrow$  No myosin head detachment  $\rightarrow$  Rigor mortis.

## Properties of Skeletal muscles :

Length - tension relationship :

Tension generated  $\propto$  Actin - myosin overlap.

|                                                                                              | Actin - myosin overlap | Tension generated |
|----------------------------------------------------------------------------------------------|------------------------|-------------------|
| Point D<br> | No overlap             | minimum           |
| Point C<br> | Optimal overlap        | maximum           |
| Point B<br> |                        |                   |
| Point A<br> | Overlap not optimal    | Decreases         |



## Types of skeletal muscle fibres :

|                        | Type I          | Type II                     |                 |
|------------------------|-----------------|-----------------------------|-----------------|
|                        |                 | IIa                         | IIb             |
| Other name             | Slow oxidative  | Fast oxidative - glycolytic | Fast glycolytic |
| Diameter               | Small           | Large                       | Large           |
| Colour                 | Red             | Red                         | White           |
| myoglobin              | ⊕ ⊕             | ⊕                           | ⊖               |
| myosin ATPase activity | Slow            | Fast                        | Fast            |
| Glycolytic capacity    | ⊖               | ⊕                           | ⊕ ⊕             |
| Oxidative capacity     | ⊕ ⊕             | ⊕                           | ⊖               |
| Fatigability           | ⊖               | ⊖                           | ⊕               |
| Recruitment order      | 1 <sup>st</sup> | 2 <sup>nd</sup>             | Last            |
| Best suited activities | Sitting         | Standing                    | Running         |

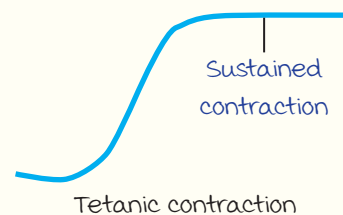
Tetanic contraction :

- Continuous stimulation at a particular frequency

State of sustained contraction (No relaxation)

Tetanic contraction.

- Tetanic frequency =  $\frac{1}{\text{Contraction period}}$



Cardiac muscle :

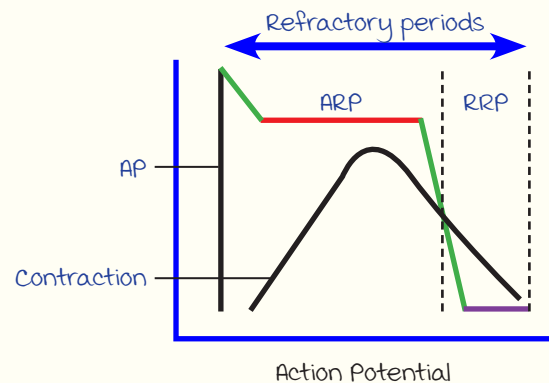
Features :

- Striated.
- Involuntary muscle (Pacemaker).
- **Functional syncytium :**
  - Synchronized contraction.
  - Due to intercalated discs containing **gap junctions** (Connexins).
- **$\text{Ca}^{2+}$  induced  $\text{Ca}^{2+}$  release (CICR) :**  
 Extracellular  $\text{Ca}^{2+}$   $\xrightarrow{\text{Triggers}}$  Intracellular  $\text{Ca}^{2+}$  release.

Absence of tetanization :

major part of contraction in **ARP**

↓  
No summation  
↓  
No tetany.

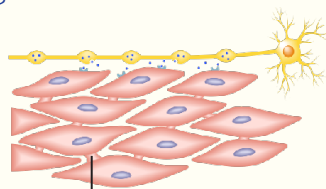
Smooth muscle :

Features :

- Striations  $\ominus$ .
- Involuntary.
- **Dense bodies**  $\oplus$ .
- Calcium binding protein : **Calmodulin**.
- Sarcoplasmic reticulum : Less developed.
- mitochondria : Less.

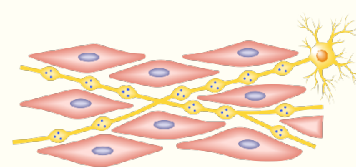
Types :

Single unit smooth muscle :



- Gap junctions  $\oplus$ .
- GIT, uterus, urinary bladder.

multi unit smooth muscle :



- Gap junctions  $\ominus$ .
- Blood vessels.

Properties :

----- Active space -----

- **Synapse en passant :**  
1 neuron  $\xrightarrow{\text{Innervates}}$  multiple smooth muscle cells.
- myosin based regulation :
  - MLCK (myosin light chain kinase) : Contraction.
  - MLCP (myosin light chain phosphatase) : Relaxation.
- **Plasticity :**
  - Stretching  $\xrightarrow{\text{Further stretch}}$  Initial  $\uparrow$  tension  $\rightarrow$   $\downarrow$  Tension.
  - Significant : Storage (E.g. : urinary bladder).
- **Latch bridge phenomenon :**  
Energy efficient : Sustained contraction for long periods with very little use of ATP.

## Synapse

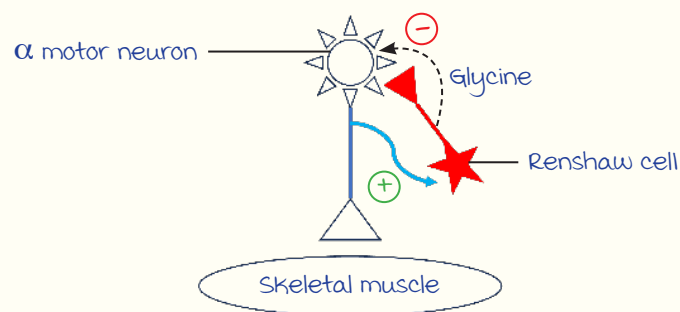
01:16:22

Postsynaptic potentials (PSP) :

|                                        | Excitatory PSP                          | Inhibitory PSP                            |
|----------------------------------------|-----------------------------------------|-------------------------------------------|
| Postsynaptic neuron membrane potential | Positive                                | Negative                                  |
| Fast                                   | $\text{Na}^+$ & $\text{Ca}^{2+}$ influx | $\text{Cl}^-$ influx, $\text{K}^+$ efflux |
| Slow                                   | $\downarrow \text{K}^+$ efflux          | $\uparrow \text{K}^+$ efflux              |
| Neurotransmitter                       | Glutamate                               | GABA                                      |

Feedback inhibition :

AKA **Renshaw cell inhibition.**



----- Active space -----

**Neurotransmitters**

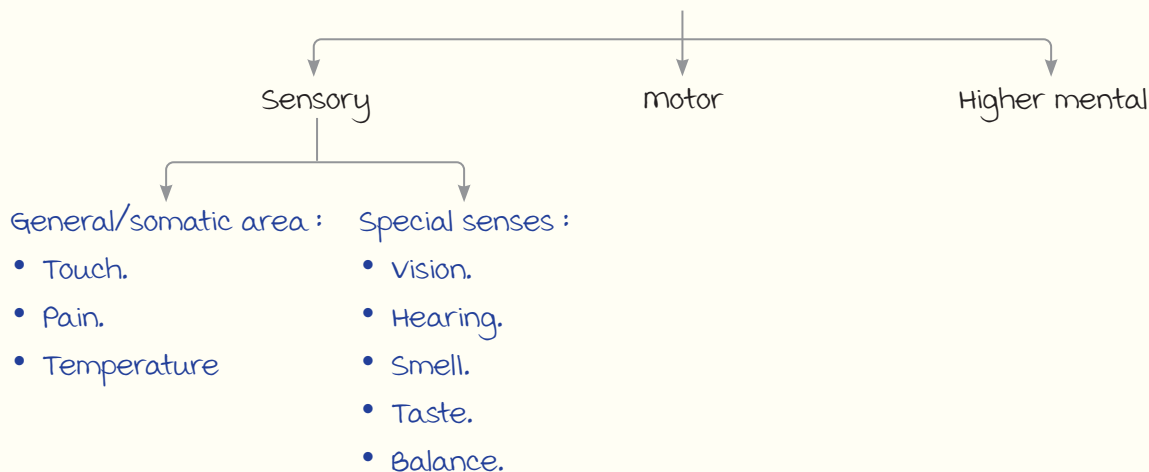
01:23:32

| Neurotransmitter                   | Location                                                                   | Function                                                                                     | Features                                                                                                                                        |
|------------------------------------|----------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Acetylcholine                      | NMJ                                                                        | muscle contraction                                                                           | <ul style="list-style-type: none"> <li>Receptors : Nicotinic &amp; muscarinic</li> <li>Botulinum toxin               <div> <div>↓</div> <div>Degrades synaptobrevin</div> <div>↓ ⊖</div> <div>Ach release</div> </div> </li> </ul> |
|                                    | Nucleus basalis of Meynert                                                 | memory                                                                                       |                                                                                                                                                                                                                                    |
|                                    | REM ON neurons                                                             | REM sleep                                                                                    |                                                                                                                                                                                                                                    |
| Dopamine                           | 1. Nigrostriatal pathway : Basal ganglia                                   | motor control                                                                                | Degeneration<br>↓<br>Parkinson's disease<br><br>-                                                                                                                                                                                  |
|                                    | 2. mesolimbic pathway :<br>• Ventral tegmental area<br>• Nucleus accumbens | → Reward<br>→ Addiction                                                                      |                                                                                                                                                                                                                                    |
|                                    | 3. mesocortical pathway                                                    | <ul style="list-style-type: none"> <li>motivation</li> <li>Regulation of emotions</li> </ul> |                                                                                                                                                                                                                                    |
|                                    | 4. Tuberoinfundibular pathway                                              | ⊖ Prolactin                                                                                  |                                                                                                                                                                                                                                    |
| Serotonin<br>(5-hydroxytryptamine) | Raphe nuclei                                                               | wakefulness                                                                                  | -                                                                                                                                                                                                                                  |
|                                    | GIT                                                                        | Peristalsis                                                                                  |                                                                                                                                                                                                                                    |
|                                    | Platelets                                                                  | Aggregation                                                                                  |                                                                                                                                                                                                                                    |
| Norepinephrine                     | Locus coeruleus                                                            | Arousal                                                                                      | Projects widely in brain                                                                                                                                                                                                           |
| Histamine                          | Posterior hypothalamus                                                     | Arousal                                                                                      | Histamine ⊖<br>↓<br>Sedation                                                                                                                                                                                                       |
| Glutamate                          | Hippocampus, subthalamic nucleus                                           | <ul style="list-style-type: none"> <li>Learning</li> <li>memory</li> </ul>                   | major excitatory NT                                                                                                                                                                                                                |
| GABA                               | -                                                                          | Hyperpolarization                                                                            | <ul style="list-style-type: none"> <li>major inhibitory NT</li> <li>Tetanus spasmin               <div> <div>↓ ⊖</div> <div>GABA</div> <div>↓</div> <div>Spastic paralysis</div> </div> </li> </ul>                                |
| Glycine                            | -                                                                          | Renshaw cell inhibition                                                                      | <ul style="list-style-type: none"> <li>Both excitatory &amp; inhibitory</li> <li>Antagonist : Strychnine</li> </ul>                                                                                                                |
| Nitric oxide                       | Hippocampus                                                                | Learning & memory<br>(↑ Glutamate release)                                                   | -                                                                                                                                                                                                                                  |
| Carbon monoxide                    | -                                                                          | <ul style="list-style-type: none"> <li>Olfaction</li> <li>Pain processing</li> </ul>         | Heme<br>Heme ↓ oxygenase II<br>CO                                                                                                                                                                                                  |

# NEUROPHYSIOLOGY : PART 1

----- Active space -----

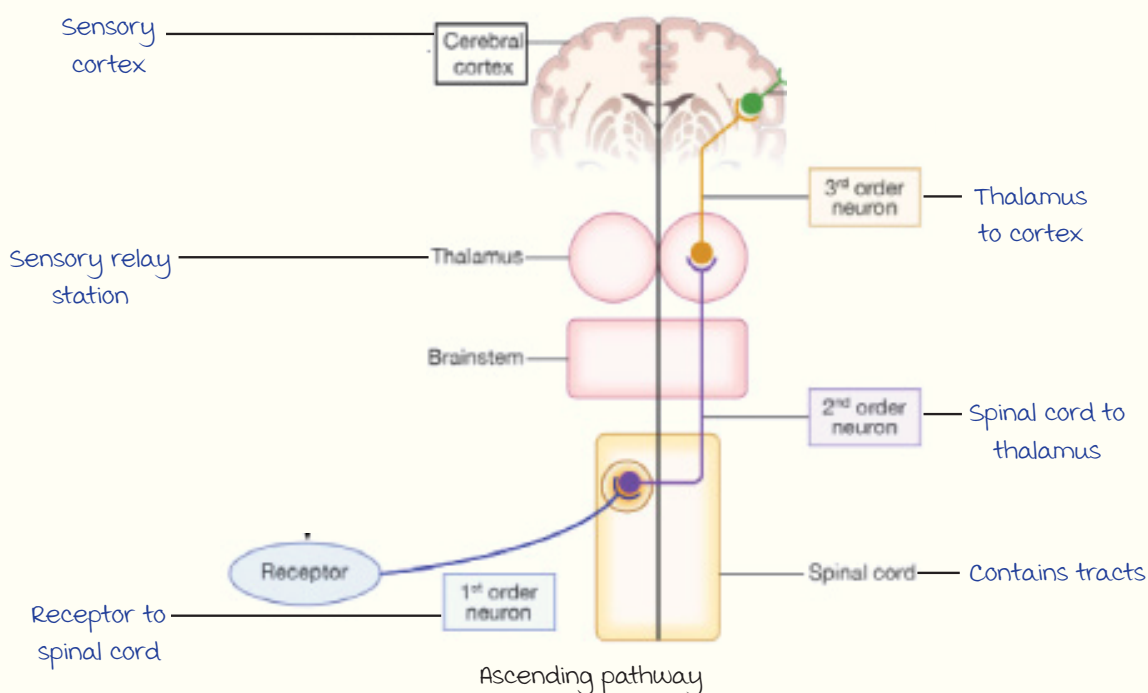
## Major Functions of Brain :



## Sensory Physiology

00:01:30

### Sensory pathways (Afferents and ascending) :



----- Active space -----

## Touch Receptors

00:03:15

|            | meissners corpuscle                                                                            | merkel cells                                                                                                           | Pacinian corpuscle                                                                                      | Ruffini endings |
|------------|------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|-----------------|
| Level      | Superficial                                                                                    |                                                                                                                        | Deep                                                                                                    |                 |
| Site       | Non hairy skin :<br>Fingertips, lips                                                           | -                                                                                                                      | -                                                                                                       | Joint capsules  |
| Stimuli    | <ul style="list-style-type: none"> <li>Light touch</li> <li>Low frequency vibration</li> </ul> | <ul style="list-style-type: none"> <li>Texture, edges :<br/>Braille reading</li> <li>2 point discrimination</li> </ul> | <ul style="list-style-type: none"> <li>Deep touch/pressure</li> <li>High frequency vibration</li> </ul> | Stretch         |
| Adaptation | Rapid                                                                                          | Slow                                                                                                                   | Rapid                                                                                                   | Slow            |

## Pain Receptors

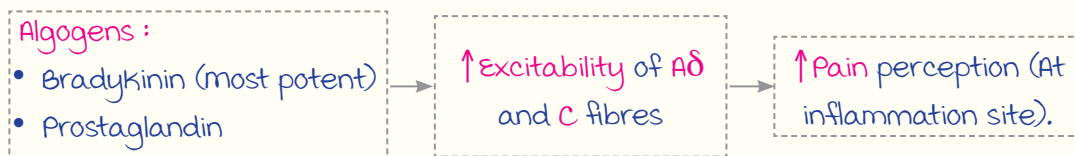
00:08:30

- AKA **Nociceptors**.
- Free nerve endings of A $\delta$  and C fibres.
- Stimulus : Pin prick (Not blunt forceps).

### A $\delta$ vs. C Fibres :

|                   | A $\delta$ fibres                       | C fibres                                  |
|-------------------|-----------------------------------------|-------------------------------------------|
| Pain              | 1 <sup>st</sup> /Fast/ <b>Epicritic</b> | 2 <sup>nd</sup> /Slow/ <b>Protopathic</b> |
| Character of pain | Sharp and pricking                      | Dull and burning                          |
| myelination       | ⊕                                       | ⊖                                         |
| Evolutionarily    | New (Neospinothalamic)                  | Old (Paleospinothalamic)                  |
| Neurotransmitters | Glutamate                               | Substance P                               |

### Hyperalgesia :



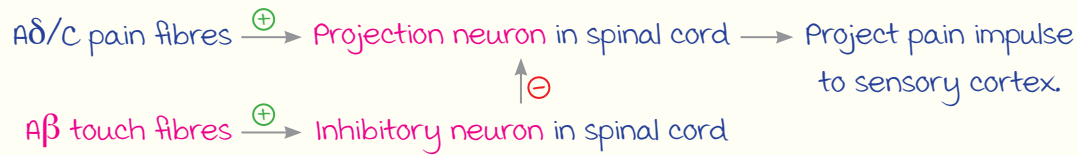
### Allodynia :

Normal non painful stimulus  $\xrightarrow{\text{Perceived as}}$  Painful stimulus.  
(E.g. : Light touch on sunburnt skin)

### Gate Control Theory :

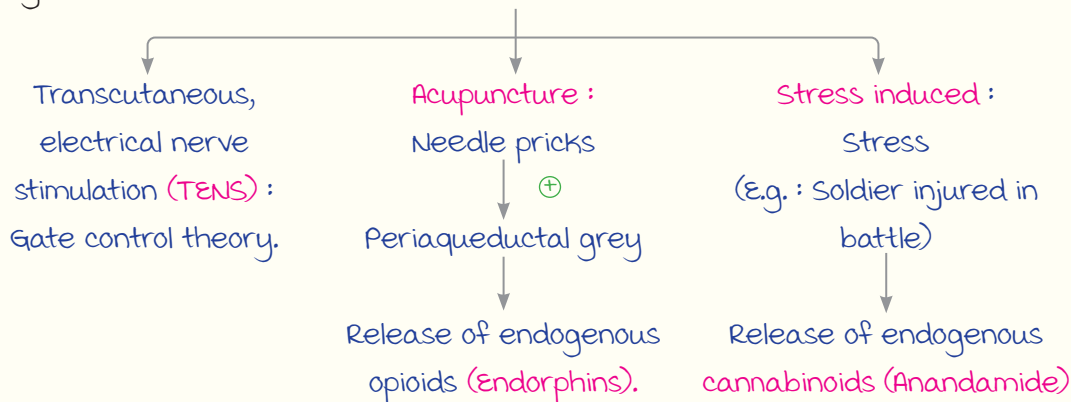
- Proposed by **melzack and wall**.
- Touch** sensation (massages)  $\ominus$   $\rightarrow$  Pain transmission.
- Gate : **Substantia gelatinosa** (Dorsal horn).

Mechanism :



### Clinical Application :

Analgesia :

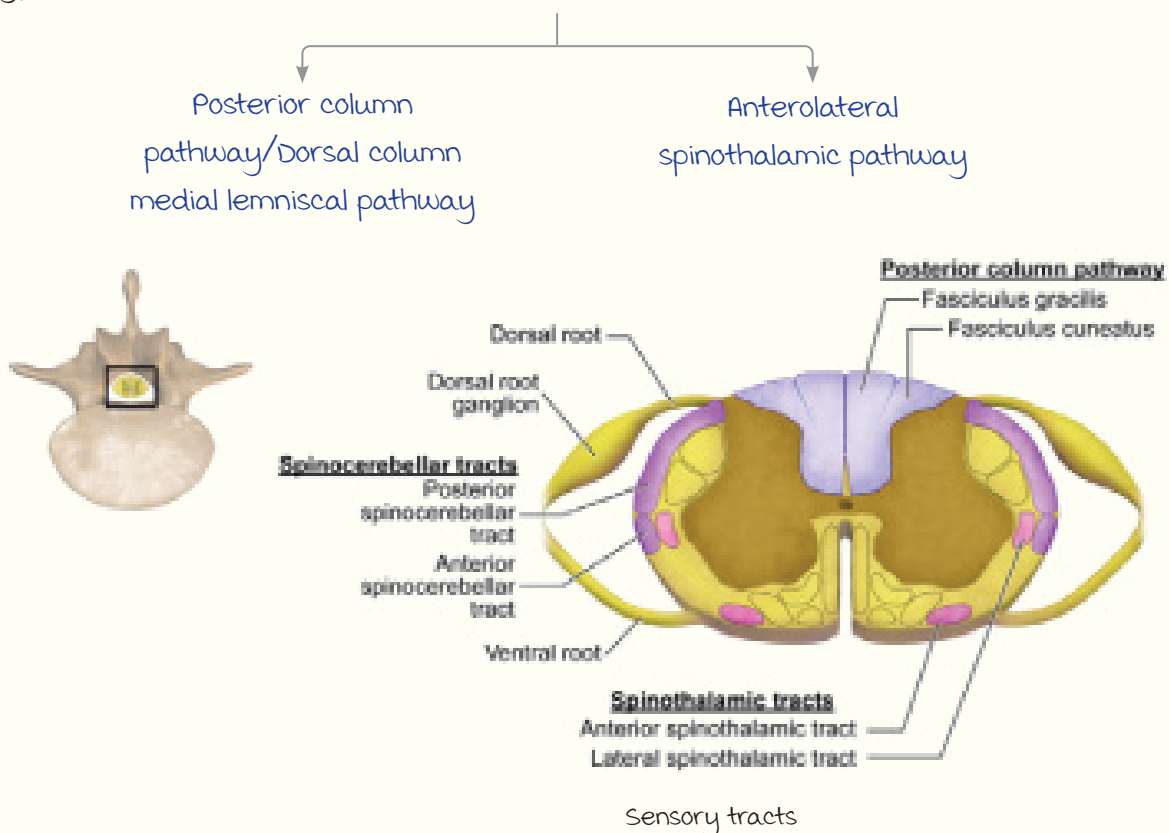


### Somatosensory Pathways

00:19:51

### Spinal Cord Tracts :

Types :

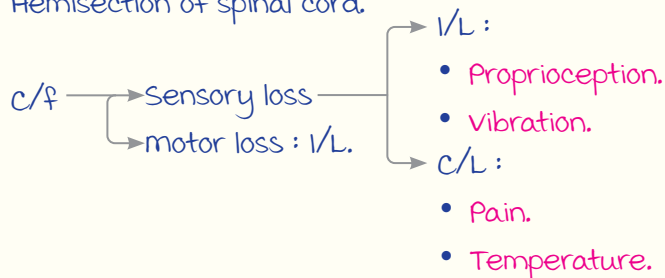


----- Active space ----- Dorsal column pathway vs. Anterolateral pathway :

|            | Dorsal column medial lemniscal pathway                                                                                                                                                                        | Anterolateral pathway                                                                                                                                                                           |
|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Sensations | <ul style="list-style-type: none"> <li>• <b>Proprioception.</b></li> <li>• Fine touch.</li> <li>• Vibration.</li> <li>• <b>Stereognosis</b> (Identification of objects by touch with closed eyes).</li> </ul> | <ul style="list-style-type: none"> <li>• <b>Pain</b> → Lateral pathway.</li> <li>• Temperature.</li> <li>• Crude touch</li> <li>• Pressure</li> <li>• Itch.</li> </ul> <p>Anterior pathway.</p> |
| Side in SC | <b>Ipsilateral</b>                                                                                                                                                                                            | <b>Contralateral</b>                                                                                                                                                                            |
| Pathway    |                                                                                                                                                                                                               |                                                                                                                                                                                                 |

Brown - Sequard syndrome :

Hemisection of spinal cord.



Thalamus :

Thalamic nuclei :

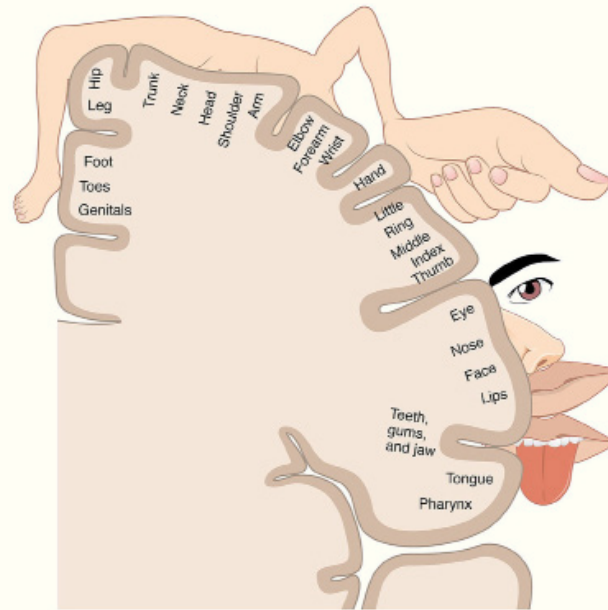
- ventral posterolateral (VPL).
- ventral posteromedial (VPM) : General senses from **face**.

## Sensory Cortex :

- Parietal (Post central).
- Brodmann areas : 3, 1, 2.

## Sensory homunculus :

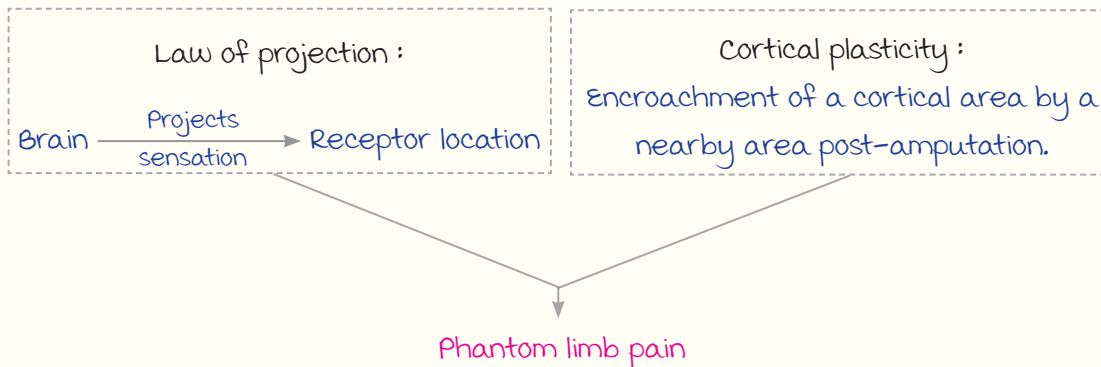
- Representation  $\propto$  use (Not size).
- maximum : Lips & thumb.
- minimum : Trunk & back.



Sensory homunculus

## Laws in Sensory Physiology

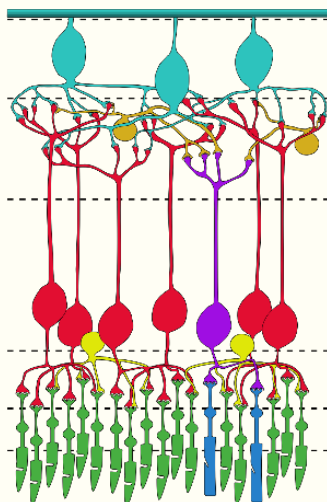
00:27:36



## Vision

00:29:08

## Retina :



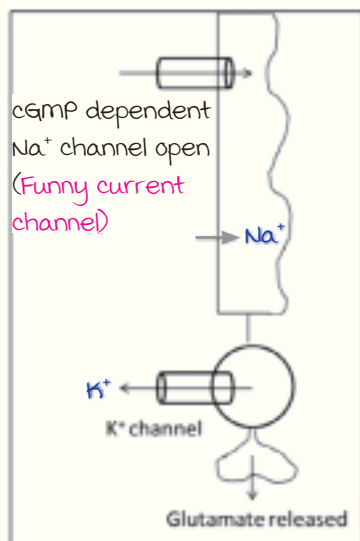
Retinal cells

| Cells                             | Features                                |                      |
|-----------------------------------|-----------------------------------------|----------------------|
| Rods                              | Dim vision, periphery, higher number    |                      |
| Cones                             | Colour vision, central, lower number    |                      |
| Bipolar                           | Connects rods & cones to ganglion cells |                      |
| Ganglion<br>(Axons : Optic nerve) | Only output cells → Action potential    |                      |
| Horizontal                        | B/w rods, cones and bipolar             | Contrast enhancement |
| Amacrine                          | B/w bipolar and ganglion cells          |                      |
| muller's/Retinal glial            | Supporting cells (glial)                |                      |

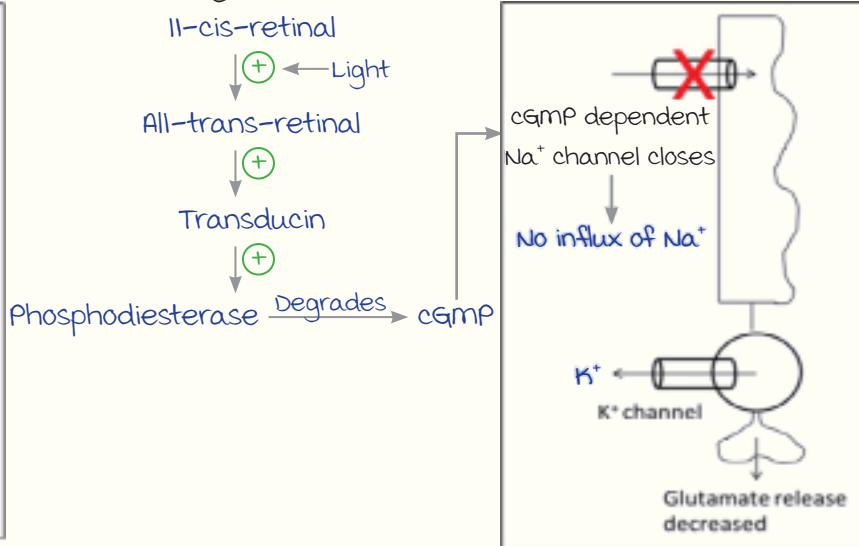
----- Active space -----

Physiology of Vision :

a) Dark state :

Rod in dark state : **Depolarisation**

b) Light state :

Rod in light state : **Hyperpolarisation**Lateral Geniculate Body (Thalamus) :

Layers : 6

- 1, 2 : magnocellular (Big cells)
- 3 - 6 : Parvocellular (Small cells)

Pathways :

|                   | magnocellular         | Parvocellular                                                                             | Koniocellular      |
|-------------------|-----------------------|-------------------------------------------------------------------------------------------|--------------------|
| Origin layer      | 1 and 2               | 3 - 6                                                                                     | -                  |
| Termination layer | 4 (Visual cortex)     |                                                                                           | -                  |
| Function          | Extraocular movements | <ul style="list-style-type: none"> <li>• Colour vision</li> <li>• Fine details</li> </ul> | Blue colour vision |

Reception of stimuli :

- Layers 2, 3, 5 : I/L eye.
- Layers 1, 4, 6 : C/L eye.

Visual Cortex :

Brodmann areas : 17, 18, 19.

Striate cortex/stria of Gennari :

- Occipital lobe (Calcarine sulcus).
- Has myelinated fibres.
- Blobs : Colour vision.

Extrastriate cortex :

- Temporal.
- Parietal.

## Colour Vision :

----- Active space -----

Retina → Parvocellular pathway (Thalamus) → Blobs (visual cortex).

Types of cones :

|            | L cone     | m cone     | S cone     |
|------------|------------|------------|------------|
| Wavelength | Long       | medium     | Small      |
| Colour     | Red        | Green      | Blue       |
| Defect     | Protanopia | Deutanopia | Tritanopia |

m/c defect : Red green.

## Hearing

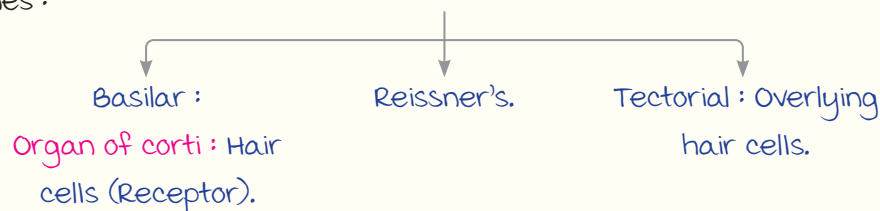
00:44:44

### Cochlea :

Parts :

|                 | Fluid     | Ions            | Resembles | Features                            |
|-----------------|-----------|-----------------|-----------|-------------------------------------|
| Scala vestibuli | Perilymph | Na <sup>+</sup> | ECF       | -                                   |
| Scala tympani   |           |                 |           |                                     |
| Scala media     | Endolymph | K <sup>+</sup>  | ICF       | Endolymphatic potential :<br>+85 mV |

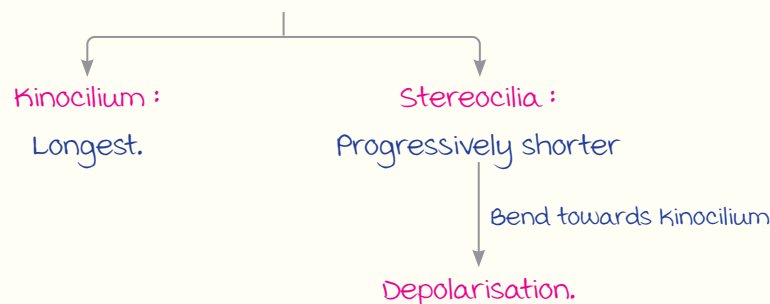
membranes :



Hair cells :

|       | Number | Innervation |
|-------|--------|-------------|
| Outer | ↑      | ↓           |
| Inner | ↓      | ↑           |

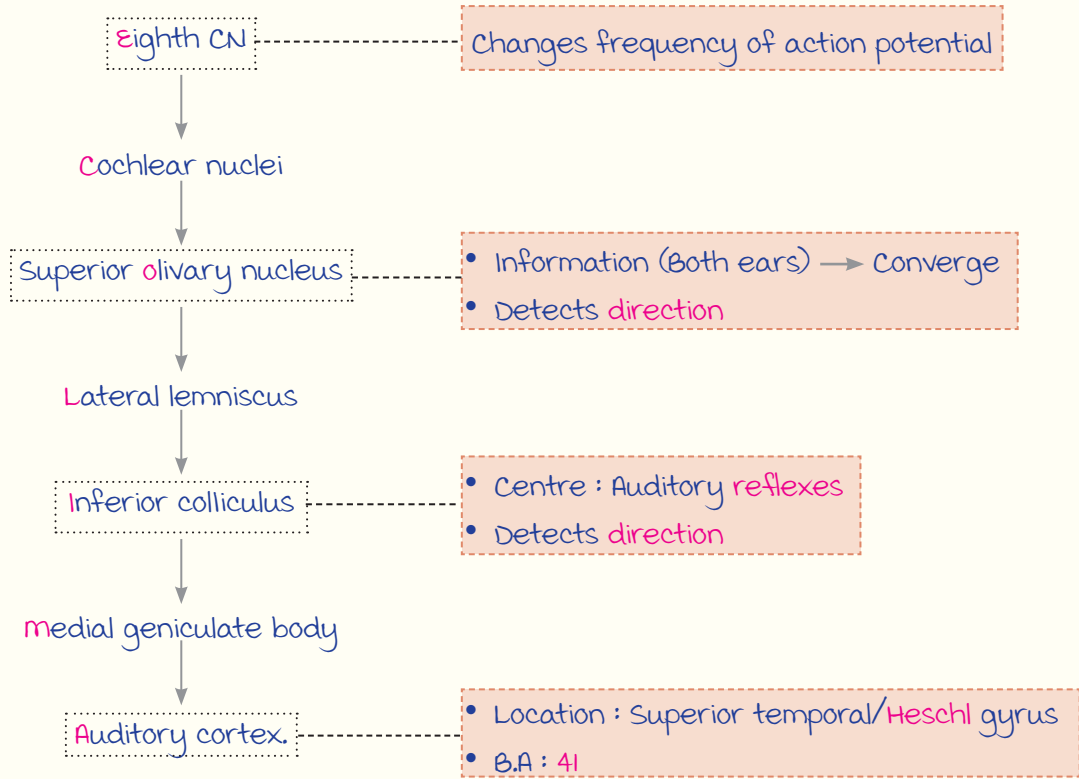
Cilia :



----- Active space -----

Auditory Pathway :

mnemonic : ECOLIMA.



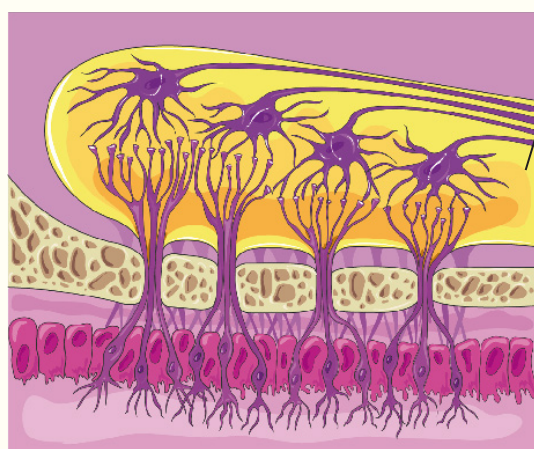
# NEUROPHYSIOLOGY : PART 2

----- Active space -----

## Olfaction

00:00:18

To detect odorants.



Olfactory sensory neurons

Bowman's gland :

Secretes → mucus.  
→ Odorant binding proteins (OBPs).

Olfactory bulb (4 cells) :

|                  | Excitatory cells                                                                                         | Inhibitory cells                                                                  |
|------------------|----------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| Neurotransmitter | Glutamate                                                                                                | GABA                                                                              |
| Example          | <ul style="list-style-type: none"> <li>mitral } <b>Olfactory glomeruli.</b></li> <li>Tufted }</li> </ul> | <ul style="list-style-type: none"> <li>Granule</li> <li>Periglomerular</li> </ul> |

Cribriform plate (Fracture : Loss of smell)

Olfactory epithelium  :

- In roof of nasal cavity
- Receptors :  
**Olfactory sensory neurons (OSNs)** → **CN I**
- **Supporting cells** : Sustentacular cells
- Regenerative cells : Basal stem cells

 Components of olfaction

## Olfactory Pathway :

| Parts                                |                          | Functions                             |
|--------------------------------------|--------------------------|---------------------------------------|
| Olfactory/orbitofrontal cortex (OFC) |                          | Conscious discrimination of odors     |
| Thalamus : Dorsomedial nucleus       |                          | Relay                                 |
| Limbic cortex (Smell brain)          | Piriform cortex          | -                                     |
|                                      | <b>Amygdala</b>          | Associates smell with <b>emotions</b> |
|                                      | <b>Entorhinal cortex</b> | Associates smell with <b>memory</b>   |

Note :

Entorhinal cortex : Earliest site of degeneration in Alzheimer's disease.

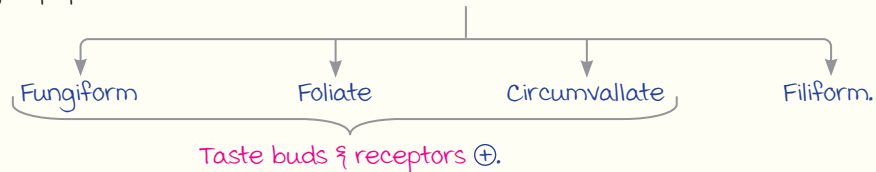
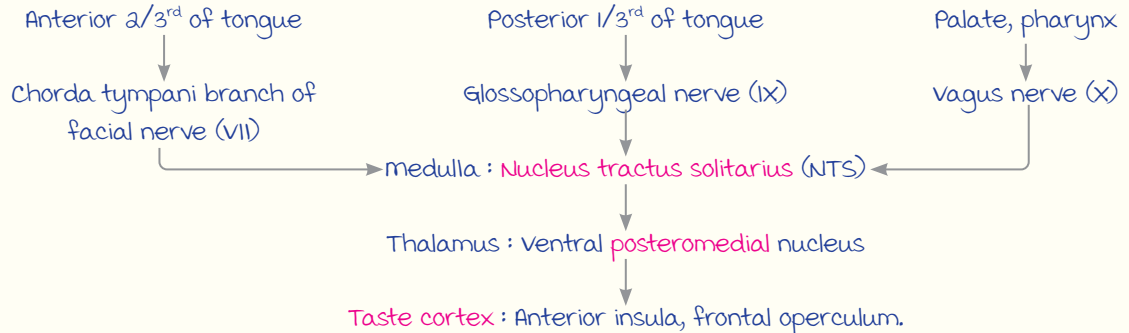
----- Active space -----

**Gustation** 

00:08:19

| Sensations | Receptor type                                                                                | Receptors                                                                                                                                               | Features                                                             |
|------------|----------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|
| Salt       | Ionotropic : $\text{Na}^+$                                                                   | Epithelial Na channels (ENaC)                                                                                                                           | -                                                                    |
| Sour       | Ionotropic : $\text{H}^+$                                                                    | <ul style="list-style-type: none"> <li>• ENaC</li> <li>• Transient receptor potential (TRP)</li> </ul>                                                  |                                                                      |
| Sweet      | metabotropic : (GPCR)                                                                        | <ul style="list-style-type: none"> <li>• <math>\text{TR}_{\text{a}}</math></li> <li>• <math>\text{TR}_{\text{s}}</math></li> </ul>                      | -                                                                    |
| Bitter     |                                                                                              | $\text{T}_{\text{a}}\text{R}$                                                                                                                           | Evolutionary adaptation for survival (Environmental toxins : Bitter) |
| Umami      | metabotropic : <ul style="list-style-type: none"> <li>• Glutamate</li> <li>• GPCR</li> </ul> | <ul style="list-style-type: none"> <li>• <math>\text{mGluR}_4</math></li> <li>• <math>\text{TR}_1</math></li> <li>• <math>\text{TR}_3</math></li> </ul> | Produced by monosodium glutamate (MSG)                               |

Tongue papillae :

**Taste Pathway :****Motor Physiology**

00:14:46

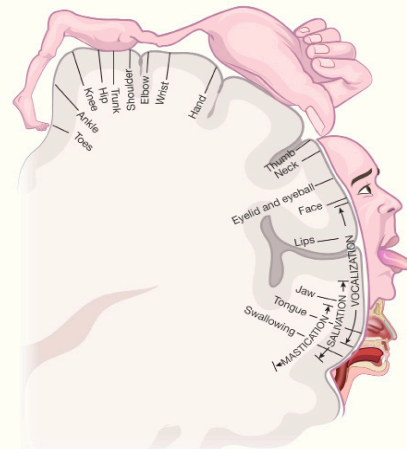
motor pathway (Efferent &amp; descending) :

Cortex & brainstem : UMNs → Spinal cord : LMNs ( $\alpha$  &  $\gamma$ ) → Skeletal muscle.**motor Cortex :**

|                            | Location                    | Brodmann area | Function                                                                                                                |
|----------------------------|-----------------------------|---------------|-------------------------------------------------------------------------------------------------------------------------|
| Primary motor cortex       | Precentral : Frontal region | 4             | Execution of motor movements                                                                                            |
| Pre motor cortex           | -                           | 6             | <ul style="list-style-type: none"> <li>• Orient the body for movement</li> <li>• Imitation by mirror neurons</li> </ul> |
| Supplementary motor cortex | -                           |               | Bimanual coordination                                                                                                   |
| Cortical association areas | Posterior parietal cortex   | -             | Generation of ideas, intention, and plan                                                                                |

motor Homunculus :

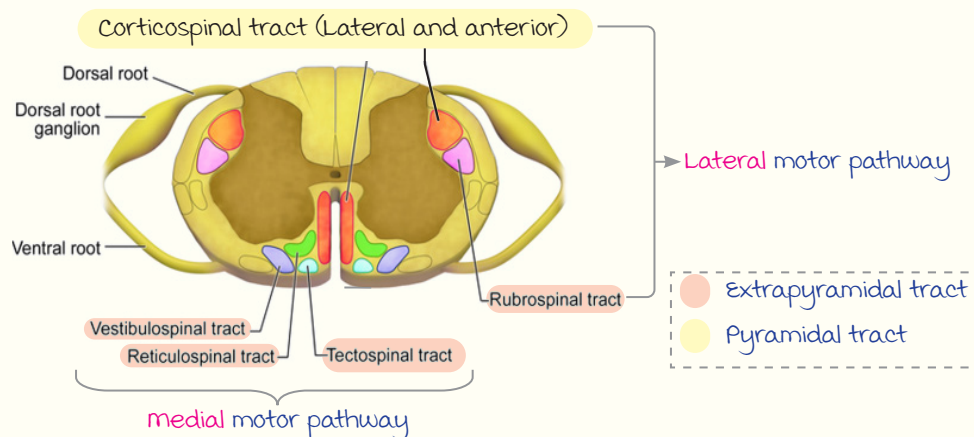
- Representation  $\propto$  use (Not size of body part).
- max, representation :
  - a. Vocalization : Lips.
  - b. Writing : Thumb.
- Arrangement :
  - a. At top : Legs.
  - b. At bottom : Face.



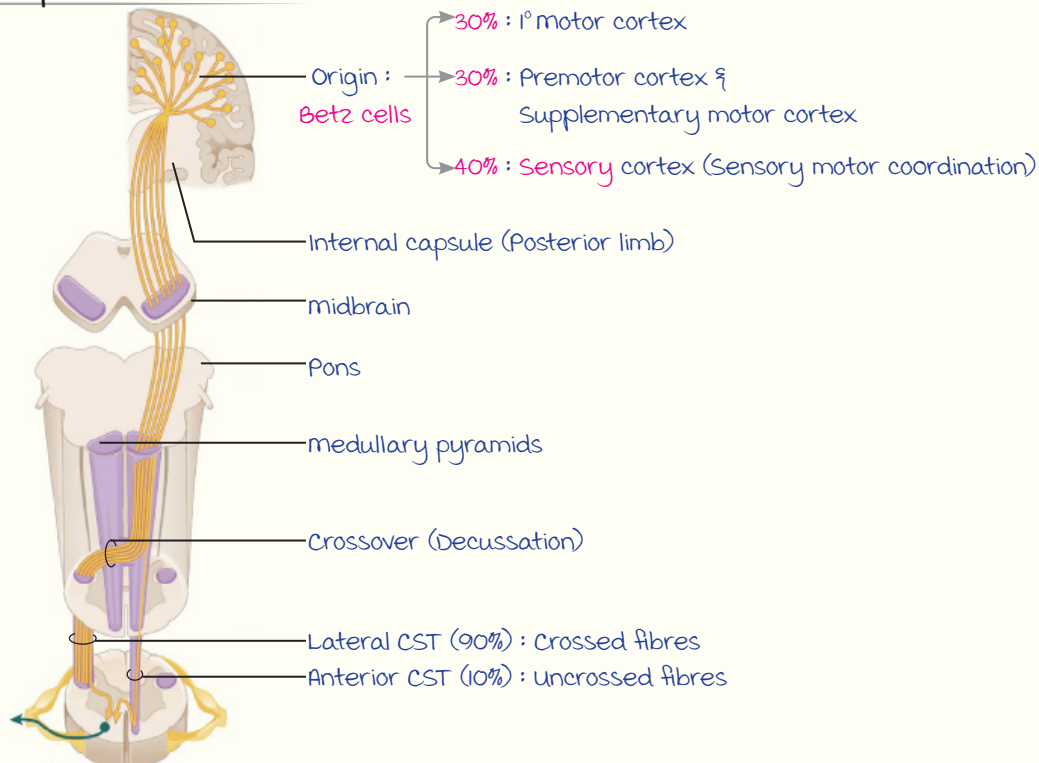
motor homunculus

## Motor Tracts

00:21:32



## Corticospinal Tract (CST) :



----- Active space -----

Lateral vs. medial motor Pathway :

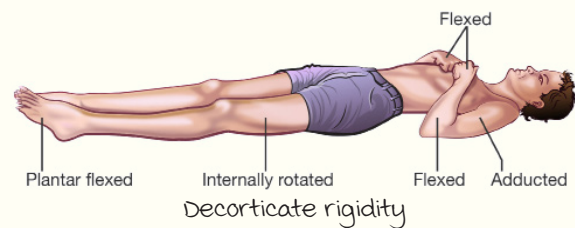
|                       | motor pathways                                                                                                          |                                                                                                                                                                                                                                                                                                  |
|-----------------------|-------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                       | Lateral                                                                                                                 | medial                                                                                                                                                                                                                                                                                           |
| Function              | <ul style="list-style-type: none"> <li>Fine</li> <li>Precise</li> <li>voluntary</li> </ul> } movements (E.g. : writing) | <ul style="list-style-type: none"> <li>Posture</li> <li>muscle tone :               <ul style="list-style-type: none"> <li>Reticulospinal tract (γ motor neurons)                   <ul style="list-style-type: none"> <li>Pontine : ⊕</li> <li>medullary : ⊖</li> </ul> </li> </ul> </li> </ul> |
| muscle group supplied | Distal flexors                                                                                                          | Proximal extensors                                                                                                                                                                                                                                                                               |

Disorders of Motor Pathway

00:30:49

Abnormal Postures :

|                                   | Lesion                                           | Pathophysiology                                                                                                                      | c/f                                                                                                                                                                                                               |
|-----------------------------------|--------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Decerebrate/<br>extensor rigidity | Transection b/w superior and inferior colliculus | Brainstem pathway<br>↓<br>Independent                                                                                                | <ul style="list-style-type: none"> <li>Lower and upper extremities <b>extended</b></li> <li>Toes pointed <b>inward</b></li> <li>Finger/forearms <b>pronated</b></li> <li>Neck and head <b>extended</b></li> </ul> |
| Decorticate/<br>flexor rigidity   | Removal of cerebral cortex                       | <ul style="list-style-type: none"> <li>CST <del>Control</del> → Lower brainstem pathway</li> <li>Rubrospinal tract intact</li> </ul> | <ul style="list-style-type: none"> <li>Lower extremities extended</li> <li>Toes pointed slightly inward</li> <li><b>upper extremities flexed</b> (Against chest)</li> <li>Head extended</li> </ul>                |

UMN vs. LMN Lesions :

| Signs         |                | UMN lesion         | LMN lesion    |
|---------------|----------------|--------------------|---------------|
| muscle        | Atrophy        | -                  | ++            |
|               | Fasciculations | -                  | ++            |
|               | Tone           | ↑↑                 | ↓             |
|               | Weakness       | Regional group     | Segmental     |
| Reflexes      |                | ↑↑ (Hyperreflexia) | ↓↓ (Sluggish) |
| Babinski sign |                | +ve                | -ve           |

Babinski sign : Dorsiflexion of great toe and fanning of other toes.

## Reflexes

00:38:40

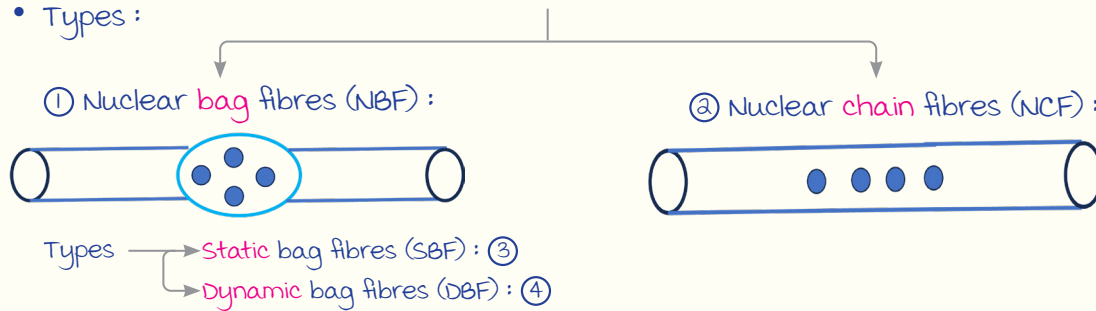
----- Active space -----

### Stretch Reflex :

muscle stretch  $\xrightarrow{\text{Induces}}$  muscle contraction.

muscle spindles (Receptors) :

- Types :



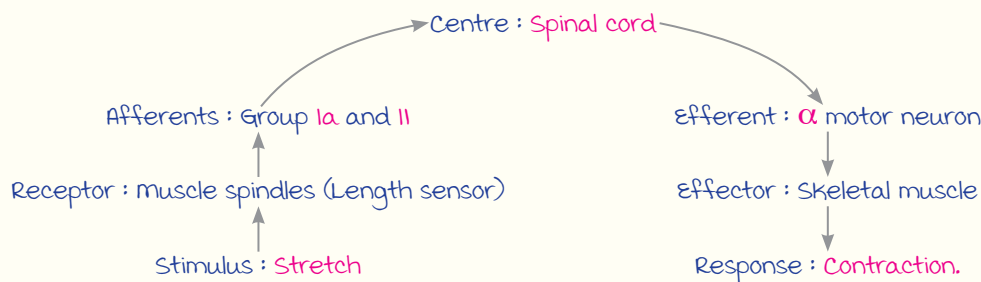
- Sensory innervation :

|            | Primary endings                                                                          | Secondary endings                                                                        |
|------------|------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| AKA        | <ul style="list-style-type: none"> <li>Group Ia</li> <li>Annulospiral endings</li> </ul> | <ul style="list-style-type: none"> <li>Group II</li> <li>Flower spray endings</li> </ul> |
| Innervates | ②, ③, ④                                                                                  | ②, ③                                                                                     |

- motor innervation :  $\gamma$  motor neurons (Periphery).

Pathway :

monosynaptic reflex :



$\alpha$ - $\gamma$  co-activation :



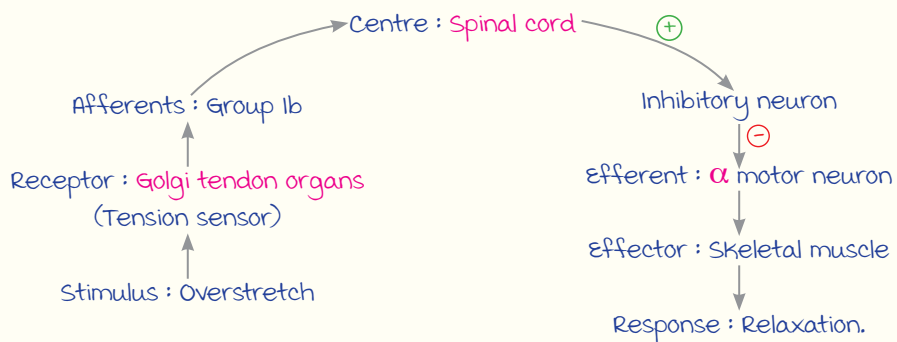
----- Active space -----

Inverse Stretch Reflex :

- Overstretch of muscle  $\longrightarrow$  muscle relaxation.
- To prevent muscle injury.

Pathway :

Disynaptic reflex :

Withdrawal Reflex :

Polysynaptic reflex.

|          | Affected limb                                                                                                                 | unaffected limb                                                                                                               |
|----------|-------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| Response | <ul style="list-style-type: none"> <li>• <math>\oplus</math> : Flexion</li> <li>• <math>\ominus</math> : Extension</li> </ul> | <ul style="list-style-type: none"> <li>• <math>\oplus</math> : Extension</li> <li>• <math>\ominus</math> : Flexion</li> </ul> |
| Reason   | To withdraw limb away from painful stimulus                                                                                   | For balance                                                                                                                   |

# NEUROPHYSIOLOGY : PART 3

----- Active space -----

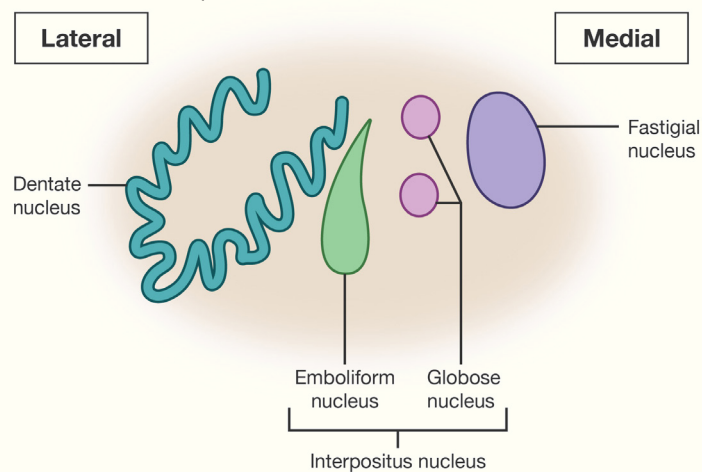
## Cerebellum

00:00:30

### Parts :

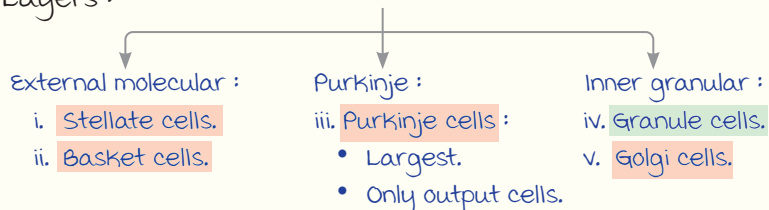
|                      |         | Known as            | Functions                 |
|----------------------|---------|---------------------|---------------------------|
| Vermis               |         | Spinocerebellum     | Coordination of movements |
| Hemisphere           | medial  |                     |                           |
|                      | Lateral | Neocerebellum       | Planning of movements     |
| Flocculonodular lobe |         | Vestibulocerebellum | Balance and equilibrium   |

### Nuclei (Deep) :



### Cells :

#### Layers :

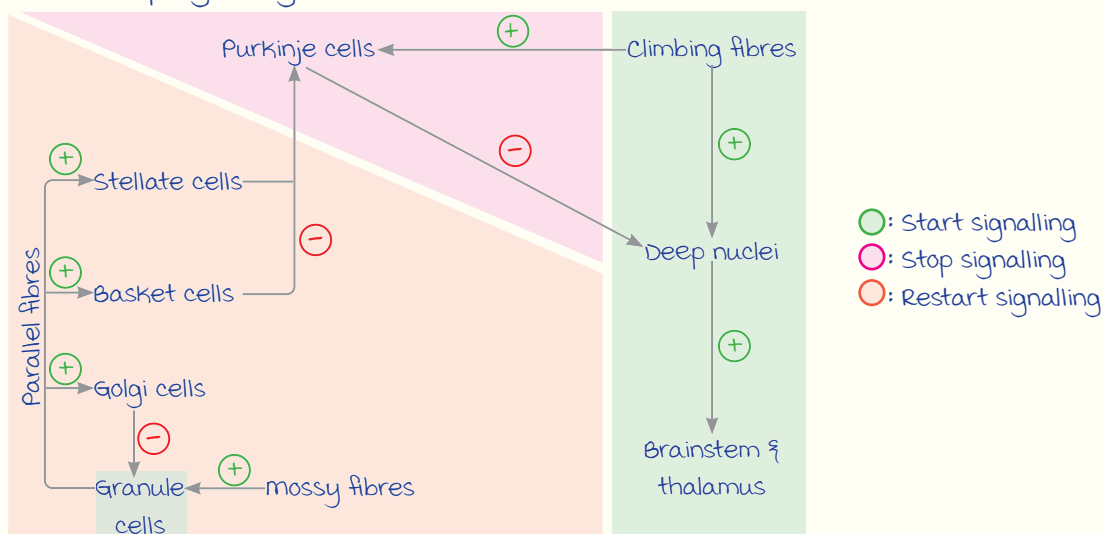


### Sensory Inputs :

|                 | Source                   | AKA             |
|-----------------|--------------------------|-----------------|
| Olivocerebellar | Inferior olivary nucleus | Climbing fibres |
| Spinocerebellar | Spinal cord              | mossy fibres    |

### Function :

#### Start - stop signalling.



----- Active space -----

**C/f of Cerebellar Lesions**  :**Speech :**

- Dysarthria.
- Scanning speech.

**Limbs :**

- Decomposition of movements.
- Past pointing.
- **Intention tremors.**
- Pendular knee jerk.
- **Dysdiadochokinesia.**
- **Rebound phenomenon.**

**Gait & posture :**

- Incoordination.
- Loss of balance.
- Tremors in head (Titubation).
- Ataxic gait.

**Eyes :**

Nystagmus.

**Basal Ganglia**

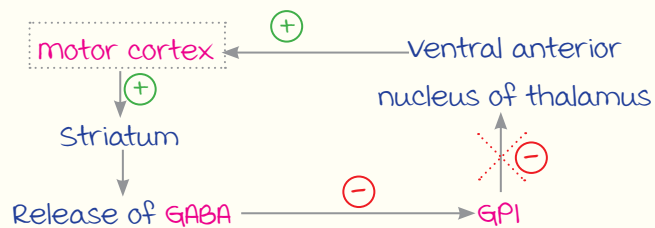
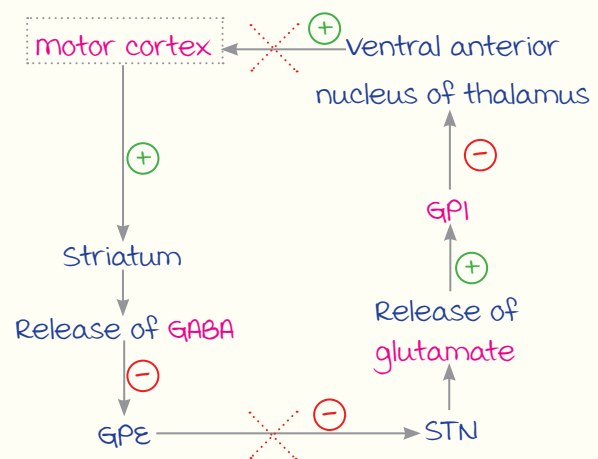
00:17:34

**Nuclei :**

|                           |                          | Neurotransmitter       |
|---------------------------|--------------------------|------------------------|
| Caudate                   | Striatum (Input)         | GABA                   |
| Putamen                   |                          |                        |
| Globus pallidus (GP)      | Interna (Output)         |                        |
|                           | Externa                  |                        |
| Substantia nigra (SN)     | Pars reticulata (Output) | Dopamine               |
|                           | Pars compacta            |                        |
| Subthalamic nucleus (STN) |                          | Glutamate (Excitatory) |

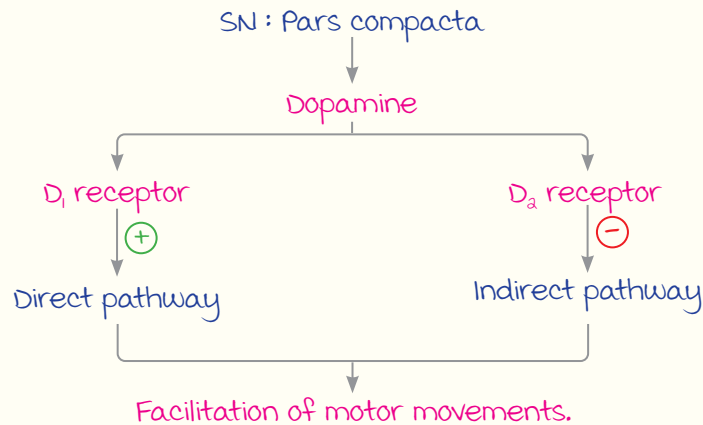
**Pathways :****At rest (Inhibitory) :**

GPI  $\ominus$  → ventral anterior nucleus (Thalamus)  $\oplus$  → motor cortex → No movement.

**Direct pathway :****Initiation of movement :****Indirect pathway :****Termination of movement :**

Action of dopamine :

----- Active space -----



C/f of Basal Ganglia Lesion  :

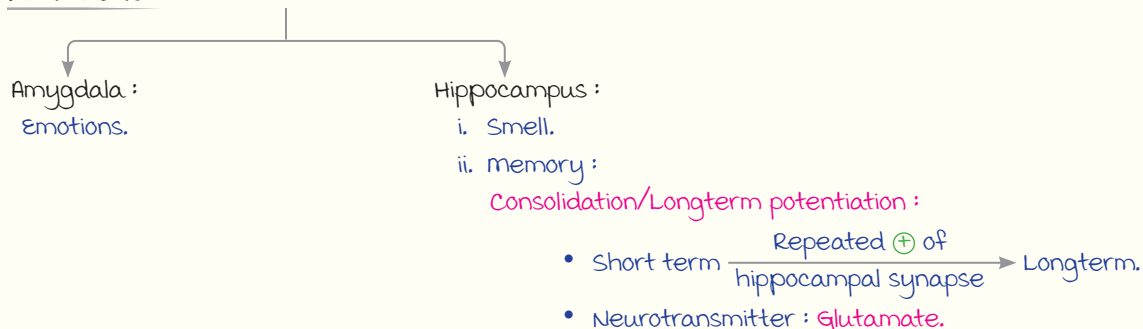
| Involuntary movements                   | Lesions                       | Features                                                         |
|-----------------------------------------|-------------------------------|------------------------------------------------------------------|
| Chorea                                  | Caudate                       | Rapid, involuntary dancing movements                             |
| Athetosis                               | Globus pallidus & putamen     | Continuous, slow writhing movements                              |
| Wing beating tremors                    | Hepatolenticular degeneration | Seen in Wilson's disease<br>(↑ Copper deposition)                |
| Hemiballismus                           | STN                           | Sudden flailing, intense, violent contraction of one large joint |
| Pill rolling tremor<br>(Resting tremor) | SN (Pars compacta)            | Seen in Parkinson's (↓ Dopamine)                                 |

## Limbic System

00:34:00

- AKA Smell brain/Rhinencephalon.
- Rim of cortical tissue (Evolution : Old).

Functions :



----- Active space -----

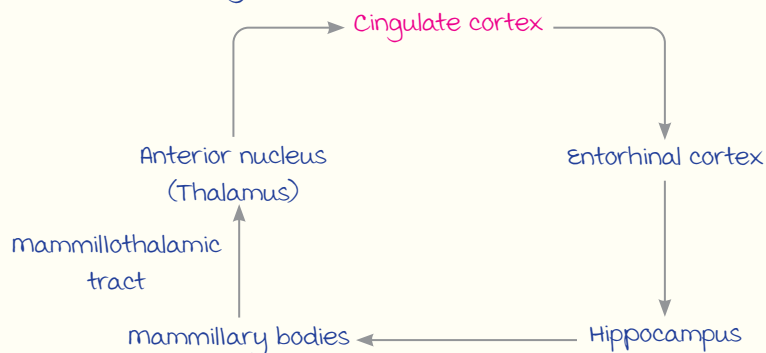
Kluver-Bucy syndrome :Lesion : **Amygdala.**

c/f :

- Visual agnosia.
- Hypermetamorphosis.
- Placidity.
- Hyperorality.
- Hypersexuality.

Papez Circuit :

memory and emotions :

Hypothalamus

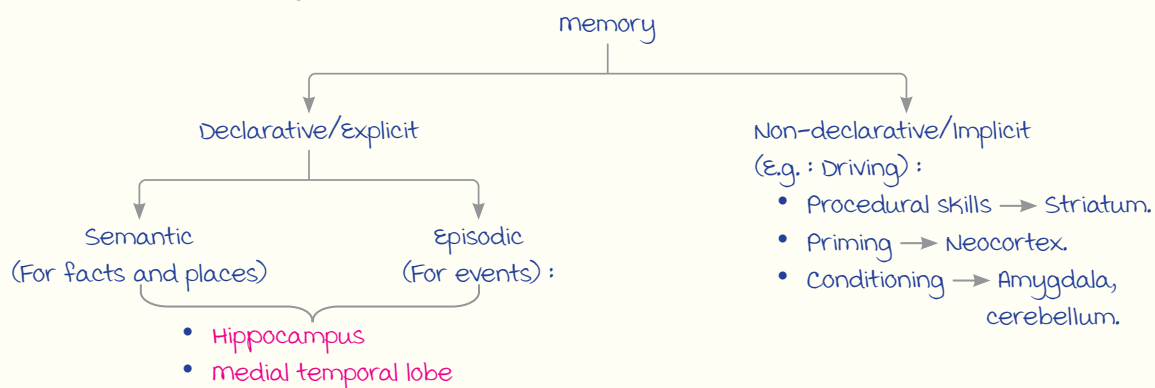
00:38:44

Nuclei :

|                               | Function                                                   |
|-------------------------------|------------------------------------------------------------|
| Anterior                      | Response to <b>heat</b>                                    |
| Posterior                     | Response to <b>cold</b>                                    |
| Lateral                       | Secretes orexin → <b>⊕ Feeding centre</b><br>→ keeps awake |
| Ventromedial                  | Satiety                                                    |
| Suprachiasmatic               | <b>Circadian rhythm</b>                                    |
| Supraoptic                    | Antidiuretic hormone (ADH)                                 |
| Paraventricular               | Oxytocin                                                   |
| Ventrolateral preoptic (VLPO) | Sleep                                                      |
| mammillary bodies             | <b>memory</b>                                              |

Cerebral Hemispheres

00:41:40

Learning & Memory 🧠 :

Lesions :

----- Active space -----

| Locations                                       | c/f                                                                                                  |
|-------------------------------------------------|------------------------------------------------------------------------------------------------------|
| Hippocampus                                     | Anterograde amnesia                                                                                  |
| mammillary bodies                               | <ul style="list-style-type: none"> <li>Wernicke-Korsakoff syndrome</li> <li>Confabulation</li> </ul> |
| Anterior nucleus of thalamus                    | Loss of recent memory                                                                                |
| Basal forebrain<br>(Nucleus basalis of Meynert) | ↓ ACh → Alzheimer's                                                                                  |
| Amygdala                                        | Loss of emotions + memory                                                                            |
| Entorhinal cortex                               | Loss of smell + memory                                                                               |

Speech area :

|               | Broca's area                         | Wernicke's area                         | Arcuate fasciculus                  | Angular gyrus               |
|---------------|--------------------------------------|-----------------------------------------|-------------------------------------|-----------------------------|
| Location      | Inferior frontal gyrus               | Superior temporal gyrus                 | Connects Broca's & Wernicke's areas | -                           |
| Brodmann area | 44                                   | 22                                      |                                     | 39                          |
| Function      | motor area for speech (vocalization) | Sensory area for speech (Comprehension) |                                     | vision - speech correlation |
| Lesion        | Broca's/<br>Non-fluent aphasia       | Wernicke's/<br>Fluent aphasia           | Conduction aphasia                  | Anomic aphasia              |

## Fluid Systems in the Brain

00:57:45

### Cerebrospinal Fluid (CSF) :

- Volume : 150 mL.
- Rate of production : 550 mL/day.
- Turnover : 3.7 times/day.
- Normal pressure :
  - 70 - 180 mm H<sub>2</sub>O.
  - Average pressure (Filtration = Absorption) : 112 mmH<sub>2</sub>O.
- Regulation : At the level of absorption.

CSF vs. Plasma concentrations :

| Conc.        | Constituents                                    |
|--------------|-------------------------------------------------|
| CSF > Plasma | Cl <sup>-</sup> , mg <sup>2+</sup>              |
| CSF = Plasma | HCO <sub>3</sub> <sup>-</sup> , osmolarity      |
| CSF < Plasma | Glucose (2/3 <sup>rd</sup> of plasma), proteins |

### Cerebral Blood Flow (CBF) :

Cerebral perfusion pressure :

- mean arterial pressure - intracranial pressure.
- 750 mL/min or 14% of CO or 54 mL/100g/min.
- Grey > white matter.
- Estimation : Kety's method (Inhaled N<sub>2</sub>O).
- O<sub>2</sub> consumption : 20% of total (At rest).

Areas sensitive to hypoxia :

- Hippocampus.
- Inferior colliculus.
- Thalamus.
- Basal ganglia.

----- Active space -----

Regulation :

- **Auto regulation** : Change in pressure (65 - 140 mm Hg)  $\rightarrow$  **Constant** (CBF).
- **Hypercarbia** :  $\uparrow \text{CO}_a \rightarrow \uparrow \text{CBF}$ .
- **Hypothermia** :  $\downarrow$  in  $1^\circ\text{C} \rightarrow \downarrow 7\%$  in CBF. (Therapeutic : Neuro-Sx)

## Sleep

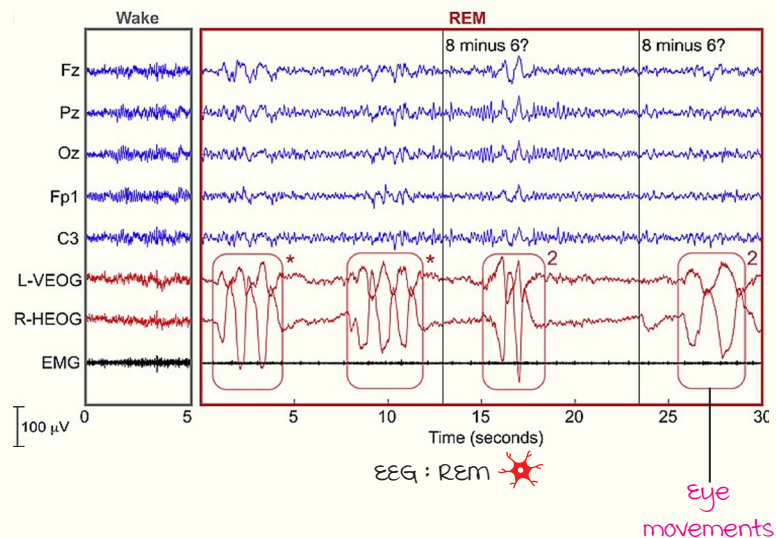
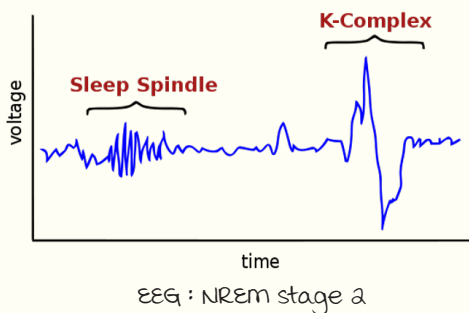
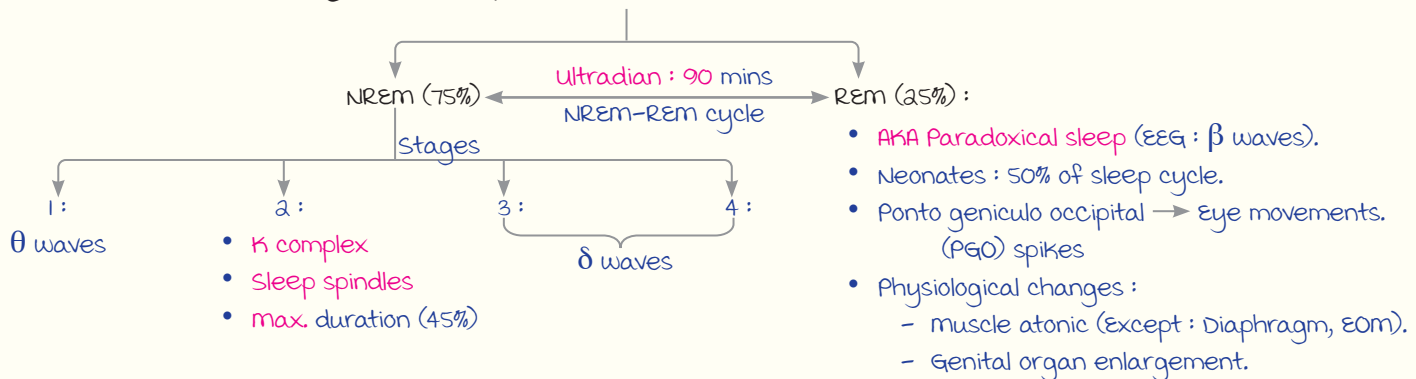
01:07:04

EEG Waves :

Frequency  $\propto \frac{1}{\text{Amplitude}}$   $\propto$  Activity state.

| EEG wave                | Frequency (Hz) | Seen during                                                                                                                                    |
|-------------------------|----------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| Alpha : $\alpha$        | 8 - 12         | <ul style="list-style-type: none"> <li>• Eye closed : Relaxed</li> <li>• Eye open : <math>\beta</math> replaces <math>\alpha</math></li> </ul> |
| Beta : $\beta$          | 13 - 30        | Attention/wakefulness                                                                                                                          |
| Theta : $\theta$        | 4 - 7          | A/w memory (Hippocampus) & sleep                                                                                                               |
| Delta (Slow) : $\delta$ | $< 4$          | <b>Sleep</b>                                                                                                                                   |
| Gamma : $\gamma$        | $> 60$         | Focused attention                                                                                                                              |

Stages of Sleep :



## Regulation of Sleep :

----- Active space -----

| Brain areas                                                     | Neurotransmitter (NT) | NT levels |     |                         |
|-----------------------------------------------------------------|-----------------------|-----------|-----|-------------------------|
|                                                                 |                       | NREM      | REM |                         |
| Cholinergic nuclei of pons - midbrain junction (REM-on neurons) | Acetylcholine         | -         | ↑   | → Promotes sleep.       |
| Locus ceruleus                                                  | Norepinephrine        | ↓         | ↓   |                         |
| Raphe nuclei                                                    | Serotonin             | ↓         | ↓   | } Promotes wakefulness. |
| Tuberomammillary nuclei                                         | Histamine             | ↓         | ↓   |                         |
| Lateral hypothalamus                                            | Orexin                | ↓         | ↓   |                         |

↑ Adenosine ~~→~~ Promotes sleep.  
 ↑ ⊖  
 Caffeine

## Parasomnias :

During NREm :

- Somnambulism : Sleep walking (Stage 3 and 4).
- Sleep talking.
- Nocturnal enuresis : Bedwetting.
- Sleep bruxism : Teeth grinding (Stage 2).
- Night terrors/Pavor nocturnus.

During REM : Nightmares, narcolepsy.

# RESPIRATORY PHYSIOLOGY : PART 1

## Airways

00:00:36

### Airway Generations :

Weibel model of airways  :

23 generations (16 + 7).

|              | Conducting airways                                                                                | Alveolar/respiratory airways                                                                            |
|--------------|---------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
| Generations  | <ul style="list-style-type: none"> <li>First 16</li> <li>Trachea → Terminal bronchiole</li> </ul> | <ul style="list-style-type: none"> <li>Last 7</li> <li>Respiratory bronchiole → Alveolar sac</li> </ul> |
| Gas exchange | ⊖ (Dead space)                                                                                    | ⊕                                                                                                       |
| Feature      | medium sized bronchi :<br>max. airway resistance                                                  | -                                                                                                       |

### Conducting Airways :

Cilia :

- Pseudostratified ciliated columnar epithelium.
- Ciliary motility (Dynein) :
  - Sputum clearance.
  - Developmental rotation of internal organs.
- Ciliary immotility :
  - Kartagener's syndrome (⊖ Dynein) : Sinusitis, bronchiectasis, situs inversus.
  - Cystic fibrosis (Thick sputum).

Stem cells : Basal cells, Clara cells.

Airway smooth muscles :

Functions :

|                       | Bronchoconstriction                                                                                                                               | Bronchodilation                                                                                                                        |
|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| muscle action         | Contraction                                                                                                                                       | Relaxation                                                                                                                             |
| Air-flow              | ↓                                                                                                                                                 | ↑                                                                                                                                      |
| Resistance            | ↑                                                                                                                                                 | ↓                                                                                                                                      |
| Clinical significance | Bronchial asthma                                                                                                                                  | Rx of bronchial asthma                                                                                                                 |
| mediators             | <ul style="list-style-type: none"> <li>Parasympathetic ⊕</li> <li>Acetylcholine</li> <li>Histamine</li> <li>Leukotrienes (most potent)</li> </ul> | <ul style="list-style-type: none"> <li>Sympathetic ⊕ (β<sub>2</sub>-receptor)</li> <li>Nitric oxide</li> <li>Prostaglandins</li> </ul> |

Alveolar Airways :

----- Active space -----

Cells :

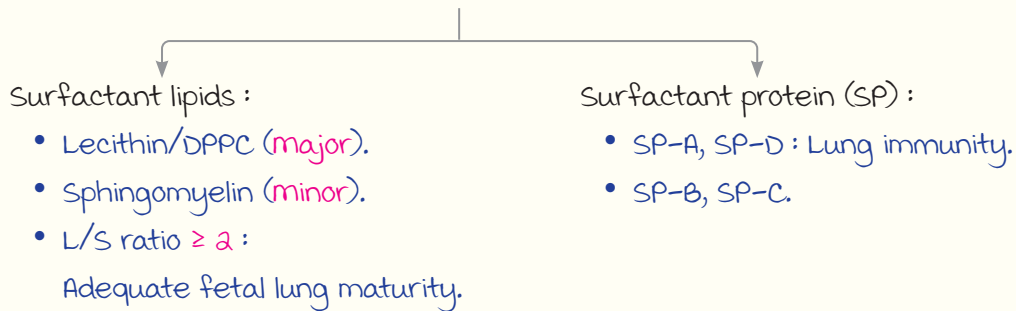
|              | Type 1 pneumocyte | Type 2 pneumocyte                                                                                    |
|--------------|-------------------|------------------------------------------------------------------------------------------------------|
| Shape        | Flat              | Round                                                                                                |
| Surface area | more              | Less                                                                                                 |
| Number       | Less              | more                                                                                                 |
| Function     | -                 | <ul style="list-style-type: none"> <li>• Stem cells</li> <li>• Produces <b>surfactant</b></li> </ul> |

Surfactant

00:15:04

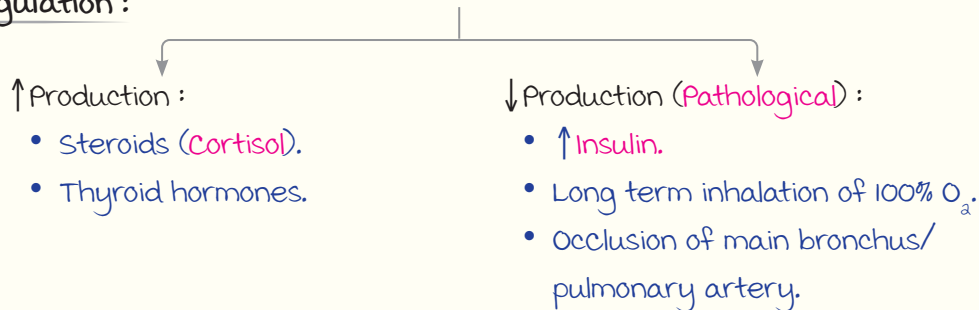
Storage :

**Lamellar bodies**  $\xrightarrow{\text{Counting}}$  Fetal lung maturity.

Composition :Function :

↓ Surface tension → Prevents alveolar collapse.

(Surfactant deficiency → Alveolar collapse → **Hyaline membrane disease** → Rx : Surfactant replacement).

Regulation :Clinical significance :

- **Betamethasone** : Rx of lung immaturity.
- Infants of **diabetic mothers** → RDS.

----- Active space -----

## Mechanics of Breathing

00:22:26

Boyle's law : Pressure (P)  $\propto \frac{1}{\text{volume (V)}}$  ; (Tidal vol. = 500 mL).

- Inspiration :  $V \uparrow$ ,  $P \downarrow$ .
- Expiration :  $V \downarrow$ ,  $P \uparrow$ .

Pressures in the lung :

1. Intrapleural pressure (IPP) :

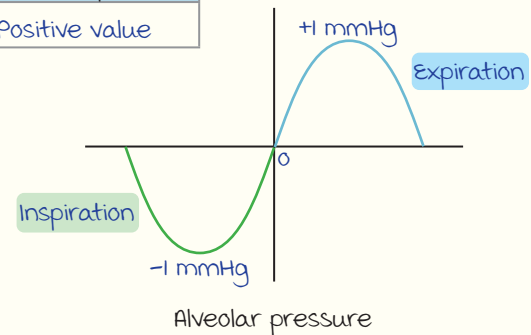
|     | At rest   | Inspiration | Expiration | Forceful expiration |
|-----|-----------|-------------|------------|---------------------|
| IPP | -2.5 mmHg | -6 mmHg     | -2.5 mmHg  | Positive value      |

1 mm Hg = 1.36 cm H<sub>2</sub>O.

2. Alveolar pressure (AP).

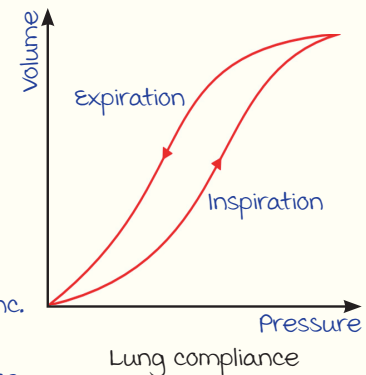
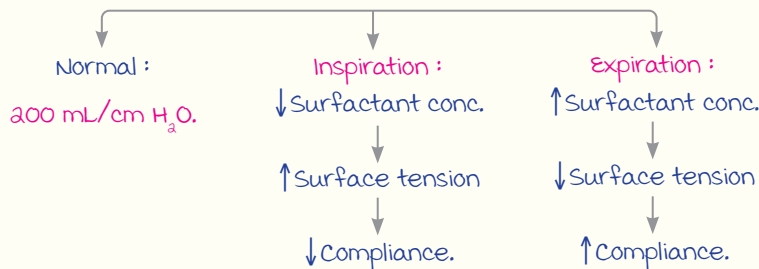
3. Transpulmonary pressure (TP) :

TP = AP - IPP (mnemonic : TAP).



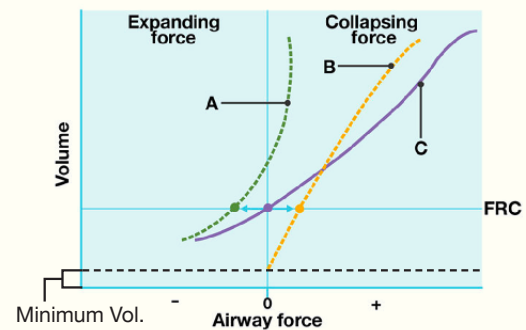
**Lung Compliance** :

- Compliance  $\propto \frac{1}{\text{Elastance, surface tension}}$ .
- **Hysteresis** : Difference between P/V curves of inspiration & expiration (Opposite).
- Compliance (C) =  $\frac{\text{Change in volume } (\Delta V)}{\text{Change in pressure } (\Delta P)}$ .



Compliance of lung, chest wall & lung + chest wall :

- **Equilibrium volume** :
  - Expanding & collapsing forces reach equilibrium.
  - At functional residual capacity (FRC).
- **minimal volume** : volume left in the lung after complete collapse.
- **Pneumothorax** :
  - ↓ Lung volume.
  - ↑ Thoracic volume.



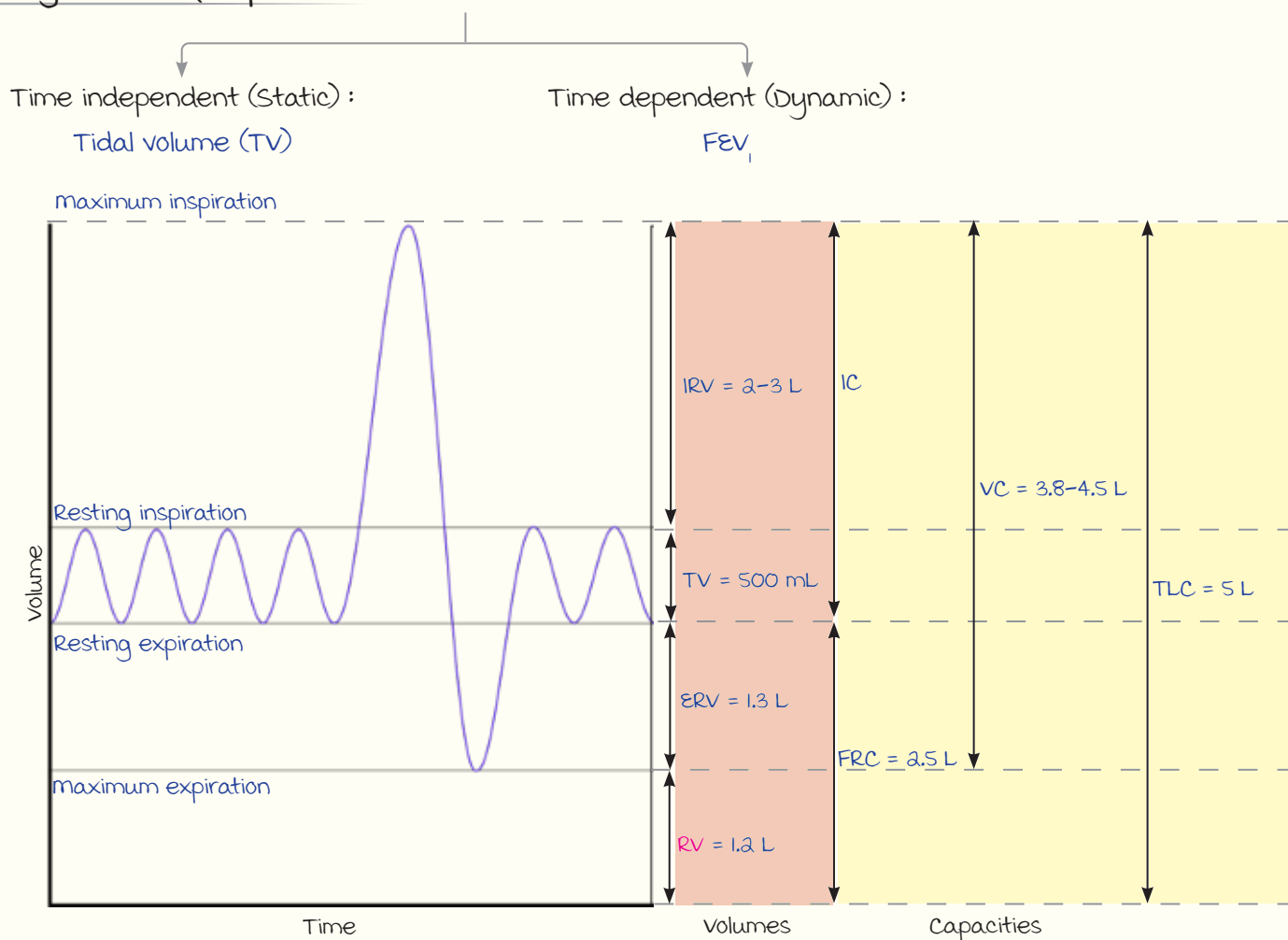
# RESPIRATORY PHYSIOLOGY : PART 2

----- Active space -----

## Spirometry

00:00:07

### Lung volumes & Capacities :



Functional residual capacity (FRC) :

- volume of air in lungs after normal expiration.
- 50% of TLC.

Closing volume (CV) :

volume of air enclosed in small airways at the end of expiration.

Closing capacity (CC) : CV + RV

FEV<sub>1</sub> : Forced expiratory volume in first second

IRV : Inspiratory reserve volume

ERV : Expiratory reserve volume

RV : Residual volume

IC : Inspiratory capacity

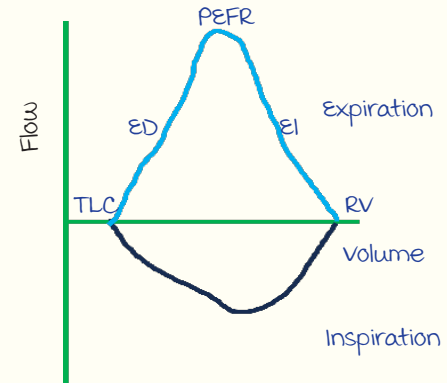
VC : Vital capacity

TLC : Total lung capacity

----- Active space -----

Flow volume loop :

- Forced inspiration & forced expiration.
- PEFR : Peak expiratory flow rate.
- FVC : TLC - RV
- FEV<sub>1</sub>/FVC :
  - (N) : 70 - 80%.
  - ↓ : Obstruction (COPD).
  - (N)/↑ : Restriction.



Flow - volume loop

Parts of expiration :

- Effort dependent part (ED) : Large airways.
- Effort independent part (EI) : medium & small airways  $\xrightarrow{\text{Affected}}$  Obstructive disease.

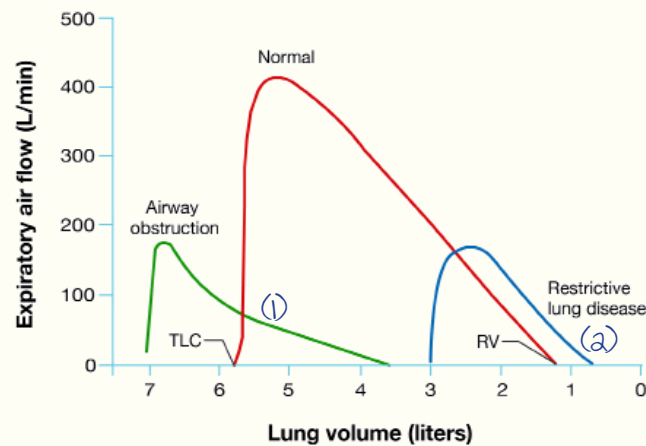
Obstructive vs restrictive lung disease :

Obstruction :

Scooped out pattern (1)

Restriction :

Normal shape, smaller (2)



Expiratory flow volume curve

Limitations of Spirometry :

- Cannot measure : RV, FRC, TLC.
- Alternative methods :
  - Helium dilution technique.
  - Nitrogen washout technique.
  - Body plethysmography.

## Ventilation and Perfusion

00:15:08

----- Active space -----

### ventilation (V) :

Partial pressure of gases :

|                         | Partial pressure                                                                |
|-------------------------|---------------------------------------------------------------------------------|
| Environmental air (dry) | $PO_a = 160 \text{ mmHg (21\% } O_a)$                                           |
| Inspired air (moist)    | $PO_a = 150 \text{ mmHg}$                                                       |
| Alveolar air            | $PO_a = 100 \text{ mmHg, } PCO_a = 40 \text{ mmHg}$                             |
| Expired air             | $P_{E O_a} = 120 \text{ mmHg (16\% } O_a) \text{ (mouth to mouth respiration)}$ |

### Tidal volume ( $V_T$ ) :

Alveolar ventilation volume ( $V_A$ ) :

- 350 mL
- $V_A / \text{min} = (V_T - V_D) \times RR$   
 $= 350 \times 12$   
 $= 4.2 \text{ L/min (useful ventilation).}$

Dead space ( $V_D$ ) :

- 150 mL
- $V_D / V_T = 30\%$ .

Dead spaces :

- Anatomical dead space (ADS) :  
**Radford formula** :  $ADS \approx \text{wt. in pounds}$
- Alveolar dead space : Inadequately perfused alveoli
- Physiological/total dead space :  
 $ADS (150) + \text{Alveolar dead space (0)} = 150 \text{ ml} = ADS$

measurement of dead spaces :

- Anatomical dead space : Single breath **nitrogen** method (**Fowler's** method).
- Total dead space :

**Bohr's equation** :  $V_D = V_T \times \frac{PA_{CO_2} - PE_{CO_2}}{PA_{CO_2}}$

$PE_{CO_2}$  :  $P_{CO_2}$  in expired air  
 $PA_{CO_2}$  :  $P_{CO_2}$  in alveolar air

### Perfusion (Q) :

Bronchial circulation : Nutrition to conducting airways.

Pulmonary circulation :

- **5.5 L/min** (Deoxygenated blood).
- **Low pressure, low resistance.**
- Functions : Blood reservoir, filters blood clots.
- Zones :

----- Active space -----

|                 | Pressure          | Blood flow                           |
|-----------------|-------------------|--------------------------------------|
| Zone 1 (Apex)   | $P_A > P_a > P_v$ | No flow                              |
| Zone 2 (middle) | $P_a > P_A > P_v$ | Intermittent flow (waterfall effect) |
| Zone 3 (Base)   | $P_a > P_v > P_A$ | Continuous flow                      |

$P_A$  : Alveolar pressure  
 $P_a$  : Arterial pressure  
 $P_v$  : Venous pressure

Effects of hypoxia  :

- Pulmonary circulation :

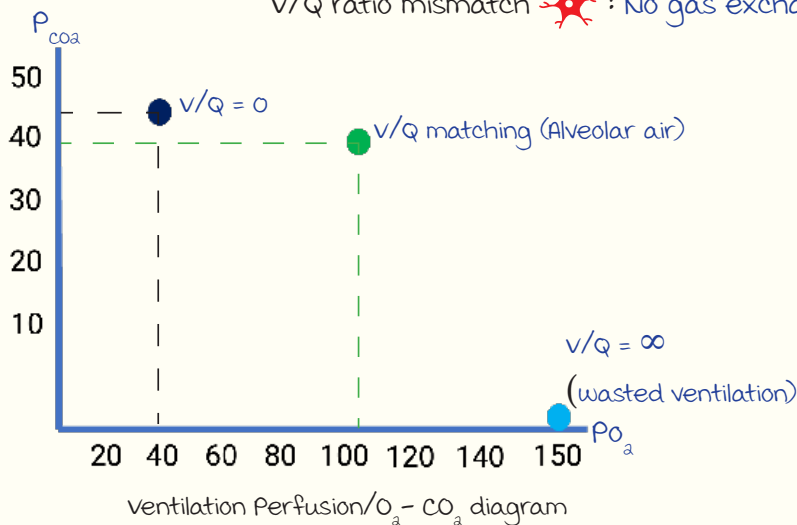
Hypoxia  $\rightarrow$  Closes  $O_2$  sensitive  $K^+$  channels  $\rightarrow K^+$  accumulation  $\rightarrow$  Depolarization  
 $\downarrow$   
 $Ca^{2+}$  channels open  
 $\downarrow$   
 $Ca^{2+}$  influx  
 $\downarrow$   
 Vasoconstriction.

- Systemic circulation : Vasodilation

ventilation-Perfusion (v/Q) ratio :

v/Q ratio :  $\rightarrow \textcircled{N} = 4.2/5.5 = 0.8$   
 $\rightarrow$  Highest : Apex  
 $\rightarrow$  Lowest : Base

v/Q ratio mismatch  : No gas exchange



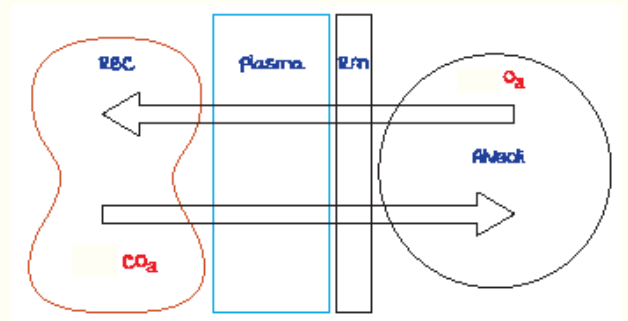
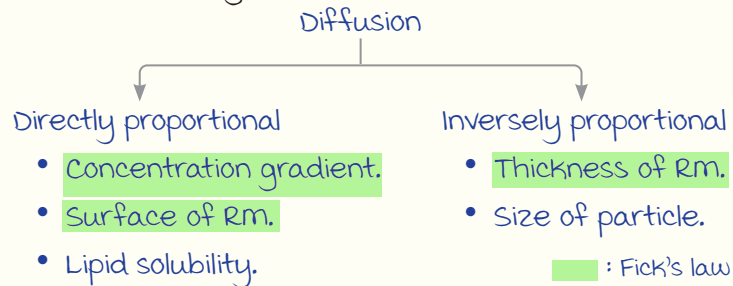
| v/Q                     | Physiological      | Pathological           |
|-------------------------|--------------------|------------------------|
| Infinity<br>( $Q = 0$ ) | Conducting airways | Pulmonary embolism     |
| Zero<br>( $V = 0$ )     | mixed venous blood | Foreign body in airway |

## Diffusion of Gases

00:41:11

----- Active space -----

Factors affecting diffusion :



Respiratory membrane (Rm)

Diffusion capacity of the lungs for carbon monoxide (DLCO) :

- CO :  $210 \times O_a$  affinity to hemoglobin.
- ① : 25 mL/min/mm Hg.

|               | ↑ DLCO       | ↓ DLCO                                                                                  |
|---------------|--------------|-----------------------------------------------------------------------------------------|
| Physiological | Exercise     | Aging                                                                                   |
| Pathological  | Polycythemia | <ul style="list-style-type: none"> <li>Emphysema</li> <li>Pulmonary fibrosis</li> </ul> |

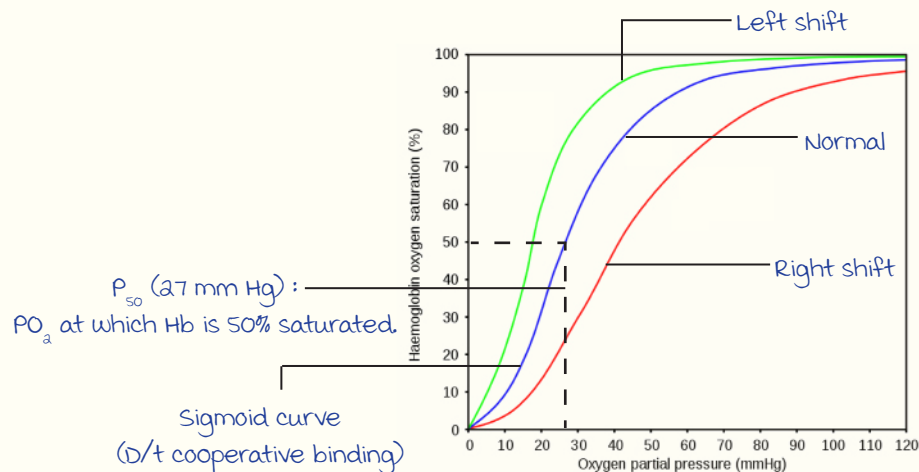
## Oxygen Transport

00:47:11

### Oxygen Dissociation Curve :

Hb saturation :

- Tissue : 75% ( $O_a = 40$  mm Hg).
- Lung : 98% ( $PO_a = 100$  mm Hg).



----- Active space ----- Right shift vs Left shift  :

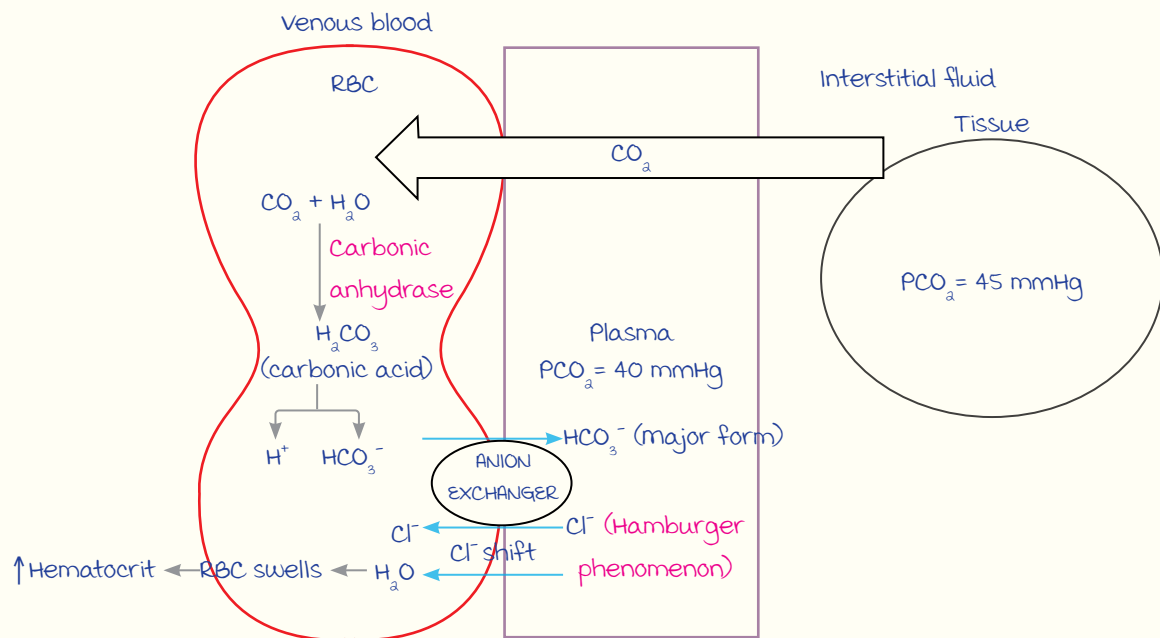
|                | Right shift                                                                                                                                                                                                                                                              | Left shift                                                                                                                                                                                                           |
|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| General points | <ul style="list-style-type: none"> <li>Release of <math>O_a</math></li> <li><math>\uparrow P_{50}</math></li> </ul>                                                                                                                                                      | <ul style="list-style-type: none"> <li>Loading of <math>O_a</math></li> <li><math>\downarrow P_{50}</math></li> </ul>                                                                                                |
| Site           | Tissue                                                                                                                                                                                                                                                                   | Lung                                                                                                                                                                                                                 |
| Seen in        | <ul style="list-style-type: none"> <li>Hypoxia</li> <li>Hypercarbia</li> <li>Acidosis</li> <li><math>\uparrow</math> 2,3 DPG (bind to <math>\beta</math>-chain of Hb)</li> <li>Thyroid hormones</li> <li>High altitude</li> <li>Exercise</li> </ul> <p>Bohr's effect</p> | <ul style="list-style-type: none"> <li>Hypocarbica</li> <li>Alkalosis</li> <li>Fetal Hb (<math>\downarrow</math> Affinity to 2,3 DPG)</li> <li>CO</li> <li>Stored blood (<math>\downarrow</math> 2,3 DPG)</li> </ul> |

Note :

Storing blood in citrate-phosphate-dextrose-adenine (CPDA)  $\rightarrow$  Less fall in 2,3-DPG.

## Carbon Dioxide Transport

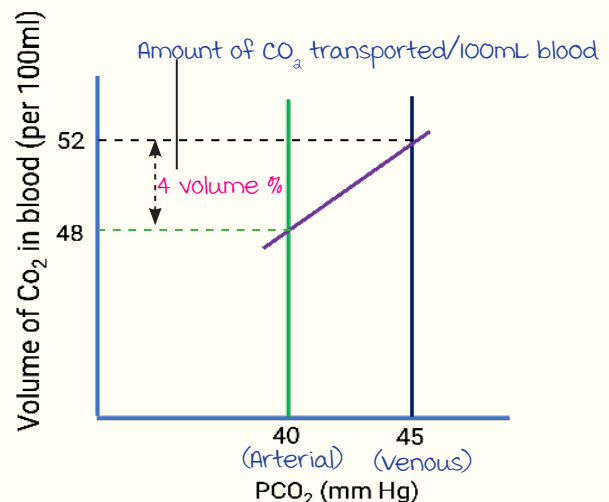
00:56:44



CO<sub>a</sub> dissociation curve :

Haldane effect :

Deoxygenation of Hb  $\rightarrow$   $\uparrow$  CO<sub>a</sub> carrying ability.



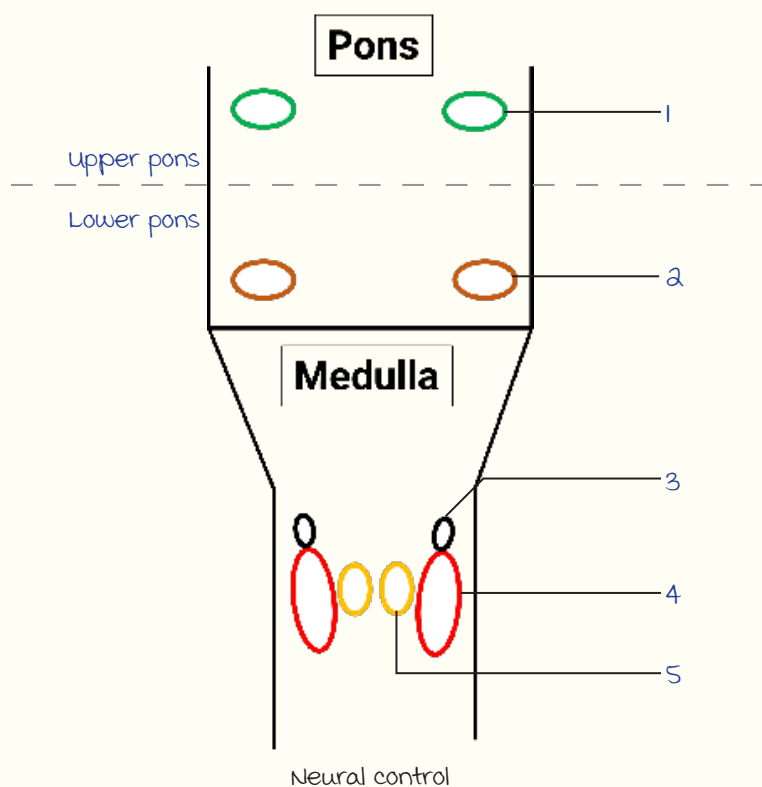
# RESPIRATORY PHYSIOLOGY : PART 3

----- Active space -----

## Neural Regulation of Respiration

00:00:06

|   | Brain area  | Function                                                                                                                                                                           |
|---|----------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | Pneumotaxic center<br>Vagus nerve                                                            | Limits inspiration ( $\ominus$ APC)                                                                                                                                                |
| 2 | Apneustic center (APC)                                                                       | Prolongs inspiration                                                                                                                                                               |
| 3 | Pre - Botzinger complex                                                                      | Pacemaker :<br><ul style="list-style-type: none"> <li>• Initiation</li> <li>• Spontaneous/automatic breathing</li> <li>• Defect <math>\rightarrow</math> Ondine's curse</li> </ul> |
| 4 | ventral respiratory group                                                                    | Active expiration (Exercise)                                                                                                                                                       |
| 5 | Dorsal respiratory group                                                                     | APC $\xrightarrow{+}$ DRG $\rightarrow$ Ramp signal $\rightarrow$ Smooth rise in tidal volume                                                                                      |

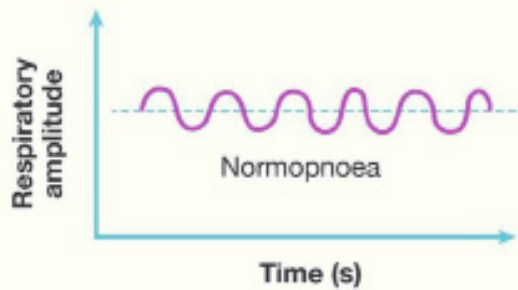


----- Active space -----

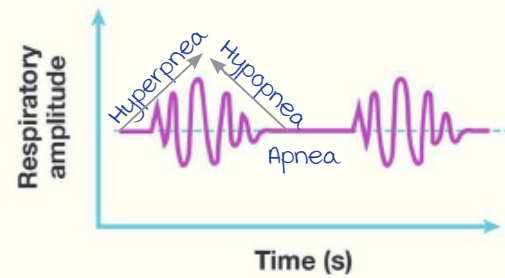
## Pulmonary Reflexes :

| Reflex                                                    | Stimulus                                | Afferent                                      | Effect                                                                                                                            |
|-----------------------------------------------------------|-----------------------------------------|-----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| Hering Breuer inflation reflex<br>(Prevents lung injury)  | Prolonged inspiration<br>(TV = 1500 mL) | myelinated<br>vagal fibres<br>(Slow adapting) | Inspiration ⊖<br>↓<br>Expiration begins                                                                                           |
| Hering Breuer deflation reflex<br>(Prevent lung collapse) | Prolonged expiration                    | myelinated<br>vagal fibres<br>(Slow adapting) | Expiration ⊖<br>↓<br>Inspiration begins                                                                                           |
| Head's paradoxical reflex                                 | Prolonged inspiration                   | -                                             | <ul style="list-style-type: none"> <li>• ↑ Inspiration</li> <li>• Replace fluid with air in newborn lungs.</li> </ul>             |
| J reflex<br>(J receptors)                                 | Pulmonary edema                         | unmyelinated<br>vagal C fibres                | <ul style="list-style-type: none"> <li>• Apnea</li> <li>• Breathlessness</li> <li>• Hypotension</li> <li>• Bradycardia</li> </ul> |

## Altered/Periodic Breathing :

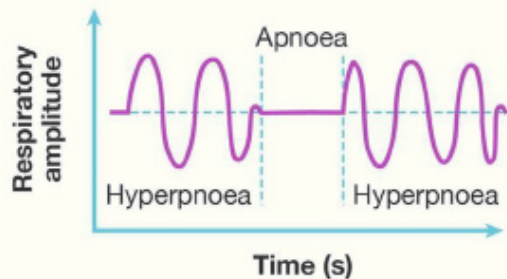


Normal respiration



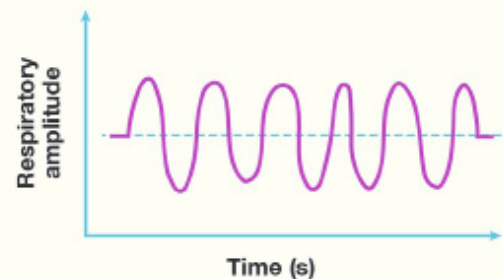
Cheyne - Stokes respiration :

- Waxing & waning.
- Crescendo decrescendo.
- CCF.



Biot's breathing :

- Ataxic breathing
- Injury to medulla, meningitis



Kussmaul's breathing :

- Rapid/acidotic breathing
- DKA

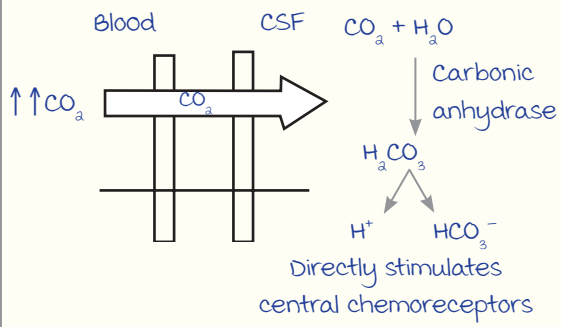
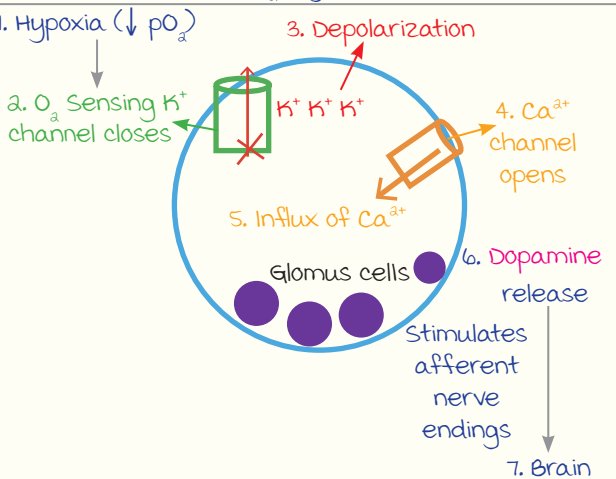
## Chemical Regulation Of Respiration

00:15:00

----- Active space -----

Chemical regulators :  $O_2$ ,  $CO_2$ ,  $H^+$ .

### Types of Chemoreceptors :

|              | Central chemoreceptors                                                                                                                                                                                                                                                                                          | Peripheral chemoreceptors                                                                                                                                                                                                                                                                                                                                                                                          |
|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Location     | Brain                                                                                                                                                                                                                                                                                                           | Neck : Carotid & aortic bodies                                                                                                                                                                                                                                                                                                                                                                                     |
| Sensitive to | $pCO_a$ (Blood)                                                                                                                                                                                                                                                                                                 | $\downarrow pO_a$ (Hypoxia)                                                                                                                                                                                                                                                                                                                                                                                        |
| MOA          |  <p>Blood <math>\uparrow \uparrow CO_a</math> enters CSF. <math>CO_a + H_2O \xrightarrow{\text{Carbonic anhydrase}} H_2CO_3 \rightarrow H^+ + HCO_3^-</math>. <math>H^+</math> directly stimulates central chemoreceptors.</p> |  <p>1. Hypoxia (<math>\downarrow pO_a</math>)<br/> 2. <math>O_a</math> Sensing <math>K^+</math> channel closes<br/> 3. Depolarization<br/> 4. <math>Ca^{2+}</math> channel opens<br/> 5. Influx of <math>Ca^{2+}</math><br/> 6. Dopamine release (from Glomus cells)<br/> 7. Stimulates afferent nerve endings<br/> 7. Brain</p> |

In asphyxia :

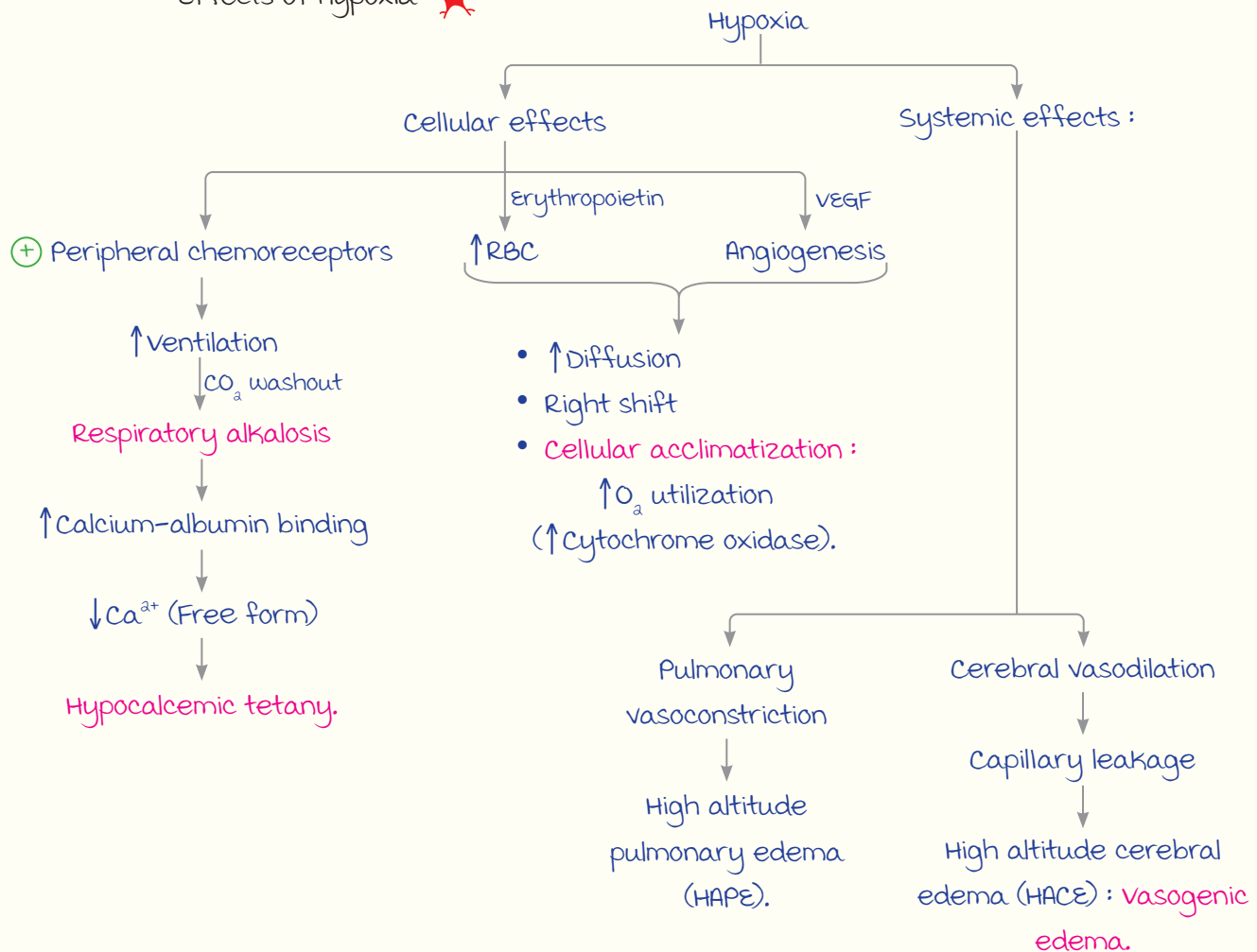
$\downarrow pO_a + \uparrow pCO_a + \uparrow H^+ \rightarrow$  Central & peripheral Chemoreceptors (+)  $\rightarrow$  Chemoreflex  $\rightarrow$  Hyperventilation.  
 $\downarrow$   
 $\uparrow pO_a, \downarrow pCO_a, \downarrow H^+$

### Types of Hypoxia :

|                           | Hypoxic hypoxia           | Anemic hypoxia                                                           | Stagnant/Ischemic hypoxia | Histotoxic hypoxia                                                            |
|---------------------------|---------------------------|--------------------------------------------------------------------------|---------------------------|-------------------------------------------------------------------------------|
| Pathophysiology           | Low $pO_a$ (Air)          | Low Hb<br>$\downarrow O_a$ carrying capacity<br>$\downarrow O_a$ content | $\downarrow$ Blood flow   | $\ominus$ Cytochrome oxidase<br>$\downarrow$<br>Cells unable to utilize $O_a$ |
| Examples                  | • High altitude<br>• COPD | • CO poisoning<br>• Anemia                                               | CCF                       | Cyanide poisoning                                                             |
| Rx                        | Oxygen therapy            | Hyperbaric oxygenation                                                   | -                         | methylene blue/<br>Nitrites                                                   |
| A -v $O_a$ difference     | -                         | -                                                                        | $\uparrow \uparrow$       | $\downarrow \downarrow$                                                       |
| Peripheral chemoreceptors | (+)                       | Not stimulated (b/t normal $pO_a$ )                                      | -                         | -                                                                             |

High Altitude :

Hypoxia (Low pressure state)

Effects of hypoxia  :Deep Sea Physiology :

High barometric pressure  $\xrightarrow{\text{Decompression}}$  Compressed gas released as bubbles.

Decompression sickness/Caisson's disease/Dysbarism  :Effects of released N<sub>a</sub> :

- Crosses BBB → Nitrogen narcosis.
- In joints : Bends.
- In lungs : Chokes.
- In blood vessels
  - Air embolism
  - Endothelial injury → Capillary leakage → Hemoconcentration
- Rx : Slow ascent

Space Physiology :

----- Active space -----

|                                      | Positive G <sub>i</sub> | Negative G <sub>i</sub> |
|--------------------------------------|-------------------------|-------------------------|
| Direction of gravitational force (G) | Head → Foot             | Foot → Head             |
| venous return                        | ↓                       | ↑                       |
| Cardiac output                       | ↓                       | ↑                       |
| Cerebral perfusion                   | ↓ (Black out)           | ↑                       |
| Blood flow to eyes                   | ↓                       | ↑ (Red eye)             |

# CARDIOVASCULAR PHYSIOLOGY : PART 1

## Properties of Heart :

Chronotropy : Heart rate.

Bathmotropy : Excitability.

D/t cardiac action potentials → SA nodal potential  
→ Ventricular potential

Inotropy : Contractility.

Dromotropy : Conduction.

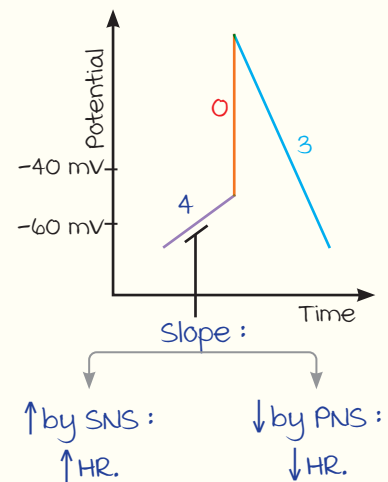
Lusitropy : Relaxation.

## Cardiac Action Potentials

00:01:49

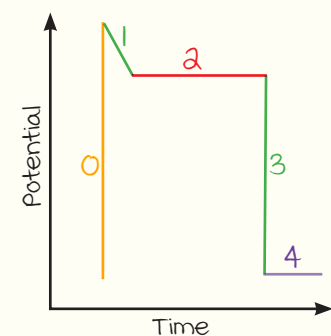
### SA Nodal Potential :

| Phase   | AKA                | Ionic activity                                                                                                                                                                                                                   |
|---------|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase 4 | Prepotential phase | <ul style="list-style-type: none"> <li><math>\text{Ca}_T^{2+}</math> influx (Transient)</li> <li><math>\text{Na}^+</math> influx (Funny currents/<math>I_f</math>)</li> <li><math>\downarrow \text{K}^+</math> efflux</li> </ul> |
| Phase 0 | Depolarization     | $\text{Ca}_L^{2+}$ influx (Long lasting)                                                                                                                                                                                         |
| Phase 3 | Repolarization     | $\text{K}^+$ efflux                                                                                                                                                                                                              |



### Ventricular Action Potential :

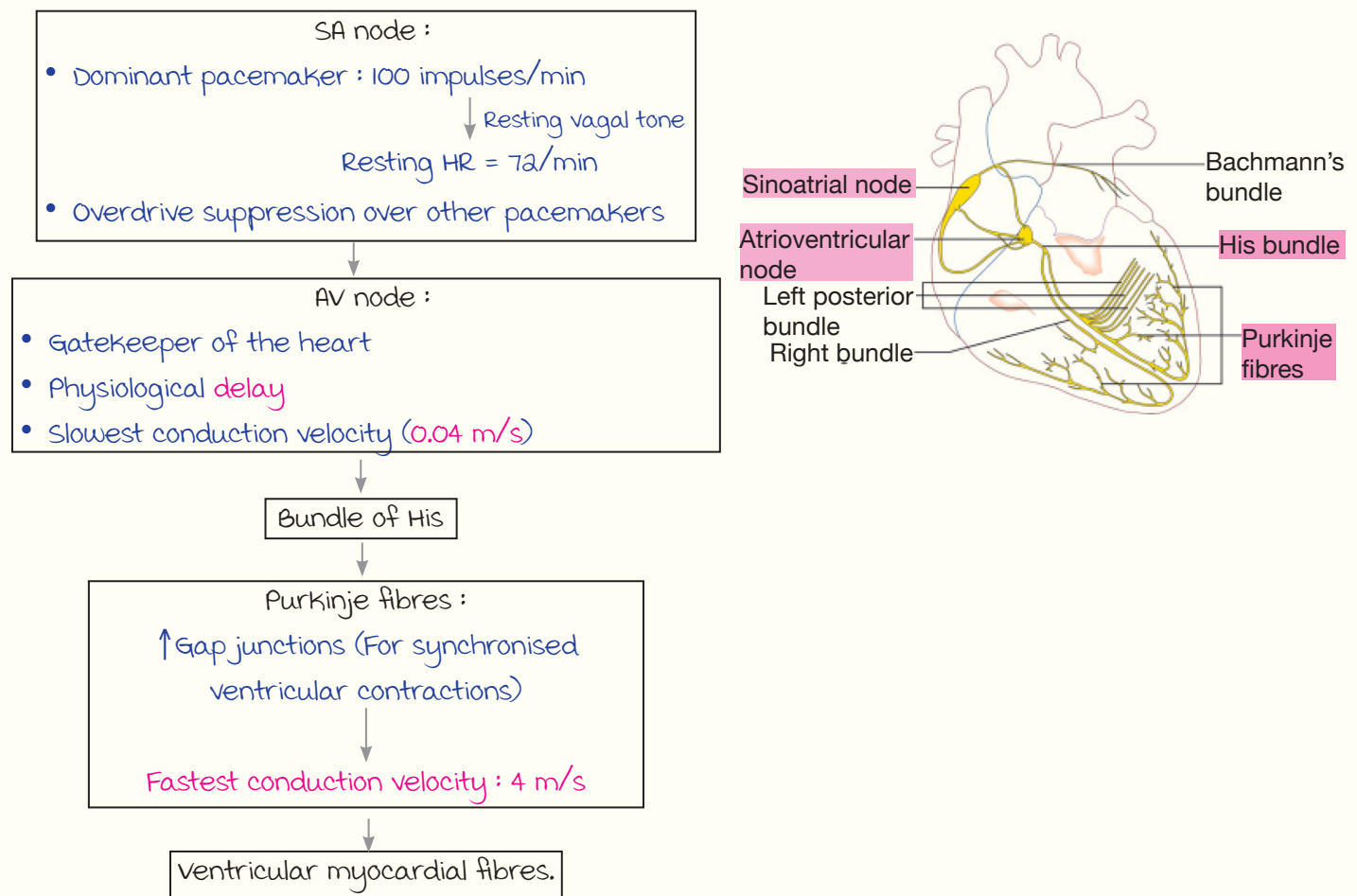
| Phase   | AKA                              | Ionic Activity                                                                  |
|---------|----------------------------------|---------------------------------------------------------------------------------|
| Phase 0 | Early depolarization             | $\text{Na}^+$ influx                                                            |
| Phase 1 | Early repolarization             | $\text{K}^+$ efflux                                                             |
| Phase 2 | Plateau                          | $\text{K}^+$ efflux = $\text{Ca}_L^{2+}$ influx                                 |
| Phase 3 | Late repolarization              | $\text{K}^+$ efflux                                                             |
| Phase 4 | Resting membrane potential (RMP) | $\text{K}^+$ influx, $\text{Na}^+$ efflux through $\text{Na}^+-\text{K}^+$ pump |



## Conducting System of Heart

00:09:41

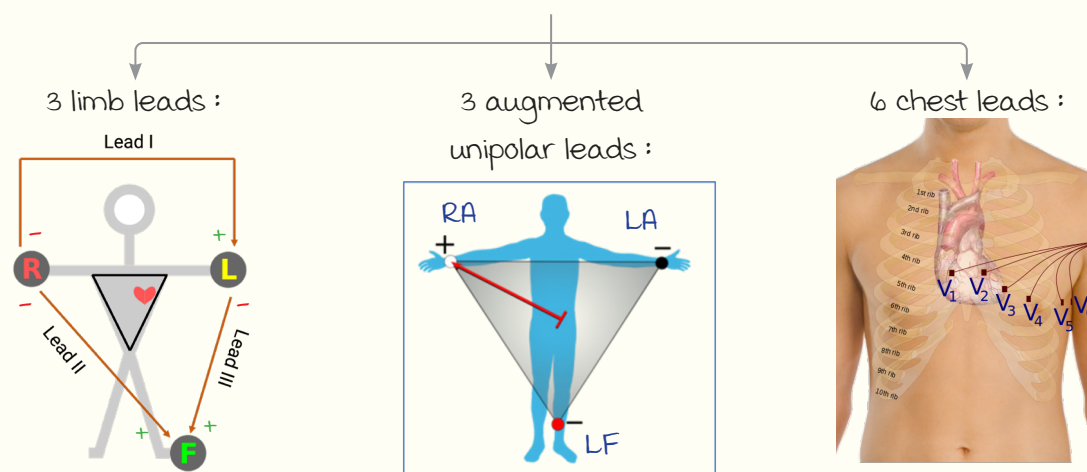
----- Active space -----



## ECG & JVP

00:13:51

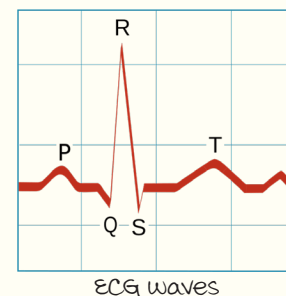
### ECG Leads :



----- Active space -----

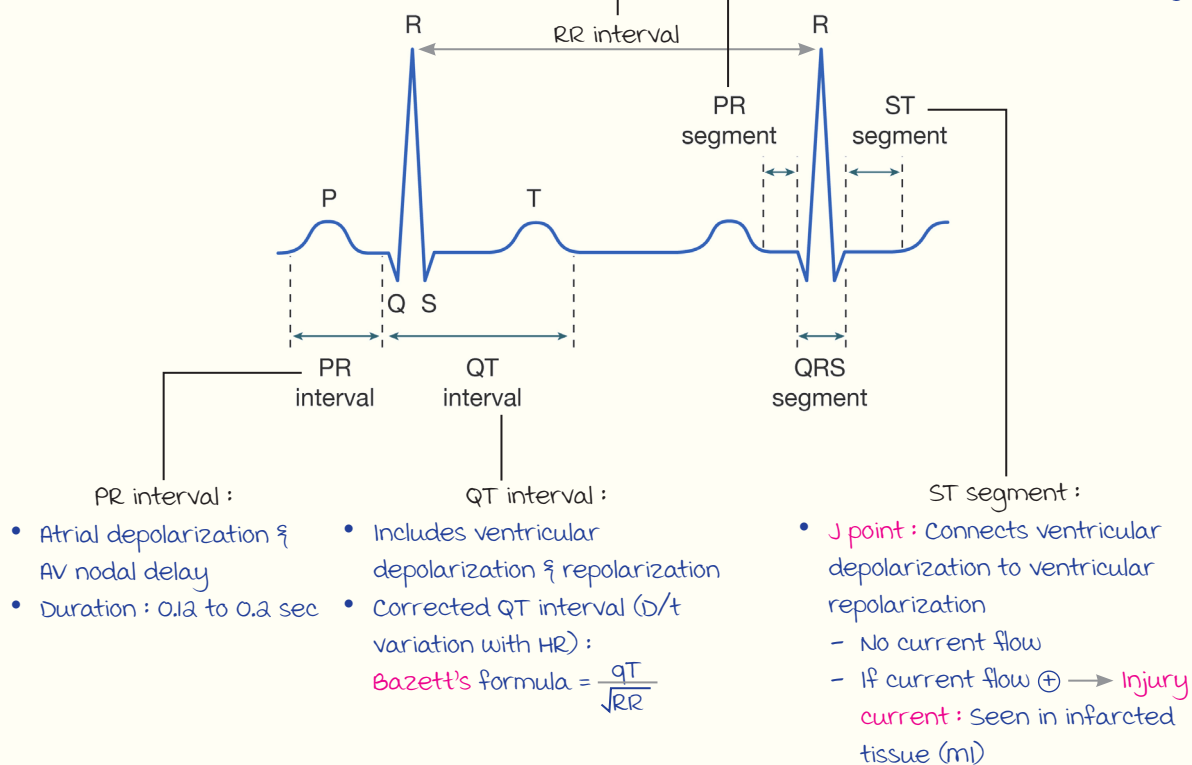
ECG Waves :

|             | Cause                                                                                                                    |
|-------------|--------------------------------------------------------------------------------------------------------------------------|
| P wave      | Atrial depolarization                                                                                                    |
| QRS complex | Ventricular depolarization                                                                                               |
| T wave      | Ventricular repolarization                                                                                               |
| U wave      | <ul style="list-style-type: none"> <li>Purkinje fibre repolarization</li> <li>Papillary muscle repolarisation</li> </ul> |

Segments & Intervals :

RR interval :

- Interval between two R waves
- HR =  $\frac{1500}{\text{No. of small squares between two R waves}}$

PR segment :  
AV nodal delay

JVP  :

----- Active space -----

A wave :

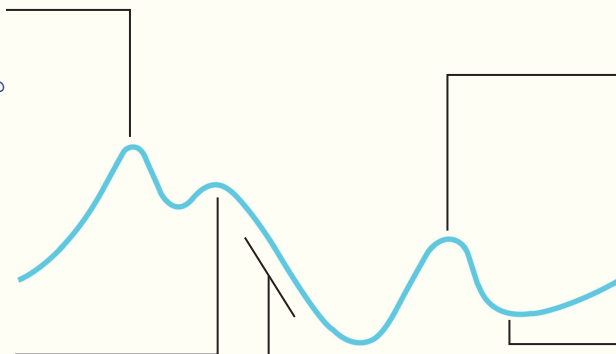
- D/t RA contraction
- Absent A wave : Afib

C wave :

- D/t RV contraction

↓  
Bulging of tricuspid valve  
into RA

- Prominent C wave :  
Tricuspid regurgitation.



V wave :

- D/t RA filling.
- Giant V wave : Severe tricuspid regurgitation

X descent :

- D/t RA relaxation
- Prominent X descent :  
Constrictive pericarditis

Y descent :

- D/t RA emptying
- Absent Y descent :  
Cardiac tamponade

## Cardiac Cycle

00:29:01

### Duration :

$$\text{Duration} = \frac{60}{\text{HR}}$$

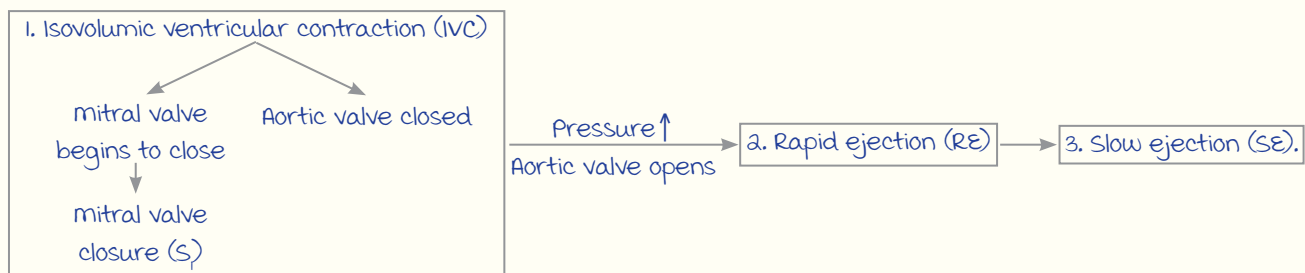
Normal duration = 0.8 s.

|                 | Atrial | Ventricular |
|-----------------|--------|-------------|
| Systole (Short) | 0.1 s  | 0.3 s       |
| Diastole (Long) | 0.7 s  | 0.5 s       |

On exercise :

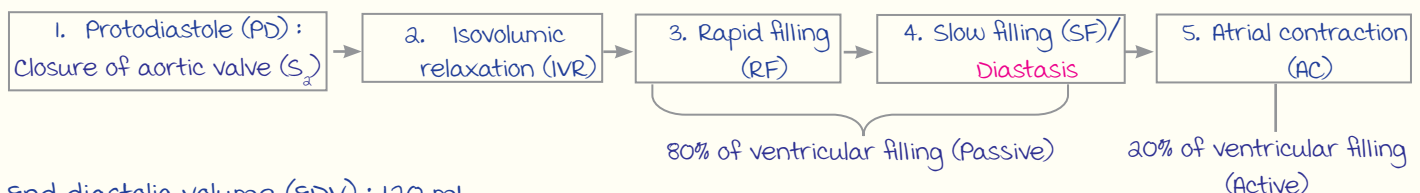
Duration : 0.3 s → ventricular systole : 0.12 s  
 → ventricular diastole : 0.18 s → Filling compromised

### Systole :



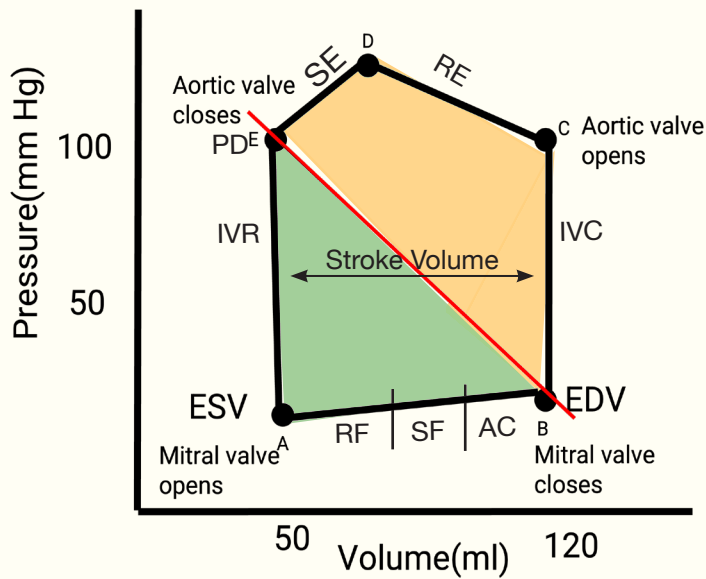
End systolic volume (ESV) : volume of blood left at the end of systole = 50 mL.

### Diastole :



End diastolic volume (EDV) : 120 mL.

----- Active space -----

Pressure Volume Graph :

$$1. \text{ Stroke volume (SV)} = \text{EDV} - \text{ESV}$$

$$= 120 - 50$$

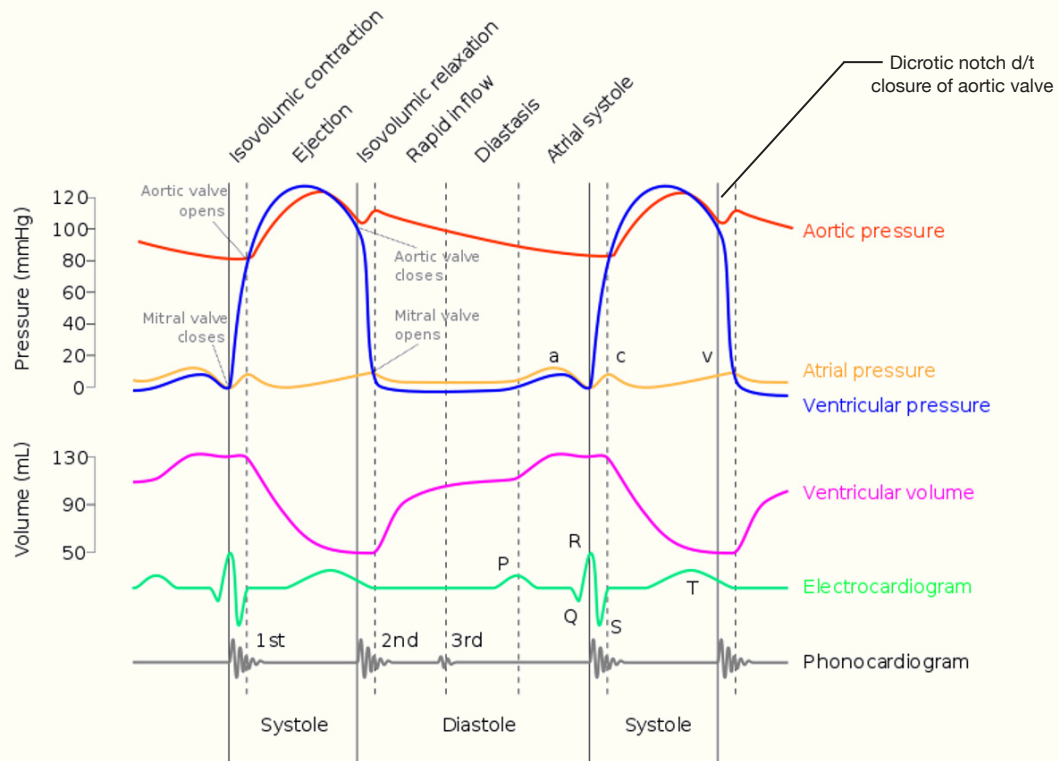
$$= 70 \text{ mL.}$$

$$2. \text{ Ejection fraction (EF)} = \frac{\text{SV}}{\text{EDV}} \times 100$$

Normal EF = 55 - 60%

● : Diastole

● : Systole

Wigger's Diagram :

Phonocardiogram : Recording of heart sounds.

----- Active space -----

| Heart sound | Cause                                               | ECG correlation |
|-------------|-----------------------------------------------------|-----------------|
| $S_1$       | Closure of mitral & tricuspid valves                | R wave          |
| $S_a$       | Closure of aortic & pulmonary valves                | End of T wave   |
| $S_3$       | Rapid filling of ventricle                          | -               |
| $S_4$       | Forceful atrial contraction against stiff ventricle | P wave          |

## Cardiac Output

00:50:17

1.  $\text{Cardiac output} = \text{HR} \times \text{SV}$

Normal CO : 5 L/min.

2. Fick's principle : measurement of CO.

$$\text{CO} = \frac{\text{O}_a \text{ consumed (mL/min)}}{\text{Arteriovenous O}_a \text{ difference}}$$

3. Cardiac index ratio : CO per  $\text{m}^2$  of body surface area/BSA  
(Normal : 3.2 L/min/ $\text{m}^2$  of BSA)

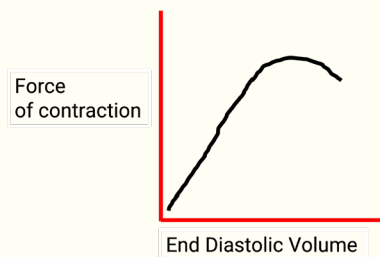
## Regulation of CO :

Mnemonic : CAP.

|                | myocardial contractility | Afterload                                   | Preload               |
|----------------|--------------------------|---------------------------------------------|-----------------------|
| Best marker    | EF                       | Aortic pressure/Total peripheral resistance | EDV                   |
| Relation to CO | Directly proportional    | Inversely proportional                      | Directly proportional |

## Frank-Starling Law :

$\uparrow \text{EDV} \rightarrow \uparrow \text{Force of contraction (within physiological limits)}$ .



Frank starling law

----- Active space -----

Pulsus alternans :



Pulsus alternans

Distribution of CO :

Highest blood flow : Liver (1500 mL/min).

Highest blood flow/100 g : Kidney (420 mL/100 g/min).

AV  $O_a$  difference : → Highest : Heart (114 mL/min).  
 → Lowest : Kidneys (14 mL/L).

Highest  $O_a$  consumption : Liver (51 mL/min).Highest  $O_a$  consumption/100 g : Heart (10 mL/100 g/min).Coronary Circulation

00:59:35

Features :

- volume : 250 mL/min (5% of CO).
- Receives blood during diastole (D/t compression of vessels during systole).
- Area prone to ischemia : Subendocardial region of LV.

Determinants of myocardial  $O_a$  demand :

- ↑ myocardial contractility.
- ↑ HR.
- ↑ Pressure work.

Regulation : By accumulation of local metabolites.

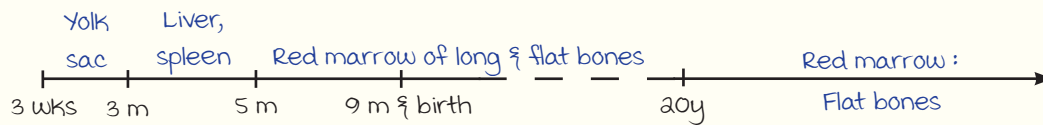


## Hematopoiesis

1:02:55

----- Active space -----

### Sites of Hematopoiesis :



### Erythropoiesis :

Steps :



### Erythropoietin : Regulation.

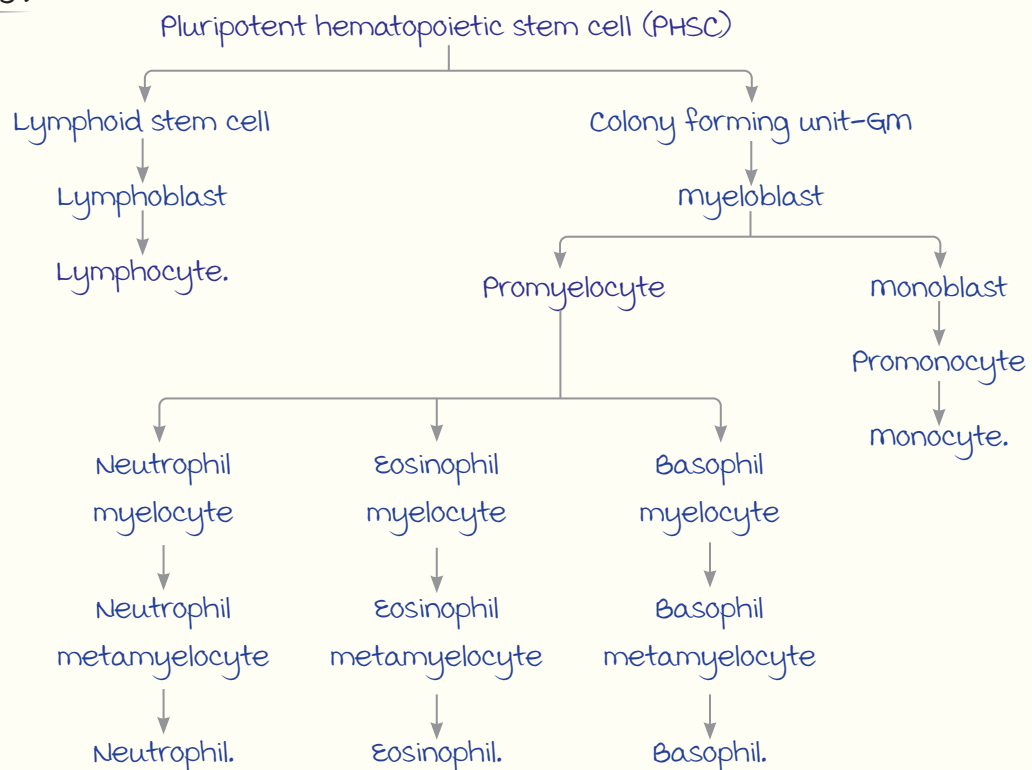
- Structure : Glycoprotein
- Sources :
  - Kidney (85%) : Peritubular capillary cells.
  - Liver (15%) : Perivenous hepatocytes.
  - minor : Brain, uterus, oviducts.
- Drug forms : Epoetin alfa, Darbepoetin alfa. Used in the treatment of anemia of CKD.
- Stimuli :
  - Hypoxia.
  - Alkalosis (High altitude).
  - Androgens.
  - Catecholamines ( $\beta$  receptor action).
- MOA : Inhibits apoptosis of RBCs.

# CARDIOVASCULAR PHYSIOLOGY : PART 2

## Leukopoiesis

00:00:06

### Steps :



Regulation : By colony stimulating factors (CSF).

- Granulocyte colony stimulating factor (G-CSF) → Drug form : **Filgrastim**.
- Granulocyte monocyte colony stimulating factor (GM-CSF) → Drug form : **Sargramostim**.
- macrophage colony stimulating factor (m CSF).
- IL-5 : **Eosinophil** development.
- IL-3 & 4 : **Basophil** development.

## Thrombopoiesis

00:04:00

Steps : PHSC → GM-CSF thrombolineage → megakaryocyte → Platelets.

Regulation :

- Thrombopoietin (Drug form : **Eltrombopag**).
- IL-11 (Drug form : **Oprelvekin**).

## Blood Grouping Systems

00:05:20

----- Active space -----

### ABO System :

| Blood group        | Antigen (RBC)    | Antibody (Plasma)      |
|--------------------|------------------|------------------------|
| A                  | A                | Anti-B                 |
| B                  | B                | Anti-A                 |
| AB                 | A & B            | Nil                    |
| O                  | No A, No B       | Anti-A, Anti-B         |
| Bombay blood group | No A, No B, No H | Anti-A, Anti-B, Anti-H |

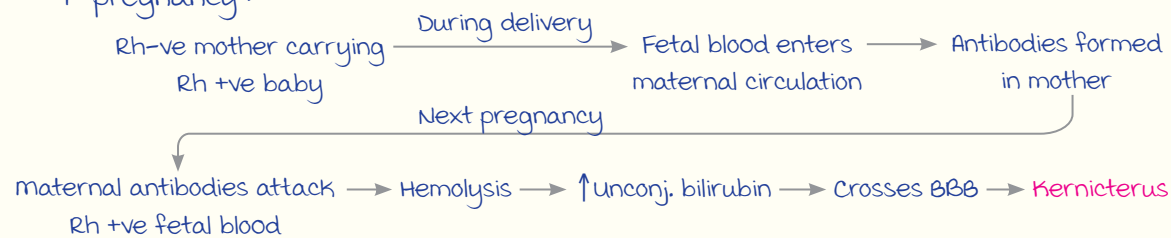
- universal donor : **O**.
- universal recipient : **AB**.
- Antibodies : Not present at birth (Develop after 3 - 6 months).
- D antigen : **Rh group**.

### Rh System :

#### Erythroblastosis fetalis :

- Pathophysiology :

1<sup>st</sup> pregnancy :



- Treatment : **Anti-Rh antibodies**.

## Blood Vessels

00:10:02

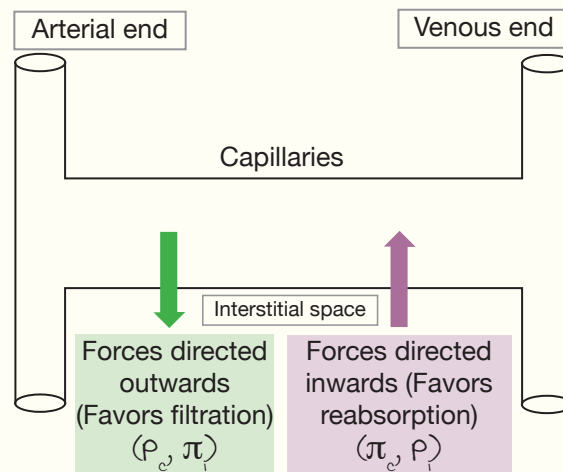
### Characteristics :

| Blood vessel | Features                                                                                                                                                                                     | Blood flow velocity | Net cross sectional area |
|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|--------------------------|
| Aorta        | <b>windkessel</b> effect (Elasticity, stretch & recoil)                                                                                                                                      | ↑                   | ↓                        |
| Arterioles   | <b>Resistance</b> vessels (D/t smooth muscles)                                                                                                                                               | -                   | -                        |
| Capillaries  | <ul style="list-style-type: none"> <li>• AKA <b>exchange</b> vessels</li> <li>• Function : Gas exchange</li> <li>• Contain 5% of blood volume</li> </ul>                                     | ↓                   | ↑                        |
| Veins        | <ul style="list-style-type: none"> <li>• AKA <b>capacitance</b> vessels</li> <li>• Contains 55% of blood volume (Storage reservoir)</li> <li>• Highly distensible &amp; compliant</li> </ul> | -                   | -                        |

----- Active space -----

maximum diameter : *Vena cava*.maximum thickness : *Aorta*.maximum cross-sectional area : *Capillaries*.maximum blood flow velocity : *Aorta*.minimum blood flow velocity : *Capillaries*.Shunt vessels : *AV anastomosis* (Temperature regulation).maximum blood pressure : *Aorta*.minimum blood pressure : *Veins*.Site of gas exchange : *Capillaries*.

### Starling Forces :



Starling forces :

1. Capillary hydrostatic pressure ( $P_c$ )
2. Capillary oncotic pressure ( $\pi_c$ )
3. Interstitial hydrostatic pressure ( $P_i$ )
4. Interstitial oncotic pressure ( $\pi_i$ )

**Net filtration pressure** = Forces favoring filtration - Forces favoring reabsorption.

$$NFP = (P_c + \pi_i) - (\pi_c + P_i)$$

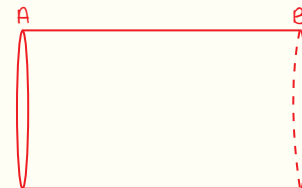
### Hemodynamics :

Poiseuille-Hagen formula :  $F$  (Blood flow) =  $(P_a - P_g) \times \pi r^4 / 8\eta L$ .

- Blood flow  $\propto$  Pressure difference ( $P_a - P_g$ ).
- Blood flow  $\propto$  radius<sup>4</sup> ( $r^4$ ).
- Blood flow  $\propto$  1/Length ( $L$ ).
- Blood flow  $\propto$  1/viscosity.

Fourth power law : Resistance  $\propto \frac{L}{\text{Radius}^4}$ 

$$\text{velocity} \propto \frac{1}{\text{Area}} \rightarrow \boxed{\text{velocity} \propto \frac{1}{r^2}}$$

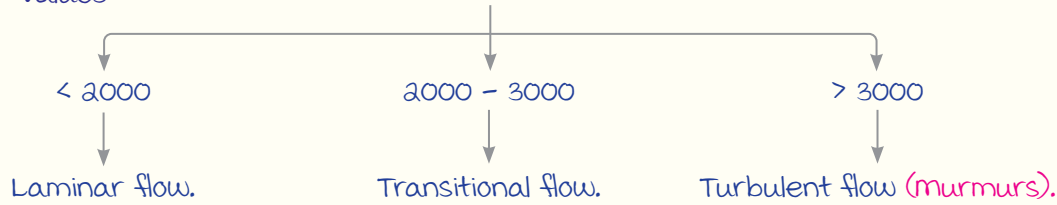


Blood pressure :

- SI unit : Pascal (mmHg is preferred).
- SBP/DBP : 120/80 mmHg.
- Pulse pressure (PP) :  $SBP - DBP = 40$  mmHg.
- mean arterial pressure (MAP) =  $DBP + 1/3 (PP) = 93 - 100$  mmHg.

Reynold's number :

- Reynold's number  $\propto \frac{\text{Density of blood, diameter, velocity of flow}}{\text{Viscosity}}$ .
- Values :



- Clinical significance :

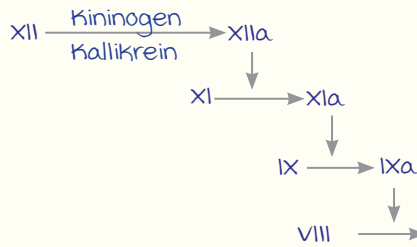
Anemia d/t pregnancy  $\rightarrow \downarrow \text{RBC} \rightarrow \downarrow \text{Viscosity} \rightarrow \uparrow \text{Reynolds number} \rightarrow \text{Flow murmur.}$

## Clotting of Blood

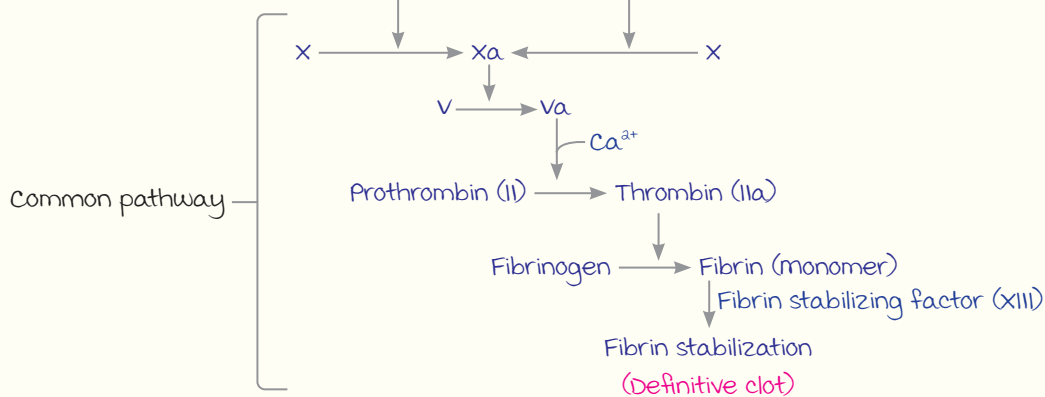
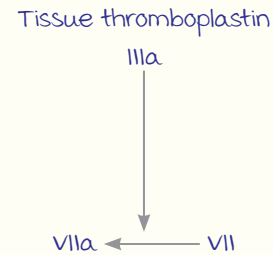
00:28:35

### Coagulation Cascade :

Intrinsic pathway :



Extrinsic pathway :



| Test | Normal value | measures                                |
|------|--------------|-----------------------------------------|
| APTT | 26 - 40 s    | Intrinsic & common coagulation pathways |
| PT   | 11 - 16 s    | Extrinsic & common coagulation pathways |

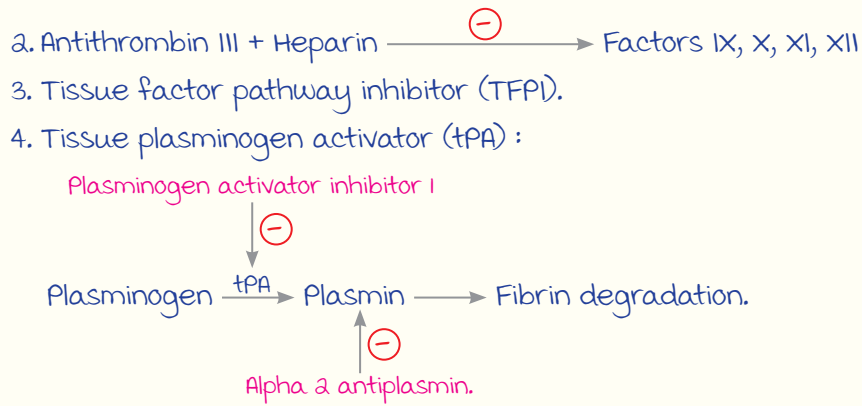
APTT : Activated partial thromboplastin time.

PT : Prothrombin time.

### Anticoagulation pathways :

1. Thrombin + Thrombomodulin  $\xrightarrow{+}$  Protein C, S  $\xrightarrow{-}$  Factor V, VIII.

----- Active space -----



### Vitamin K Dependent Factors :

- Factor II, VII, IX, X.
- Protein C & S.

Vitamin K antagonist : Warfarin ( $\ominus$  vitamin K epoxide reductase).

### Regulation of Blood Pressure

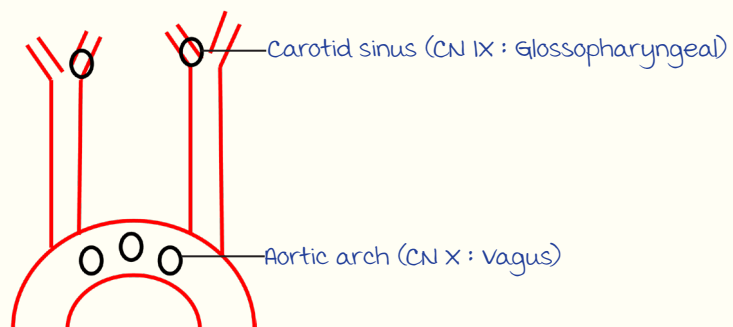
00:35:38

Regulatory mechanisms :

|                           | Short term regulation                                                                                        | Intermediate term regulation                     | Long term regulation                                                                              |
|---------------------------|--------------------------------------------------------------------------------------------------------------|--------------------------------------------------|---------------------------------------------------------------------------------------------------|
| Time taken for regulation | <ul style="list-style-type: none"> <li>• Within seconds</li> <li>• Under <b>neural</b> regulation</li> </ul> | Within hours                                     | Within days                                                                                       |
| Regulatory mechanisms     | 1. Baroreflex<br>2. Chemoreflex<br>3. CNS ischemic response                                                  | 1. Capillary fluid shift<br>2. Stress relaxation | <b>Renal regulation :</b><br>Pressure diuresis and natriuresis<br>$\downarrow$<br>$\downarrow$ BP |

### Baroreflex :

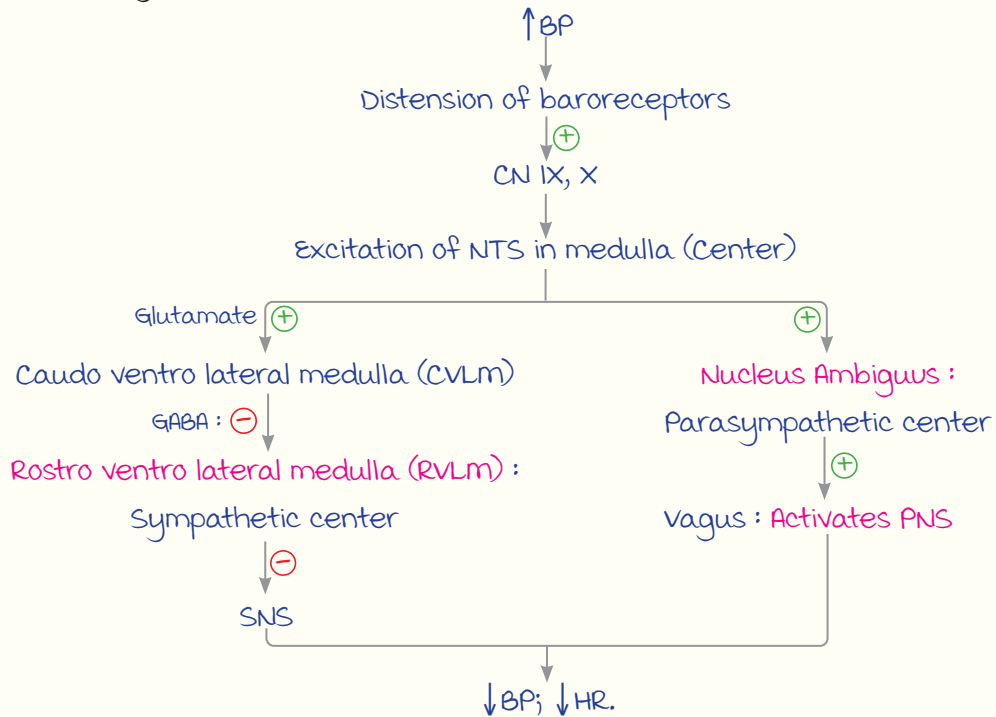
- 1<sup>st</sup> line of BP control  
(MAP : 70 - 140 mmHg).
- Negative feedback mechanism.
- Located in **adventitia** of the vessel.



Baroreceptor location & nerve supply

Reflex pathway :

----- Active space -----



### Chemoreflex :

Second line for BP control.

Range : 40 - 70 mmHg.

Effects : Variable effect.

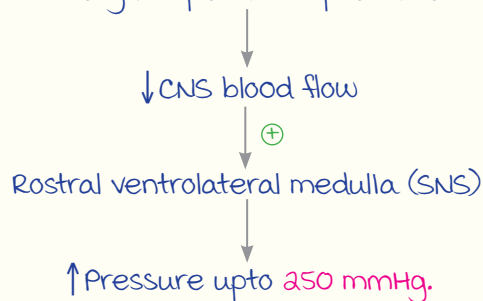
- Direct effect : PNS activation → ↓ HR.
- Indirect effect : SNS activation → ↑ HR.

### CNS Ischemic Response :

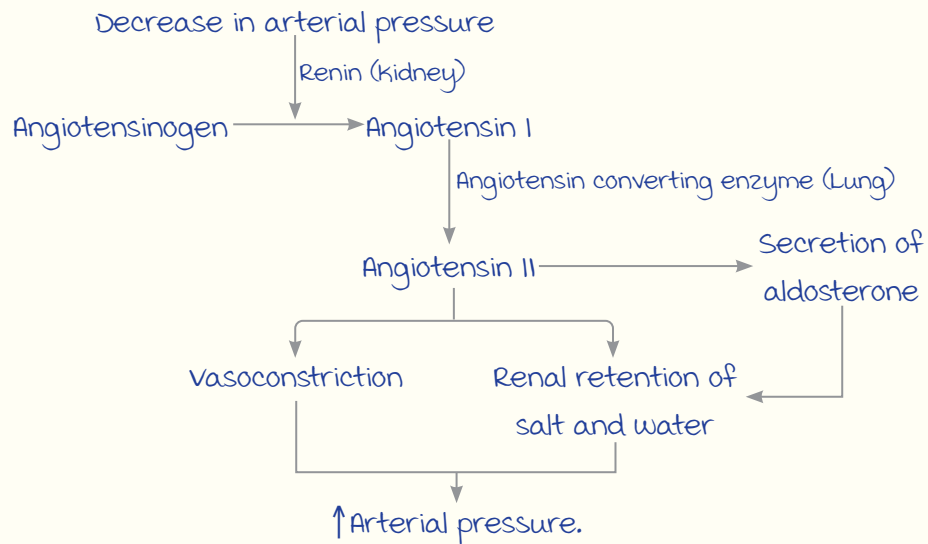
Last line of control for BP/last ditch stand mechanism.

Range : If MAP < 40 mmHg.

Effects : Very low perfusion pressure



----- Active space -----

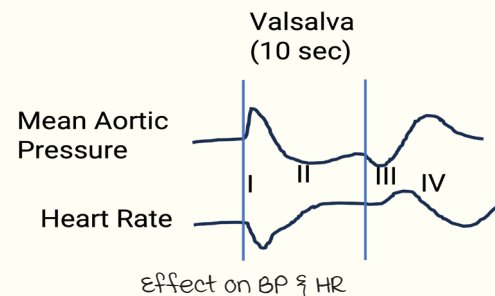
**Renin Angiotensin Aldosterone System :****Chemical Control of Blood Pressure :**

| vasoconstrictors (↑BP)                                                                                  | vasodilators (↓BP)                                                                                      |
|---------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
| 1. Urotensin : most potent<br>2. Norepinephrine<br>3. Endothelin<br>4. Vasopressin<br>5. Angiotensin II | 1. Calcitonin gene related peptide (CGRP)<br>2. Nitric oxide<br>3. Natriuretic peptide<br>4. Bradykinin |

**Valsalva Maneuver** 

00:48:31

- Forced expiration against a closed glottis.
- mayer's law :  $BP \propto \frac{1}{HR}$



| Phase                 | Physiological change                | Effect on BP | Effect on HR |
|-----------------------|-------------------------------------|--------------|--------------|
| 1. Onset of straining | ↑↑↑ Intrathoracic pressure          | ↑            | ↓            |
| 2. maintain straining | ↓ venous return<br>↓ Cardiac output | ↓            | ↑            |
| 3. Release straining  | ↓ Intrathoracic pressure            | ↓            | ↑            |
| 4. Overshoot          | ↑ venous return<br>↑ Cardiac output | ↑            | ↓            |

## Cardiac Reflexes

00:52:56

----- Active space -----

|                      | Reflex                | Cause & pathophysiology                                                                                                                                       |
|----------------------|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Tachycardic reflex   | Bainbridge reflex     | <p><b>Congestive cardiac failure :</b></p> <p>Fluid overload</p> <p>↓</p> <p>↑ Right atrial stretching</p> <p>↓</p> <p>↑ HR</p>                               |
| Bradycardic reflexes | Bezold-Jarisch reflex | <p><b>myocardial infarction :</b></p> <p>Infarcted tissue</p> <p>↓</p> <p>Release of <b>serotonin</b></p> <p>↓</p> <p>Bradycardia (↓ HR)      Hypotension</p> |
|                      | Cushing's reflex      | <p><b>Head injury :</b></p> <p>↑ Intracranial tension (ICT)</p> <p>↓</p> <p>↓ HR</p>                                                                          |
|                      | Czermak-Hering reflex | <p>• Carotid sinus massaging</p> <p>↓</p> <p>⊕ PNS</p> <p>↓</p> <p>↓ HR</p> <p>• Rx : PSVT</p>                                                                |

# GASTROINTESTINAL PHYSIOLOGY

## Layers of GIT :

mucosa (Innermost) → Submucosa → muscularis externa → Serosa (Outermost).

## Enteric Nervous System :

|          | meissner's plexus/Submucosal plexus | myenteric plexus/Auerbach's plexus |
|----------|-------------------------------------|------------------------------------|
| Location | Submucosa                           | B/w circular & longitudinal muscle |
| Function | Secretion                           | motility                           |

## Daily Water Turnover in GIT :

| Source/Location                      |                   | volume (mL)    |
|--------------------------------------|-------------------|----------------|
| Ingested fluid                       |                   | 2000           |
| GI secretions                        | Salivary glands   | 1500           |
|                                      | Stomach           | 2500 (maximum) |
|                                      | Exocrine pancreas | 1500           |
|                                      | Biliary           | 500            |
|                                      | Intestine         | 1000           |
| Total (Secretions + ingested fluid)  |                   | 9000           |
| Reabsorption                         | Jejunum           | 5500           |
|                                      | Ileum             | 2000           |
|                                      | Colon             | 1300           |
|                                      | Total reabsorbed  | 8800           |
| Remaining 200 mL : Excreted in stool |                   |                |

## GIT Secretions

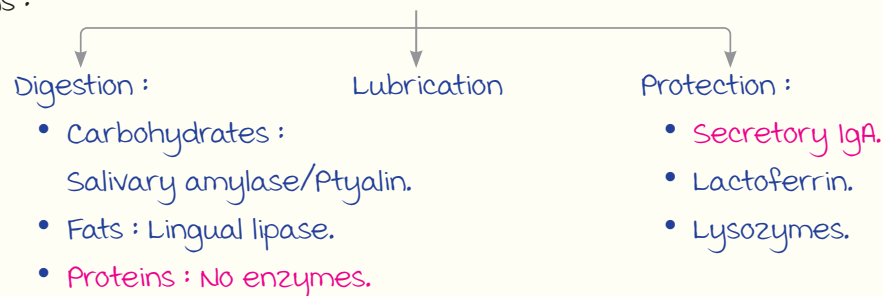
00:04:25

### Saliva :

Salivary glands :

- Parotid.
- Sublingual.
- Submandibular.
- minor glands (Tongue, palate).

Functions :



Types of secretions :

----- Active space -----

|                          | Features                                                                                                                                                                                        | Tonicity (w.r.t plasma)                |
|--------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|
| Acinar cells secretion   | <ul style="list-style-type: none"> <li>Primary secretion</li> <li>Rich in NaCl</li> </ul>                                                                                                       | Isotonic                               |
| Ductal secretion (Final) | <ul style="list-style-type: none"> <li>Reabsorbs <math>\text{Na}^+</math> &amp; <math>\text{Cl}^-</math></li> <li>Secretes <math>\text{K}^+</math> &amp; <math>\text{HCO}_3^-</math></li> </ul> | Hypotonic<br>(Impermeability to water) |

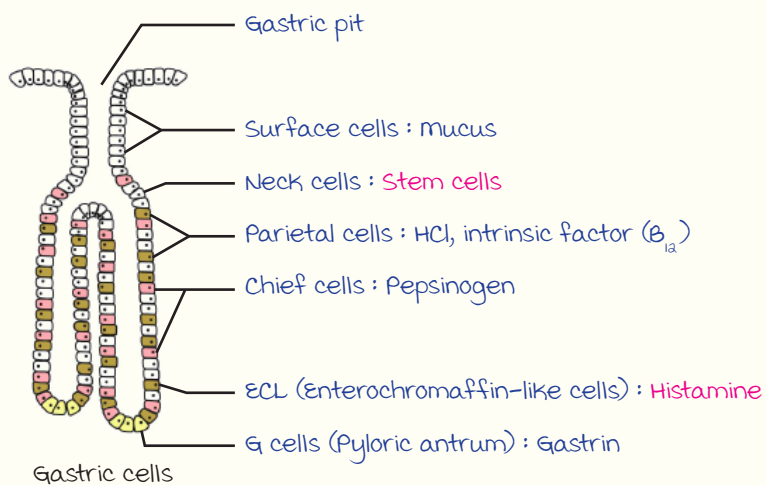
High salivary flow rate :

- $\uparrow \text{Na}^+$  &  $\text{Cl}^-$
  - $\downarrow \text{K}^+$
- } D/t less time for reabsorption & secretion.

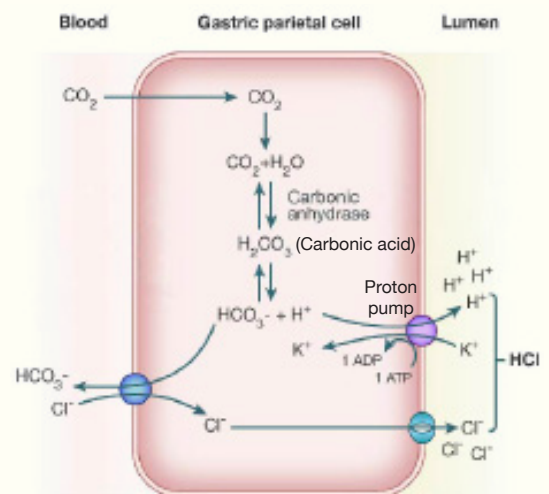
Gastric Secretion :

pH : 1.5 - 3.

Gastric cells :



Parietal cells :



Function of parietal cells

Post prandial alkaline tide :  $\uparrow \text{HCO}_3^-$  levels post food intake.

Factors affecting gastric acid secretion :

| $\uparrow$ Secretion                                                                                                                                                                                                                                                                                 | $\downarrow$ Secretion                                                                                                                                |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>Gastrin <math>\rightarrow</math> CCK-B receptor</li> <li>Histamine <math>\rightarrow</math> <math>\text{H}_2</math> receptor</li> <li>Acetylcholine <math>\rightarrow</math> <math>\text{m}_3</math> receptor</li> </ul> <p>Synergism <math>\oplus</math></p> | <ul style="list-style-type: none"> <li>Gastric acid (-ve feedback)</li> <li>Somatostatin (D-cell)</li> <li>Prostaglandins (Cytoprotective)</li> </ul> |

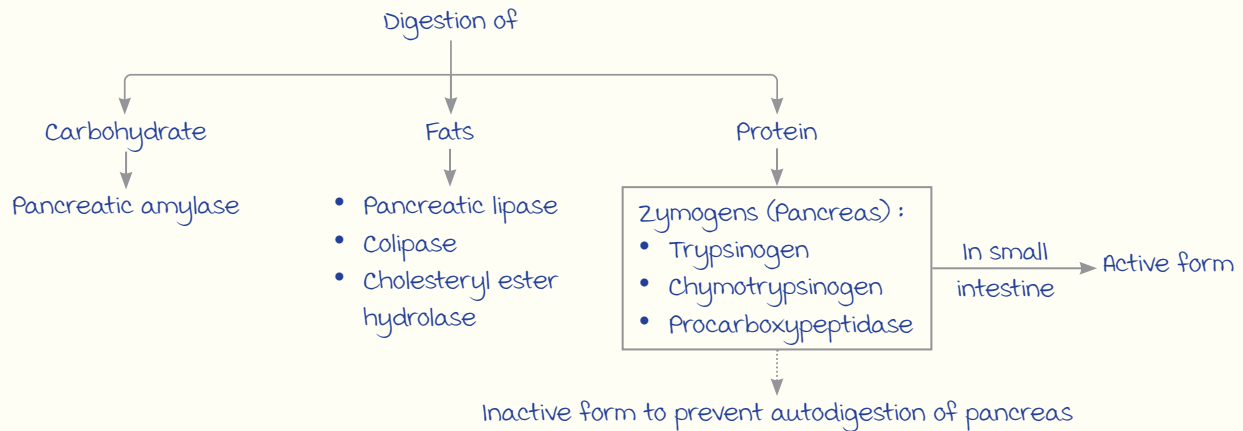
Note :

NSAID  $\rightarrow$   $\downarrow$  Prostaglandin  $\rightarrow$  Peptic ulcer.

----- Active space -----

Exocrine Pancreatic Secretion :pH : 8 (Rich in  $\text{HCO}_3^-$ ).

Enzymes :



Hormones involved in regulation:

|                             | Cholecystokinin-pancreozymin | Secretin                                 |
|-----------------------------|------------------------------|------------------------------------------|
| Site of action              | Acinar cell                  | Pancreatic duct                          |
| Function                    | ↑ Digestive enzymes          | ↑ $\text{HCO}_3^-$ (Acid neutralization) |
| Stimuli                     | Fat/Protein diet             | Acid                                     |
| mechanism of enzyme release | Exocytosis                   | -                                        |
| Second messenger            | $\text{Ca}^{2+}$             | cAMP                                     |

Biliary Secretion :

pH : 7.8.

Bile acid → Primary : Cholic acid & chenodeoxycholic acid.  
 → Secondary (Intestine) : Deoxycholic acid & lithocholic acid.

Conjugation → In liver : Bile acids + glycine/taurine.

Bile salts : Conjugated bile acids + sodium/potassium.

Regulation :

- Choleretics (↑ es bile secretion) : Bile salts.
- Cholagogues (⊕ Gall bladder contraction) : Cholecystokinin.

Small Intestinal Secretions  :

Epithelial cells (Enterocytes).

Goblet cells : mucus secretion.

Paneth cells (Base) : Store zinc, defensins, lysozymes.

} Present in crypts of  
Lieberkuhn.

Stem cells : *Base.*

----- Active space -----

**GI Hormones**

00:21:16

| Hormone                                                | Stimulus                                                                           | Source                    | Function/Key Features                                                                                                                                                                                                                                                                                                                                                                                               |
|--------------------------------------------------------|------------------------------------------------------------------------------------|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Gastrin                                                | <ul style="list-style-type: none"> <li>Distension</li> <li>Protein diet</li> </ul> | G cells                   | <ul style="list-style-type: none"> <li>↑ Gastric acid secretion               <ul style="list-style-type: none"> <li>Pepsinogen → Pepsin</li> </ul> </li> <li>Lower esophageal sphincter : Contraction</li> <li>↑ Gastric motility</li> <li>Trophic action : ⊕ GI mucosal growth</li> </ul>                                                                                                                         |
| Cholecystokinin (CCK)                                  | Fat/protein diet                                                                   | I cells                   | <ul style="list-style-type: none"> <li>Receptor : CCK-A</li> <li>↑ Digestive enzymes in pancreatic secretion</li> <li>Cholagogue + relaxation of sphincter of Oddi</li> </ul> <p style="text-align: center;">Bile release</p> <ul style="list-style-type: none"> <li>⊖ Gastric emptying</li> <li>⊕ Intestinal motility</li> <li>Augments action of secretin</li> <li>Satiety signal</li> </ul>                      |
| Secretin<br>(1 <sup>st</sup> hormone to be discovered) | Acid                                                                               | S cells                   | <ul style="list-style-type: none"> <li>↑ <math>\text{HCO}_3^-</math> in pancreatic secretion               <ul style="list-style-type: none"> <li>Acid neutralization</li> </ul> </li> <li>⊖ Gastric emptying</li> <li>↓ Gastric acid</li> <li>Augments action of CCK</li> </ul>                                                                                                                                    |
| GIP                                                    | -                                                                                  | K cells                   | Incretin (↑ Insulin)                                                                                                                                                                                                                                                                                                                                                                                                |
| Vasoactive intestinal peptide (VIP)                    | -                                                                                  | -                         | <ul style="list-style-type: none"> <li>Smooth muscle relaxation</li> <li>↑ Fluid &amp; electrolyte secretion</li> <li>Vasodilatation</li> </ul>                                                                                                                                                                                                                                                                     |
| motilin                                                | -                                                                                  | m <sub>0</sub> cells      | <ul style="list-style-type: none"> <li>↑ motility</li> <li>Regulates migratory motor complex</li> </ul>                                                                                                                                                                                                                                                                                                             |
| Peptide YY                                             | -                                                                                  | -                         | ⊖ Ileal motility (Ileal brake)                                                                                                                                                                                                                                                                                                                                                                                      |
| Ghrelin                                                | -                                                                                  | Oxyntic region of stomach | <ul style="list-style-type: none"> <li>Levels :               <ul style="list-style-type: none"> <li>Peak : Fasting</li> <li>Low : Obesity</li> <li>High : Anorexia nervosa</li> </ul> </li> <li>Function :               <ul style="list-style-type: none"> <li>Orexinergic</li> <li>↑ Gastric motility</li> <li>↑ Gastric acid secretion</li> <li>↑ Adipogenesis</li> <li>↑ Growth hormone</li> </ul> </li> </ul> |

GIP : Gastric inhibitory polypeptide/Glucose dependant insulinotropic polypeptide.

----- Active space -----

**Digestion & Absorption**

00:30:50

**Carbohydrates, Lipids & Proteins :**Digestion : **Enzymes.**

|                 | Carbohydrate                           | Lipid                                                                                                                                                           | Protein                                           |
|-----------------|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|
| Saliva          | Salivary amylase                       | Lingual lipase                                                                                                                                                  | No enzyme                                         |
| Stomach         | No enzymes                             | Gastric lipase                                                                                                                                                  | Pepsin                                            |
| Pancreas        | Pancreatic amylase                     | <ul style="list-style-type: none"> <li>• Pancreatic lipase</li> <li>• Colipase</li> <li>• Cholesterol esterase</li> </ul> Triglycerides (TG) ↓ Free fatty acids | Trypsin, chymotrypsin, carboxypeptidase, elastase |
| Small Intestine | Disaccharides : (Brush border enzymes) | No enzymes                                                                                                                                                      | No enzyme                                         |

Absorption :

|          | Carbohydrate                                                                                                                                        | Lipid                                                                                                                                                                                                                                          | Protein                                                                                                                                                      |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Site     | Small intestinal mucosa                                                                                                                             |                                                                                                                                                                                                                                                |                                                                                                                                                              |
| Features | Transport : <ul style="list-style-type: none"> <li>• Glucose : <b>SGLT-1</b></li> <li>• Fructose : <b>GLUT-5</b></li> </ul> } <b>GLUT-2</b> → Blood | Fatty acid emulsified by bile salts<br>↓<br>Via micelles (Transport)<br>Reach brush border cells of intestine<br>↓<br>Via simple diffusion<br>Absorbed into enterocyte (mucosal cell)<br>↓<br>Re-esterified to TG → <b>Chylomicron</b> → Blood | Dipeptides & tripeptides<br>↓<br><b>Peptide transporter</b><br>Dipeptides & tripeptides (mucosal cells)<br>↓<br><b>Proteases</b><br>Amino acid<br>↓<br>Blood |

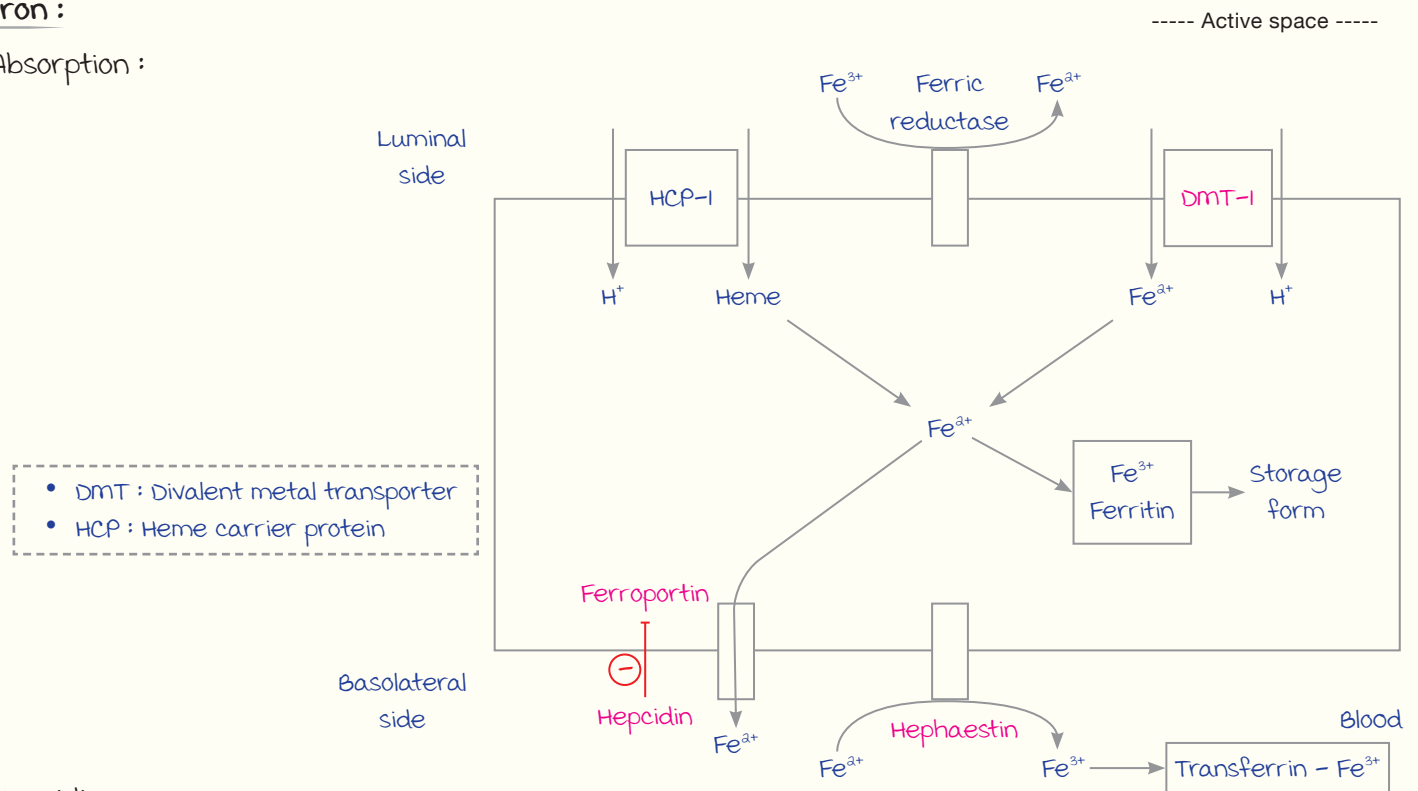
Note :

SGLT-1 (Secondary active transport) :

- Basis for  $\text{Na}^+$  absorption along with glucose in ORS.
- mutation → Congenital glucose galactose malabsorption (CGGM).

Iron :

Absorption :

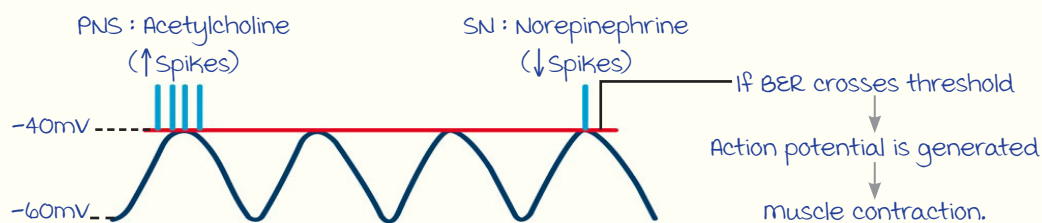


Hepcidin :

- Source : Liver.
- Levels regulated by **HFE gene** (mutation  $\rightarrow$  Hemochromatosis).
- Function :
  - $\ominus$  Basolateral transport of iron into blood : Binds to **ferroportin**.
  - Links body iron stores with iron absorption.
  - Inhibits Fe release from macrophage & other cells by binding to ferroportin.

GI Motility

00:45:10

BER :Pacemaker (**Interstitial cells of Cajal**)  $\rightarrow$  Has RMP.

Restless membrane potential (RMP)/Basal electrical rhythm (BER)

----- Active space -----

Motility Patterns :

## Deglutition :

## • Phases :

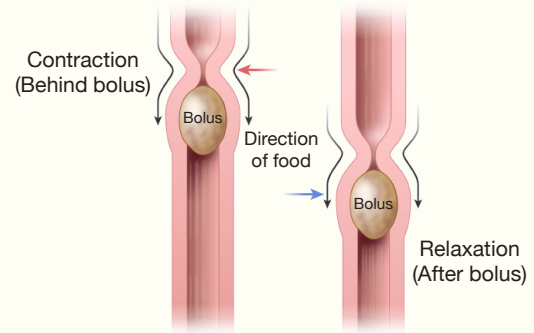
- Oral : Voluntary.
- Pharyngeal } Involuntary.
- Esophageal }

## • Deglutition reflex :

- Afferent : Cranial nerves V, IX, X.
- Center : Nucleus tractus solitarius (medulla).
- Efferent : Cranial nerves V, VII, XII.

## Esophageal motility :

- Stretch induced peristalsis.
- Segments :
  - Contracting : Substance P & acetylcholine.
  - Relaxing : Nitric oxide & VIP.
- Clinical significance : Achalasia cardia  
(LES :  $\ominus$  NO &  $\uparrow$  ACh activity).



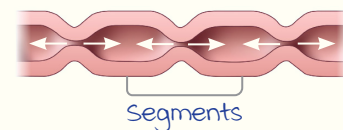
## Gastric motility :

- Receptive relaxation.
- Peristalsis.
- Retropulsion  $\rightarrow$  Chyme.
- Gastric emptying
  - $\rightarrow$  Fastest : Carbohydrates.
  - $\rightarrow$  Slowest : Fats.

| $\uparrow$ Gastric emptying                                                       | $\downarrow$ Gastric emptying  |
|-----------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>• Distension</li> <li>• Gastrin</li> </ul> | Duodenum $\rightarrow$ Fats, carbohydrates & acids $\rightarrow$ enterogastric reflex                               |

## Small intestinal motility :

- Peristalsis.
- Segmentation :  
Both ends contract  $\xrightarrow{\text{Slows food movement}}$  Adequate digestion & absorption.



## Large intestinal motility :

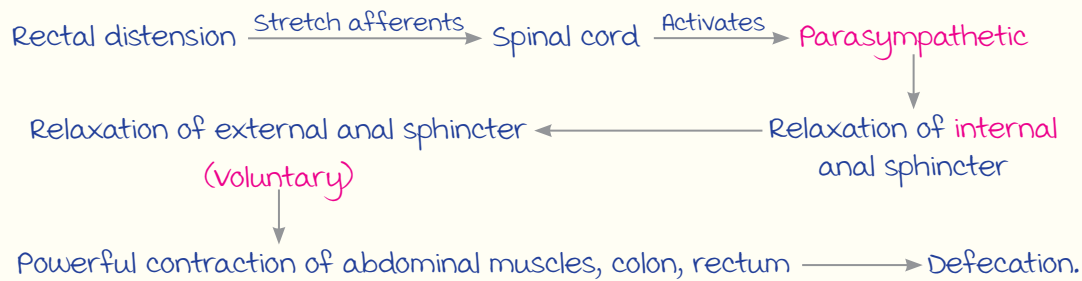
- Proximal : Haustrations  $\rightarrow$  Combined contraction of circular & longitudinal muscles  $\rightarrow$  Absorption of water & electrolytes.
- Distal :
  - Mass movements.
  - movement from one segment to another  $\rightarrow$  Rectal distension  $\rightarrow$  Defecation.

## Defecation

00:56:31

----- Active space -----

### Reflex :



First urge to defecate : Rectal pressure of 18 mmHg.

maximum rectal pressure : 55 mmHg → Beyond which involuntary defecation occurs.

## Migratory Motor Complex (MMC)

00:58:55

- Seen during fasting.
- migrates from stomach to colon : Clears GIT b/w meals.
- 1 MMC once every 90 minutes : Regulated by motilin.
- Erythromycin :
  - Rx of gastroparesis (m/c d/t diabetics).
  - Action : ↑ MMC via motilin ( $m_o$  receptor).

# RENAL PHYSIOLOGY

----- Active space -----

Functional unit : **Nephrons** (1 million/kidney).

Functions :

1. Filtration.
2. Reabsorption.
3. Secretion.
4. Excretion [(1-2) + 3].

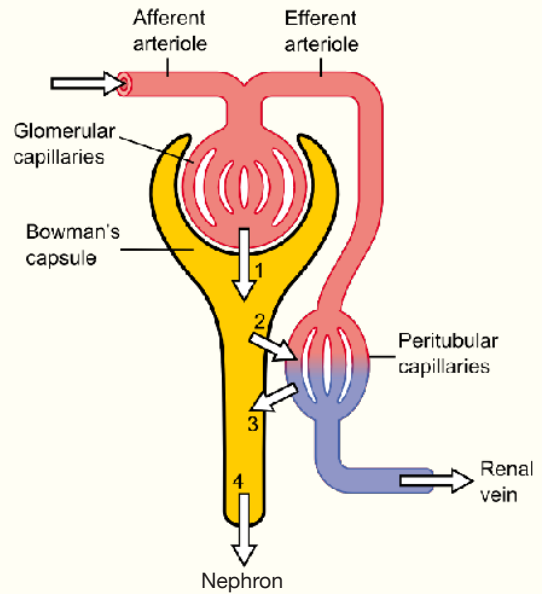
**Glomerular Filtration Rate (GFR) :**

Normal : **125 mL/min** (or) **180 L/day**.

Ultrafiltration :

**Proteins** not excreted d/t :

- Charge : **Heparan sulfate** (-ve).
- Size : Fenestration (**4-8 nm**).
- Podocytes : Foot process → **Slit diaphragm**.  
(Integrity : By nephrin, podocin,  $\alpha$ -actinin)



**Filtration forces :**

Favoring :

Hydrostatic pressure in glomerular capillaries ( $P_c$ ) : **60 mmHg**.

Opposing :

- Oncotic pressure in glomerular capillaries ( $\pi_c$ ) : **32 mmHg**
- Hydrostatic pressure in Bowman space ( $P_b$ ) : **18 mmHg**.

Net filtration pressure :  $P_c - (\pi_c + P_b) \approx$  **10 mmHg**.

**Filtration Co-efficient ( $K_f$ )** :

- Depends on capillary → Permeability.
- $GFR = K_f \times NPF$ . → Surface area.
- $K_f =$  **12.5 mL/min per mmHg**.

**Clearance :**

$$C = \frac{U \times V}{P}$$

U : Urine concentration  
V : Volume of urine  
P : Plasma concentration

markers to estimate :

- Gold standard : **Inulin**
  - m/c : **Creatinine**
  - New marker : **Cystatin c**
- } Freely filtered,  
no reabsorption/secretion

----- Active space -----

### Renal Circulation :

Renal blood flow (RBF) : 1100 mL/min (22% of C.O).

Renal plasma flow (RPF) : 625 mL/min.

Filtration fraction :

- $GFR/RPF = 125/625 = 20\%$
- Substances used to measure :
  - GFR : Inulin.
  - RPF : Para-aminohippuric acid.

### Regulation of GFR

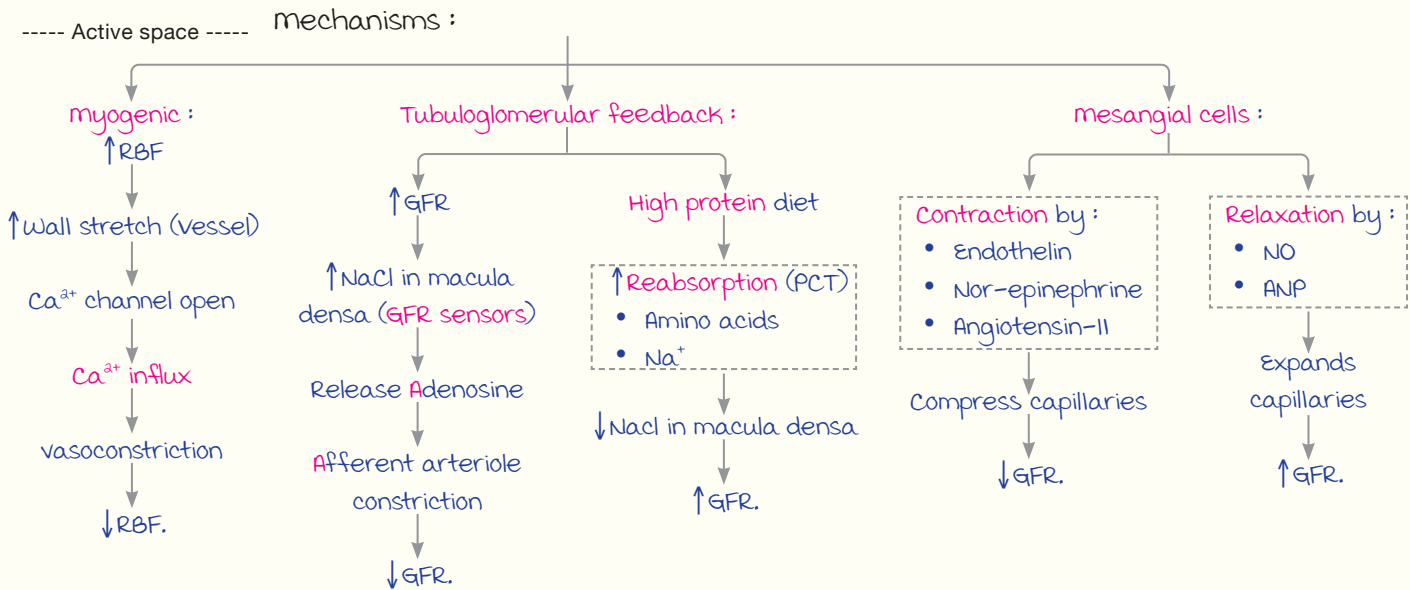
00:11:11

Factors  :

|                             |                              | RBF | GFR                                                                            |
|-----------------------------|------------------------------|-----|--------------------------------------------------------------------------------|
| Constriction<br>(Arteriole) | Afferent<br>(Norepinephrine) | ↓   | ↓                                                                              |
|                             | Efferent<br>(Angiotensin-II) |     | Biphasic :<br>• Initial ↑ ( $\uparrow P_c$ )<br>• Later ↓ ( $\uparrow \pi_c$ ) |
| Dilation<br>(Arteriole)     | Afferent<br>(Prostaglandins) | ↑   | ↑                                                                              |
|                             | Efferent<br>(ACE inhibitors) |     | ↓                                                                              |

### Autoregulation :

- Independent of neural regulation.
- Pressure maintained at **80-180 mmHg** → RBF & GFR constant.



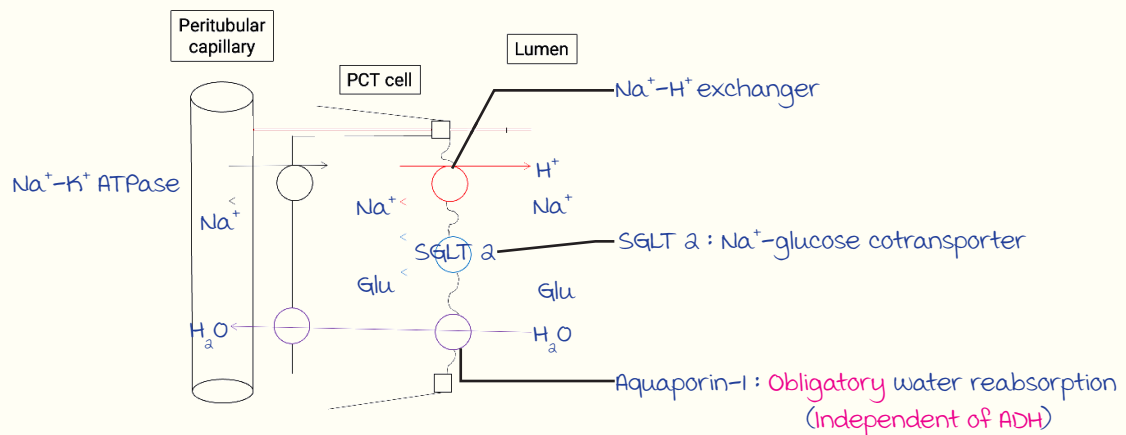
## Proximal Convoluted Tubule (PCT)

00:21:32

maximum reabsorption d/t :



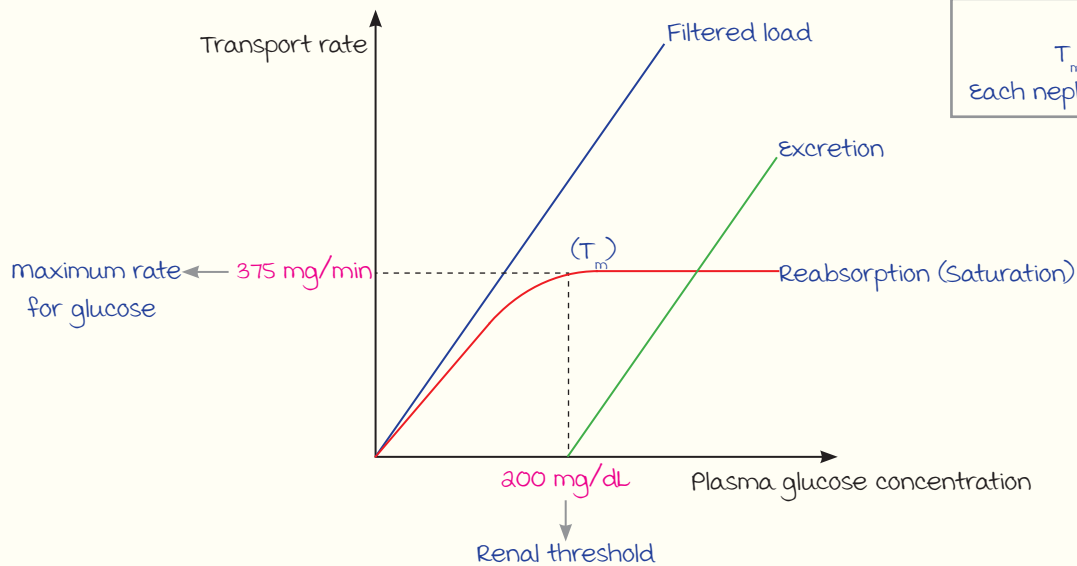
## Reabsorption :



Glucose reabsorption :

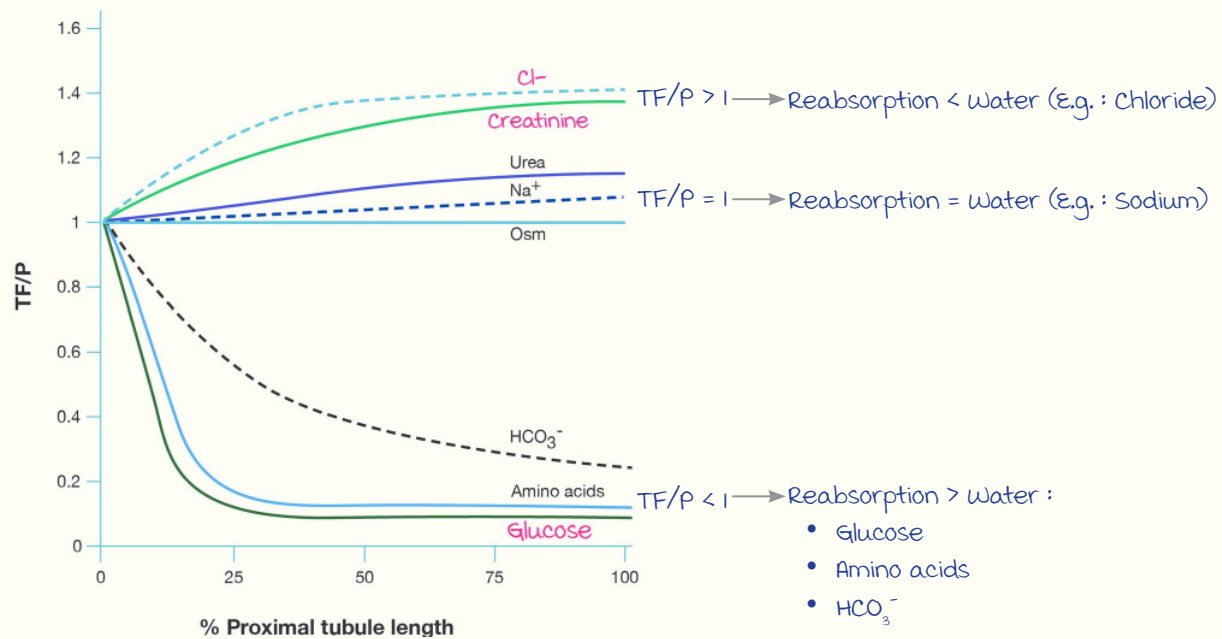
Via SGLT-2.

----- Active space -----



Tubular fluid plasma ratio (TF/P) ☼ :

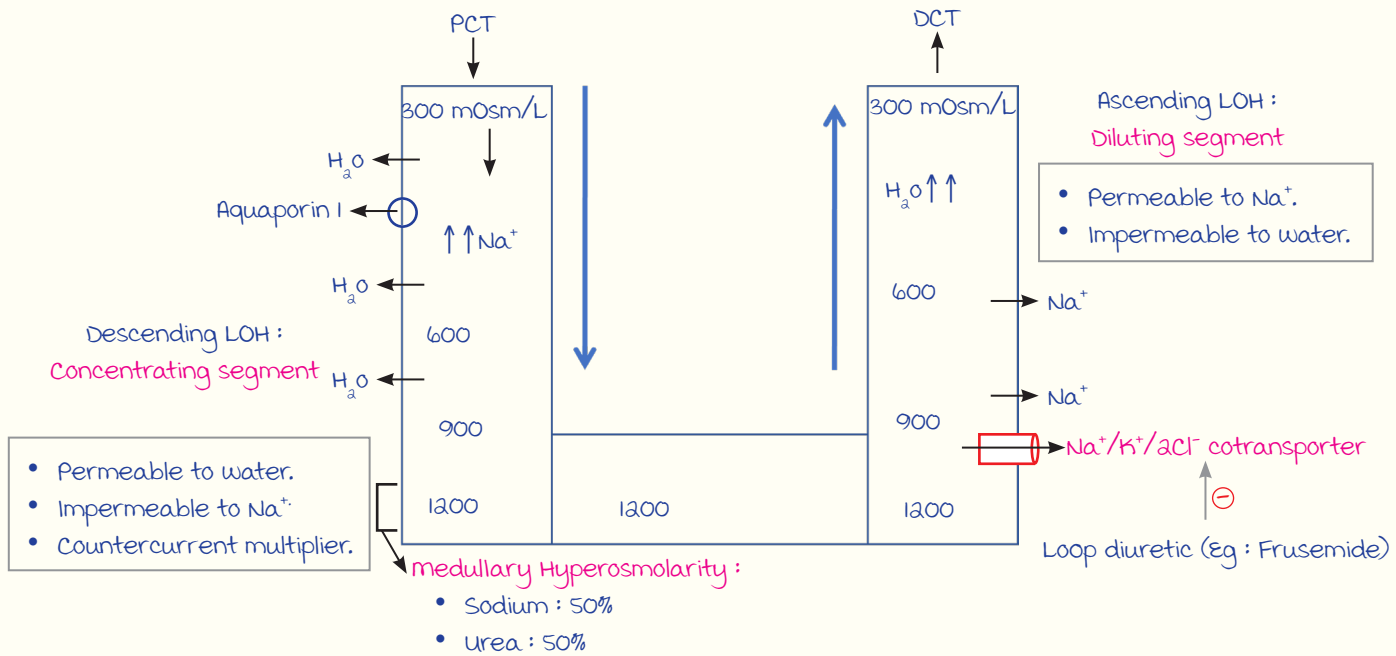
water reabsorption : 70%.



----- Active space -----

**Loop of Henle (LOH)**

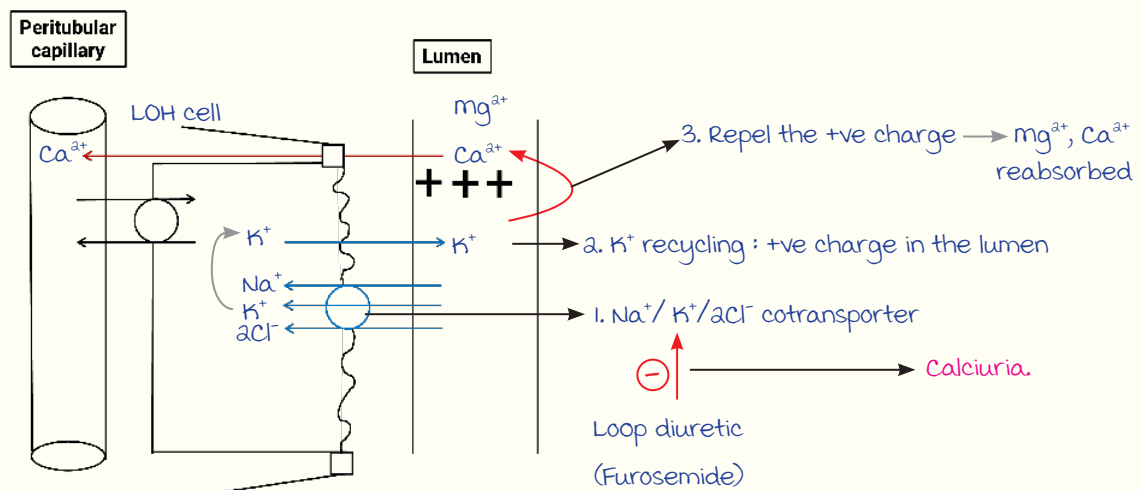
00:30:31

**Counter Current System :****Note :**

Counter current system is also seen in kidney, limbs, testis, intestine.

**Vasa Recta :**

- AKA Counter current exchanger.
- Blood vessels along LOH.
- Prevent solutes washout → maintain medullary hyperosmolarity.

**Ca<sup>2+</sup> & Mg<sup>2+</sup> Reabsorption :**

## Distal Convoluted Tubule (DCT)

00:38:10

----- Active space -----

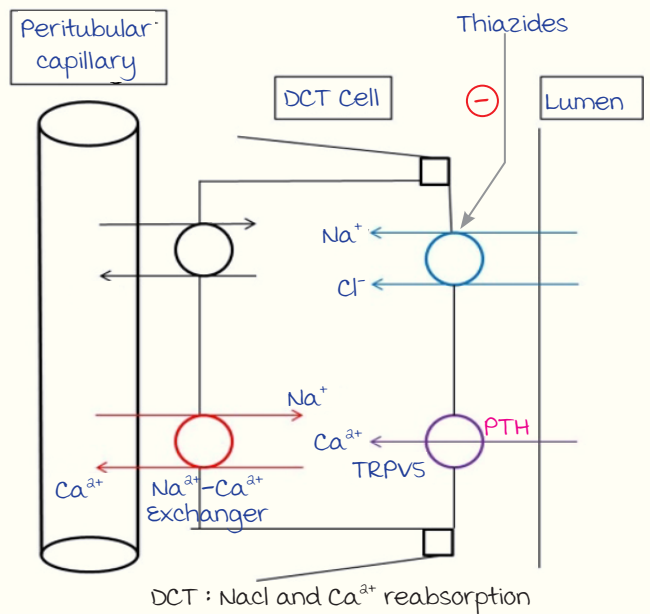
NaCl re-absorption :

NaCl co-transporter.

Ca<sup>2+</sup> re-absorption :

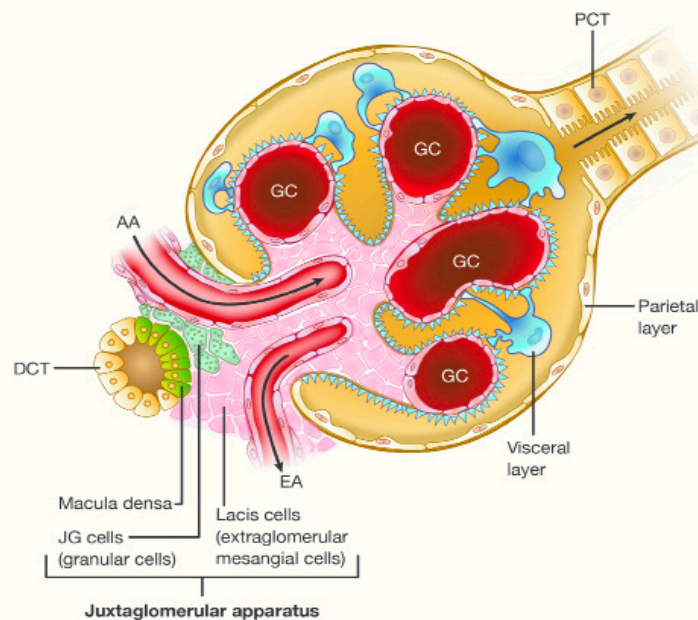
TRPV5 (Transport receptor potential vanilloid channel) :

Regulated by PTH.



## Juxtaglomerular Apparatus ✨ :

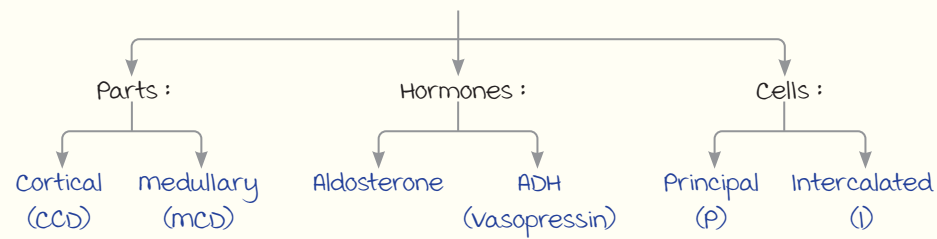
|                                                | Features                                                                                                                                                                                                                        |
|------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Juxta glomerular cells (Afferent arteriole) | <ul style="list-style-type: none"> <li>• AKA Granule cells</li> <li>• Storage &amp; release of renin               <ul style="list-style-type: none"> <li>↓ Renal perfusion pressure → (+) → RAAS → ↑ BP</li> </ul> </li> </ul> |
| 2. macula densa                                | GFR sensors → Tubuloglomerular feedback                                                                                                                                                                                         |
| 3. Lacis cells                                 | <ul style="list-style-type: none"> <li>• Extraglomerular mesangial cells</li> <li>• Supporting role</li> </ul>                                                                                                                  |



----- Active space -----

**Collecting Duct (CD)**

00:43:02



Functions :

## • Reabsorption :

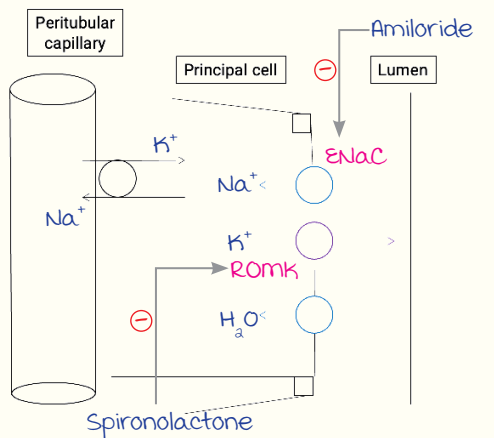
- $\text{Na}^+$ .
- $\text{H}_2\text{O}$ .

## • Excretion :

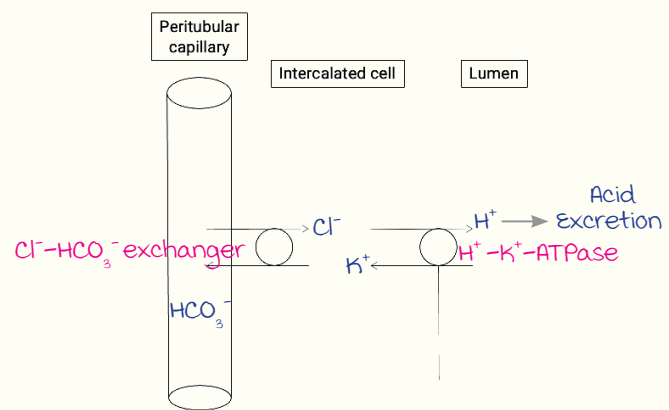
- $\text{K}^+$ .
- $\text{H}^+$ .

Actions of Aldosterone :

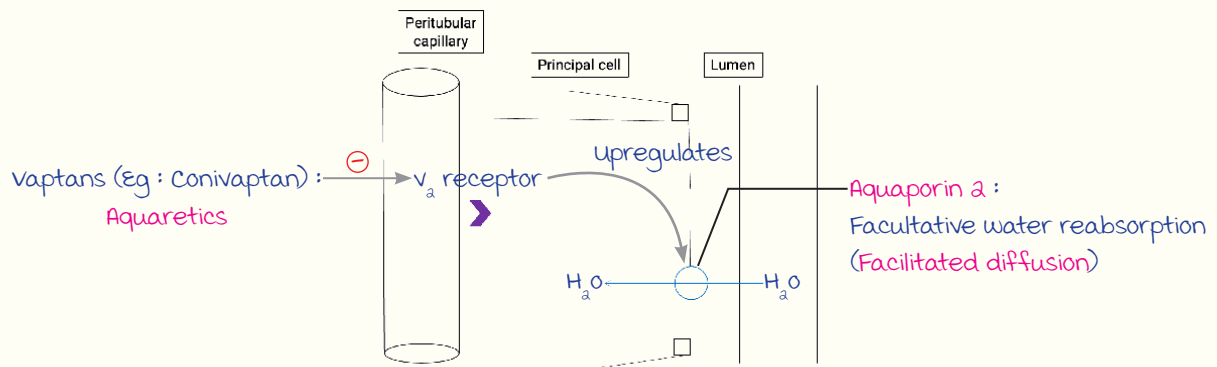
In principal cells :



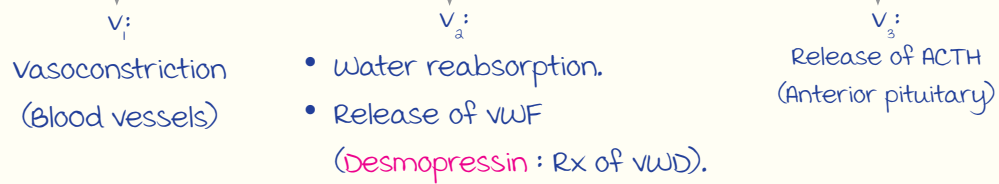
In intercalated cells :

•  $\text{ENaC}$  (Epithelial Na<sup>+</sup> channels) :

Overactivity → Liddle syndrome

(Hypertension d/t  $\text{Na}^+$  &  $\text{H}_2\text{O}$  retention).•  $\text{ROMK}$  : Renal outer medullary K<sup>+</sup> channels.Actions of ADH :

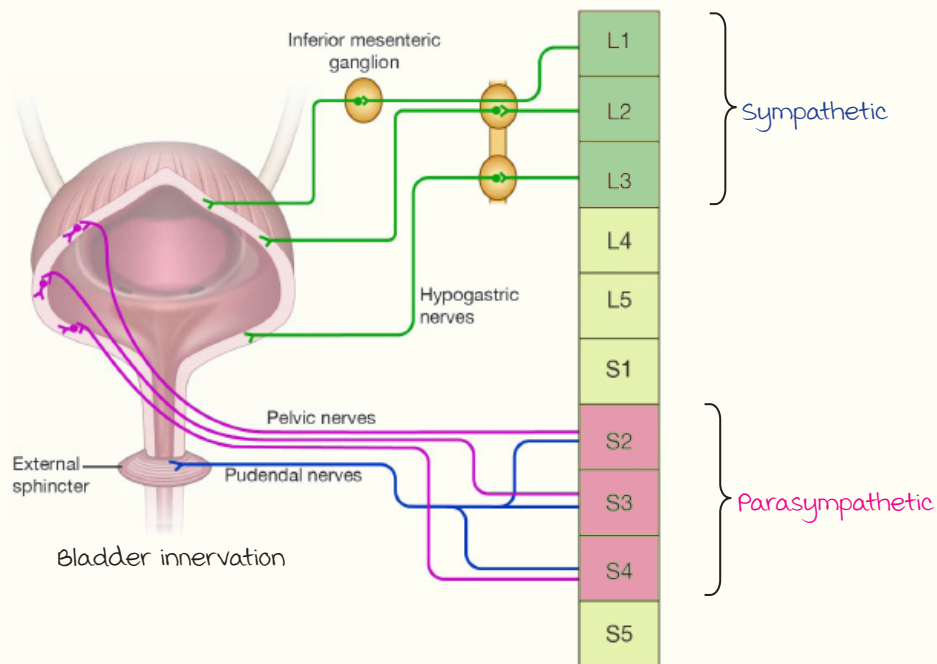
Vasopressin receptors :



## Micturition

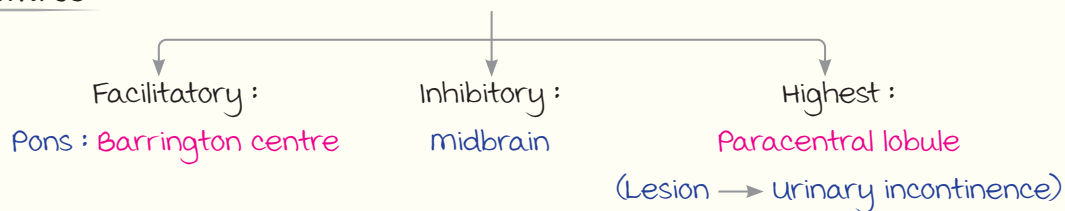
00:50:49

Nerve Supply :



Somatic supply : Pudendal nerves (voluntary).

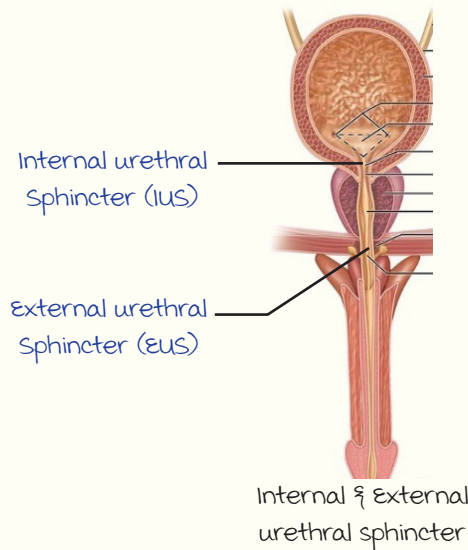
Centres :



----- Active space -----

**Micturition Reflex :**

Bladder filling (Stretch) → Afferent to spinal cord (Pelvic nerve)



↓ Parasympathetic ⊕

Contraction of detrusor & relaxation of IUS

↓

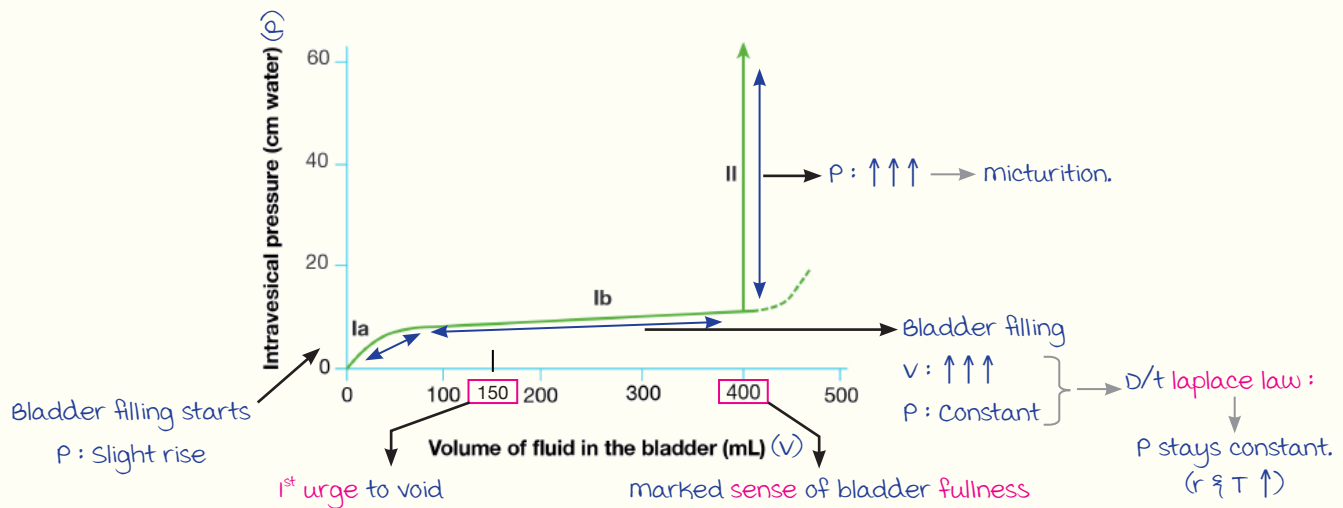
Relaxation of EUS (voluntary)

↓

micturition

**Cystometrogram :**

Study of volume (V) &amp; pressure (P) changes in the bladder.

**Acid-Base Balance :**

00:56:30

• Normal pH : 7.35 to 7.45 (&lt; 7.35 / &gt; 7.45 → Fatal).

• Deranged pH :

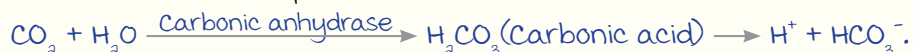
- Denaturation of proteins (Enzymes)

- CNS :

Alkalosis : muscle twitches.

Acidosis : ↓ Excitability.

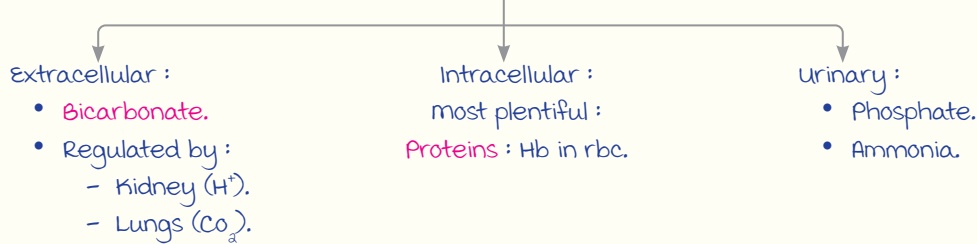
Acid base balance equation :



## First Line Defense :

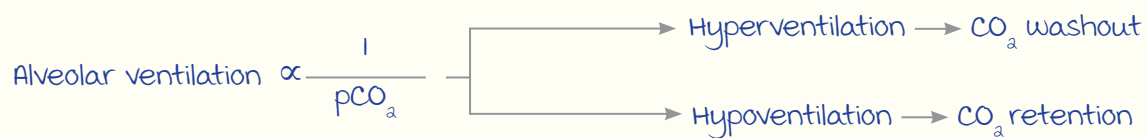
----- Active space -----

### Buffers :



## Second Line Defense :

### Lungs (75% correction) :

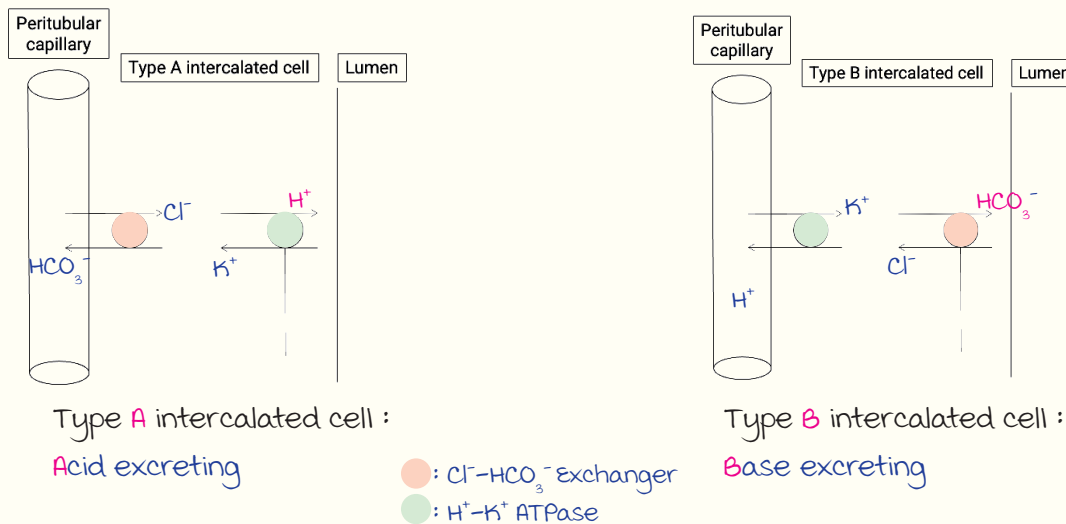


## Third Line Defense :

### Kidneys (100% correction)

- Excretion of  $H^+$  &  $HCO_3^-$ .
- Reabsorption of  $HCO_3^-$ .
- Generation of new  $HCO_3^-$  : uses phosphate & ammonia buffer in place of bicarbonate.

### Role of CD :



----- Active space -----

## Acid Base Disorder

01:05:18

- Normal PH : 7.35 - 7.45
- Normal  $pCO_2$  : 35-45 mmHg.
- Normal  $HCO_3^-$  : 22-26 mEq/L.

|                                       | Causative agents                                                                                                                                                                                                                                                                                                  |                                               |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------|
| metabolic acidosis<br>(↓pH)           | $\uparrow H^+$ : <ul style="list-style-type: none"> <li>• Lactic acid</li> <li>• Ketoacids</li> <li>• Salicylates</li> </ul>                                                                                                                                                                                      | ↓ $HCO_3^-$ : Diarrhea                        |
| metabolic alkalosis<br>(↑pH)          | $\downarrow H^+$ : <ul style="list-style-type: none"> <li>• Vomiting</li> <li>• Hyperaldosteronism (<math>\uparrow H^+</math> excretion)</li> <li>• Diuretics <math>\xrightarrow{+RAAS}</math> <math>\uparrow</math> Aldosterone</li> </ul> <p style="text-align: center;">↓<br/>volume contraction alkalosis</p> | ↑ $HCO_3^-$ :<br>Sodium bicarbonate ingestion |
| Respiratory acidosis<br>(↑ $pCO_2$ )  | <ul style="list-style-type: none"> <li>• Respiratory depression : Opioids</li> <li>• Respiratory muscle paralysis : Polio</li> <li>• Airway obstruction : COPD</li> </ul>                                                                                                                                         | } Renal compensation                          |
| Respiratory alkalosis<br>(↓ $pCO_2$ ) | <ul style="list-style-type: none"> <li>• Hyperventilation</li> <li>• Hypoxia (Eg : High altitude)</li> </ul>                                                                                                                                                                                                      |                                               |

Common binding protein (Albumin) :

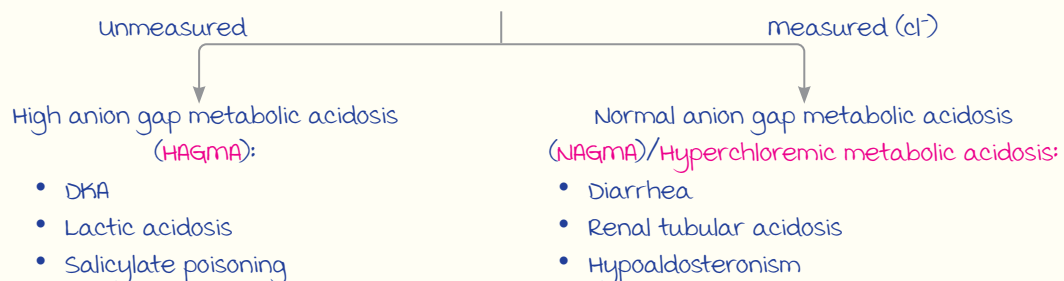
- Binds both  $H^+$  and  $Ca^{2+}$ .
- In respiratory alkalosis :  
 $\uparrow CO_2$  washout  $\rightarrow \downarrow H^+ \rightarrow$ 

- $\uparrow Ca^{2+}$  bound form
  - $\downarrow Ca^{2+}$  free form

 $\rightarrow$  Hypocalcemic tetany.

Anion Gap (AG) :

- D/t unmeasured anions (Protein, sulfate, lactate, salicylate).
- $AG = Na^+ - (HCO_3^- + Cl^-)$ .
- $\textcircled{N} = 8-16 \text{ mEq/L}$ .
- $\uparrow$  anions :



# ENDOCRINE PHYSIOLOGY

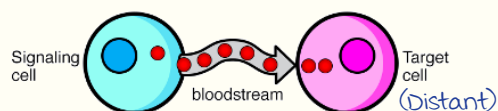
----- Active space -----

Forms of cell signaling :

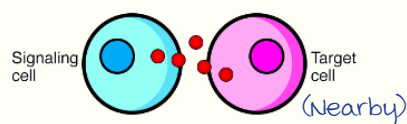
Autocrine : T lymphocytes  $\rightarrow$  IL-2.



Endocrine : Hormones.



Paracrine : Histamine  $\rightarrow$  Parietal cells  $\rightarrow$  ↑ Gastric acid.



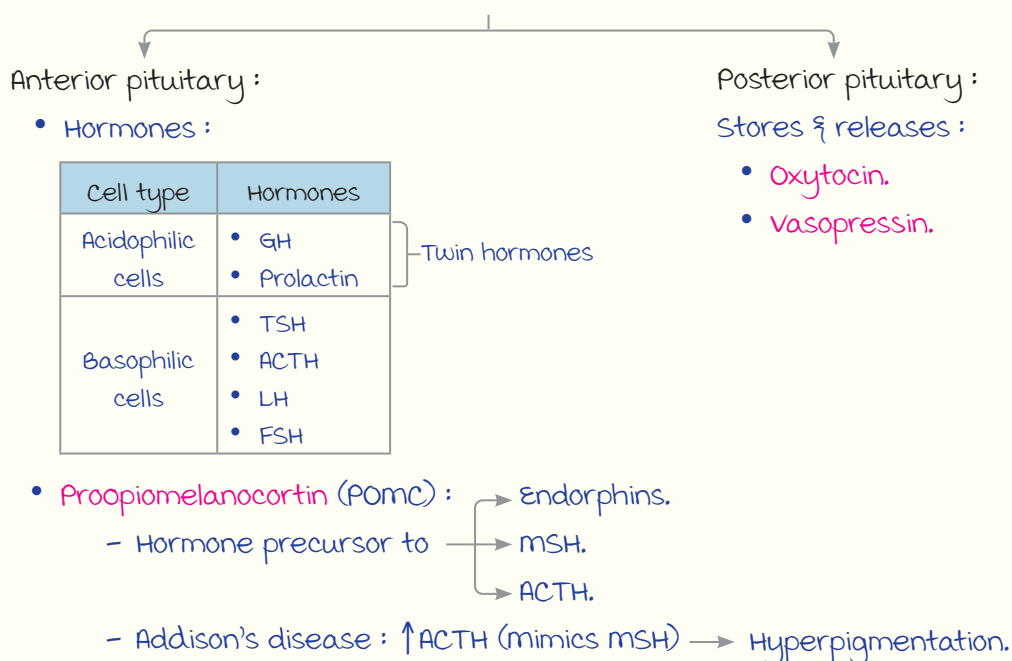
Typical negative feedback control of endocrine hormone release :



## Pituitary Gland & Hypothalamus

00:03:42

### Pituitary Gland :



----- Active space -----

Hypothalamus  :

| Hypothalamic hormones |                   | Anterior pituitary hormone |
|-----------------------|-------------------|----------------------------|
| Stimulatory hormones  | TRH               | TSH                        |
|                       | CRH               | ACTH                       |
|                       | GnRH              | FSH, LH                    |
|                       | GHRH              | GH                         |
| Inhibiting hormones   | PIH/Dopamine      | Prolactin                  |
|                       | GHIH/Somatostatin | GH                         |

**Growth Hormone**

00:09:19

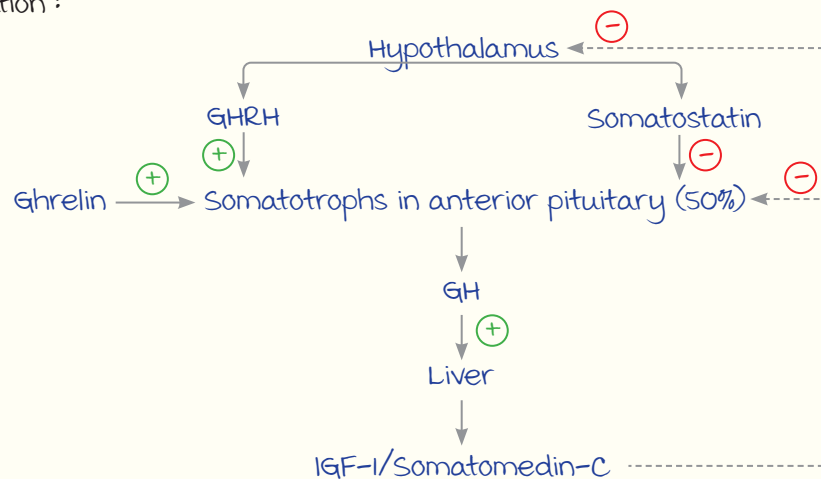
Source : Somatotrophs.

Receptor : JAK-STAT pathway.

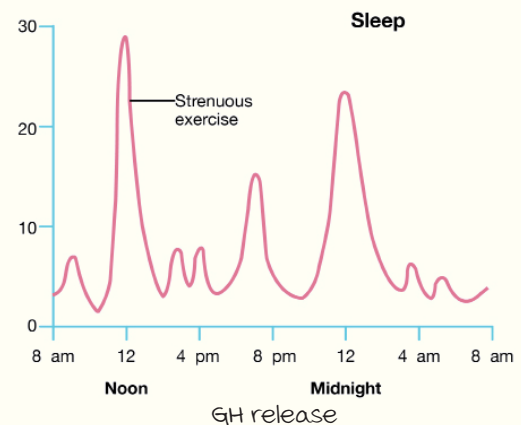
Levels :

- Highest : Adolescence.
- Lowest : Elderly.

Regulation :

Factors affecting Secretion  :

| Factors that increase growth hormone secretion                                                                                                                                                         | Factors that inhibit growth hormone secretion                                                                                                                     |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>• Hypoglycemia</li> <li>- Exercise</li> <li>- Fasting</li> <li>- Stress</li> <li>• Deep sleep (NREM)</li> <li>• Protein rich meal</li> <li>• Ghrelin</li> </ul> | <ul style="list-style-type: none"> <li>• Hyperglycemia</li> <li>• Somatostatin</li> <li>• REM sleep</li> <li>• Somatomedin</li> <li>• Free fatty acids</li> </ul> |

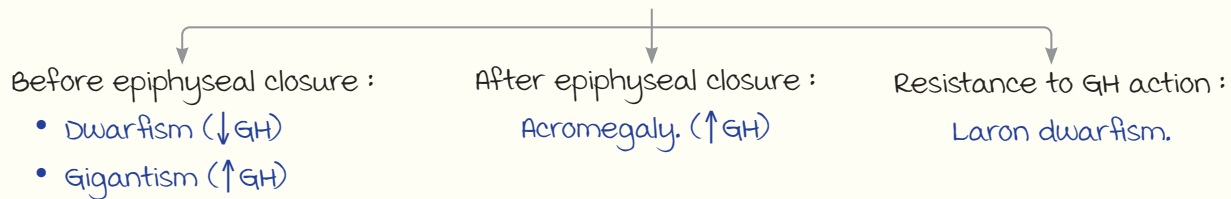


Actions :

----- Active space -----

|             | Direct (Anti-insulin like)                                                                                                         | Indirect (Insulin like)                                                                                                                  |
|-------------|------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| mediated by | GH                                                                                                                                 | Somatomedins                                                                                                                             |
| Actions     | <ul style="list-style-type: none"> <li>• Prechondrocytes → Chondrocytes</li> <li>• ↑ Blood glucose</li> <li>• Lipolysis</li> </ul> | <ul style="list-style-type: none"> <li>• Chondrocyte proliferation</li> <li>• Linear bone growth</li> <li>• Protein synthesis</li> </ul> |

### GH Disorders :



Acromegaly :

- c/f :
  - ↑ Hand & foot size.
  - Carpal tunnel syndrome.
  - mandibular enlargement.
  - (Prognathism)
  - Diabetes mellitus.

- mx → medical :
  - Somatostatin analogues : Octreotide.
  - GH receptor antagonist : Pegvisomant.
- Surgical : Transsphenoidal surgical resection.

Laron dwarfism :

- AKA GH insensitivity syndrome.
- ⊕/↑ GH.
- ↓ IGF-1.

### Prolactin

00:19:30

Source : Lactotrophs.

Receptor : JAK STAT pathway.

Factors affecting secretion :

| ↑ Secretion                                                                                                                     | ↓ Secretion                                                                           |
|---------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>• Pregnancy &amp; lactation</li> <li>• Suckling</li> <li>• Sexual intercourse</li> </ul> | <ul style="list-style-type: none"> <li>• Dopamine</li> <li>• Bromocriptine</li> </ul> |

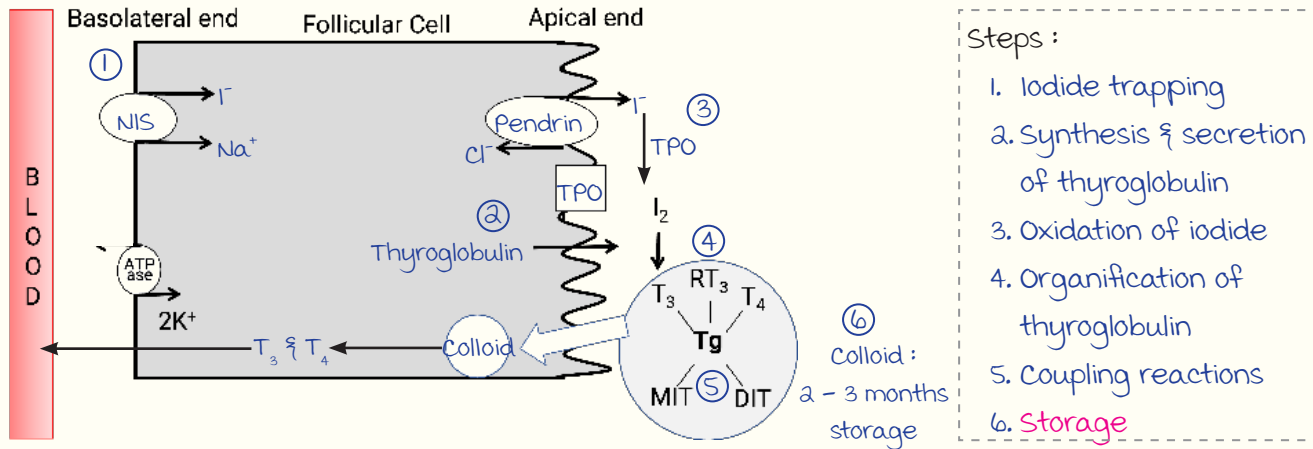
Actions :

- milk production
- ⊖ GnRH → Lactational amenorrhea.

----- Active space -----

**Thyroid Gland**

00:21:24

**Thyroid Hormone Synthesis**  :

MIT : monoiodotyrosine

DIT : Diiodotyrosine

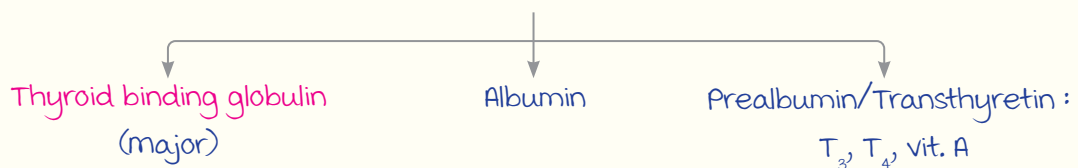
Sodium iodide symporter (NIS)  :

- Location : Thyroid follicular cell.
- **2° active transport.**
- **Other locations :** Salivary gland, mammary glands, placenta.
- **Wolff Chaikoff effect :**  
Excess iodide moves through NIS →  Organification → ↓ T<sub>3</sub>, T<sub>4</sub>.

Iodide chloride antiport (Pendrin)  :

- Locations : Thyroid gland, inner ear.
- mutation → **Pendred** syndrome : Goiter, SNHL.

Thyroid hormone binding proteins :

**Triiodothyronine (T<sub>3</sub>) vs. Thyroxine (T<sub>4</sub>) :**

|                        | T <sub>3</sub> (Physiological form) | T <sub>4</sub> (Pharmacological/drug form) |
|------------------------|-------------------------------------|--------------------------------------------|
| Potency                | <b>more</b>                         | Less                                       |
| Half Life              | 1 day                               | 7 days                                     |
| Plasma protein binding | Less                                | more                                       |
| Action                 | Rapid                               | Long lasting                               |
| In colloid             | Less                                | more                                       |

## Thyroid Hormone Receptors :

----- Active space -----

- Nuclear receptors.
- membrane transport proteins :
  - mono carboxylase transporter (MCT).
  - Organic anion transporting polypeptide (OATP).

## Actions & Clinical Significance :

| Actions                                  | Hypothyroidism                     | Hyperthyroidism                           |
|------------------------------------------|------------------------------------|-------------------------------------------|
| BMR (Energy expenditure)                 | ↓ (Weight gain)                    | ↑ (Weight loss)                           |
| Physiological uncoupler<br>(↑ Heat)      | ↓ Heat<br>(Cold intolerance)       | ↑ Heat<br>(Heat intolerance)              |
| Carbohydrates<br>(↑ Blood glucose)       | Hypoglycemia                       | Hyperglycemia                             |
| Lipid (Lipolysis, ↓ cholesterol)         | ↑ Cholesterol                      | ↓ Cholesterol                             |
| Protein                                  | Protein synthesis<br>↓<br>myxedema | Protein catabolism<br>↓<br>muscle wasting |
| Development (Bone, brain)                | Growth & mental retardation        | Accelerated growth                        |
| CNS (Alertness)                          | Lethargy                           | Tremors                                   |
| CVS (↑ HR, ↑ contractility)              | ↓                                  | ↑                                         |
| RS (↑ ventilation)                       | ↓                                  | ↑                                         |
| GIT (↑ Digestive juices,<br>GI motility) | Constipation                       | Diarrhea                                  |
| Reproduction<br>(Ⓝ Sexual functions)     | menorrhagia                        | Oligomenorrhea                            |

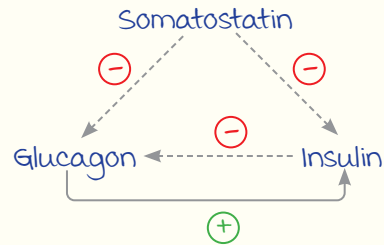
## Endocrine Pancreas

00:37:23

Type of cells :

| Cell type                       | Secretion                                                                     |
|---------------------------------|-------------------------------------------------------------------------------|
| A cells (or) Alpha cells        | Glucagon (↑ Blood glucose, lipolysis,<br>↑ ketone bodies)                     |
| B cells (or) Beta cells (major) | Insulin, amylin, C peptide (Insulin marker)<br>↓<br>Ratio = 1 : 1 (Equimolar) |
| D cells (or) Delta cells        | Somatostatin                                                                  |
| F cells                         | Pancreatic polypeptide                                                        |

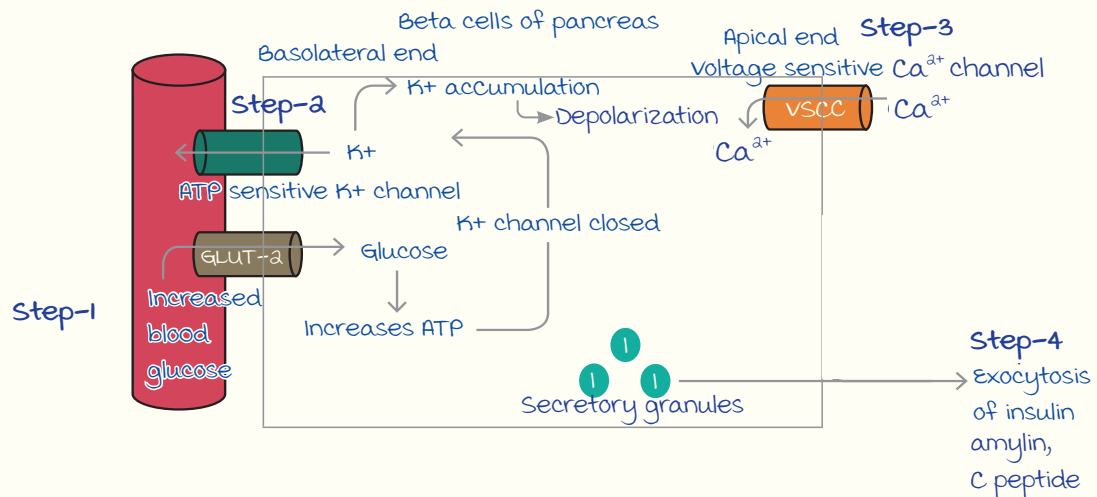
----- Active space ----- Paracrine regulation :



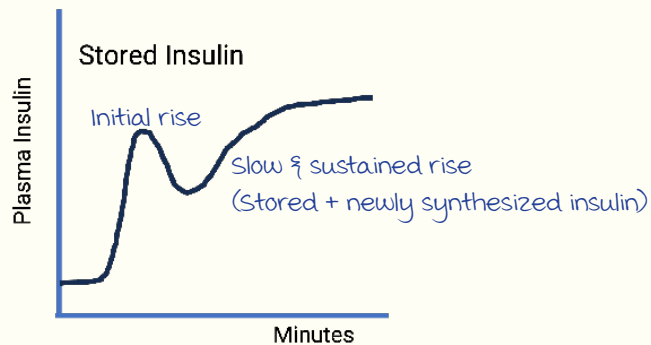
Blood flow : Center to periphery.

Insulin :

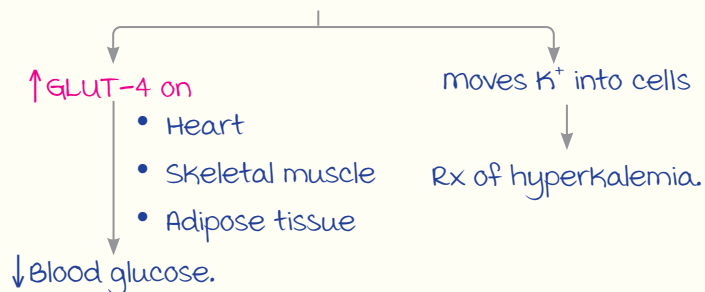
Release :



Biphasic release :



Actions :

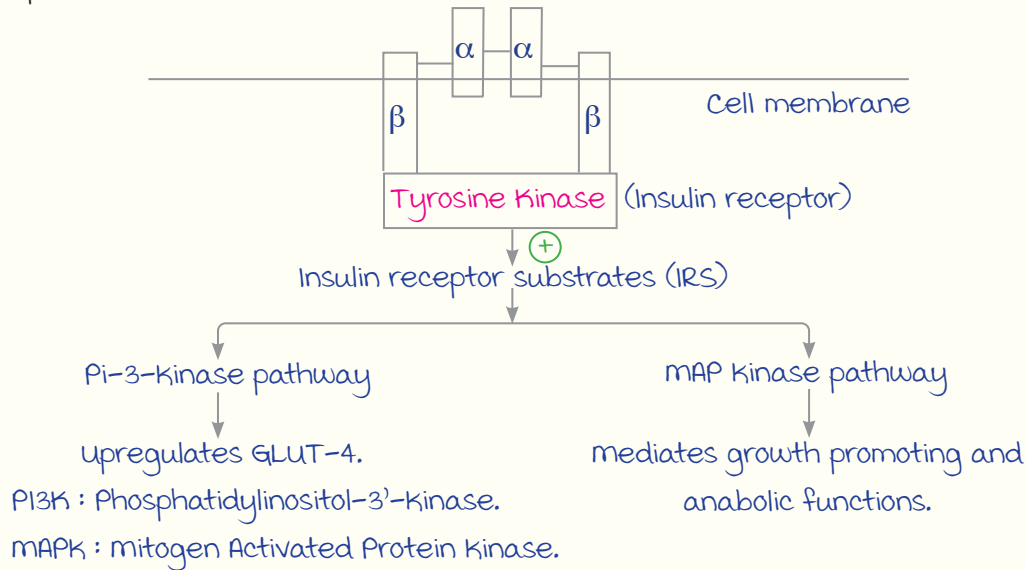


Note :

Exercise (+) AMP activated kinase pathway → upregulates GLUT-4.  
Insulin independent

Receptors :

----- Active space -----



Factors affecting insulin secretion :

| Factors ↑ insulin secretion                                                                                                                                                                                                                                   | Factors ↓ insulin secretion                                                                                                                              |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| High blood glucose                                                                                                                                                                                                                                            | Somatostatin                                                                                                                                             |
| <ul style="list-style-type: none"> <li>Glucagon</li> <li>Amino acids like arginine, leucine</li> </ul>                                                                                                                                                        | <ul style="list-style-type: none"> <li>Toxins : Streptozocin, Alloxan</li> <li>Sympathetic : <math>\alpha</math> adrenergic receptor mediated</li> </ul> |
| ANS { Parasympathetic : vagal<br>Sympathetic : $\beta$ receptor mediated                                                                                                                                                                                      |                                                                                                                                                          |
| Gastrin, cholecystokinin, secretin                                                                                                                                                                                                                            |                                                                                                                                                          |
| Incretin hormones : <ul style="list-style-type: none"> <li>Glucagon like peptide - 1 (GLP - 1)</li> <li>Glucose dependent insulinotropic peptide (GIP)</li> <li>Incretin effect : _____</li> </ul> Oral glucose stimulates insulin much more than IV glucose. |                                                                                                                                                          |

Glucose Homeostasis :

| ↓ Blood glucose | ↑ Blood glucose/<br>Counter regulatory hormones                                                                                                    |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| Insulin         | <ul style="list-style-type: none"> <li>Glucagon</li> <li>Cortisol</li> <li>Growth hormone</li> <li>Thyroid hormone</li> <li>Epinephrine</li> </ul> |

Dawn phenomenon : Early morning hyperglycemia.

Somogyi phenomenon : Overnight hypoglycemia

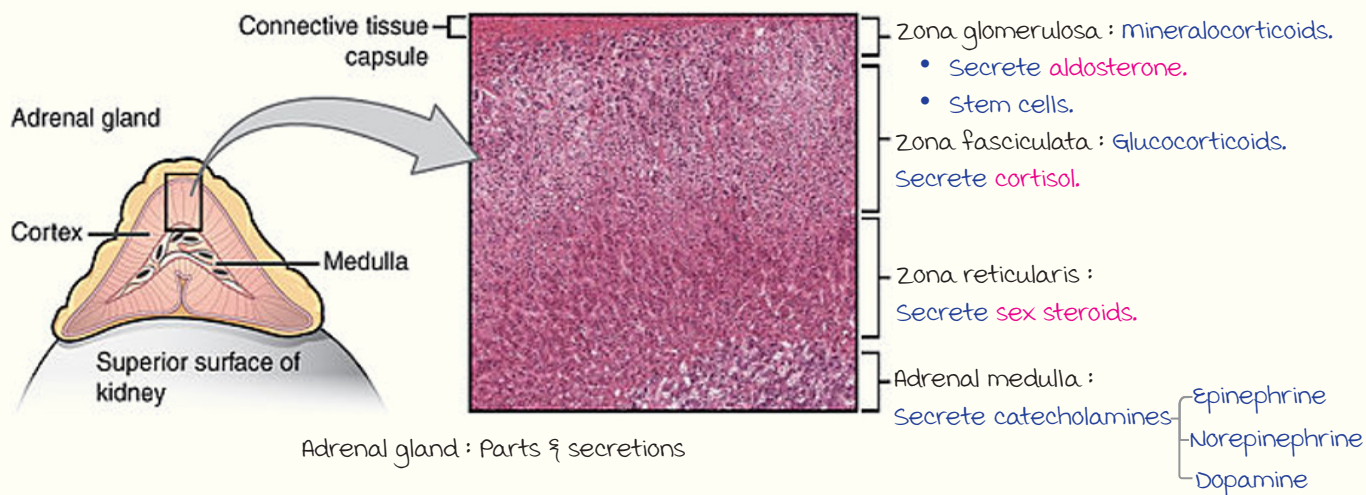
↓

Early morning hyperglycemia.

----- Active space -----

## Adrenal Gland

Layers  :



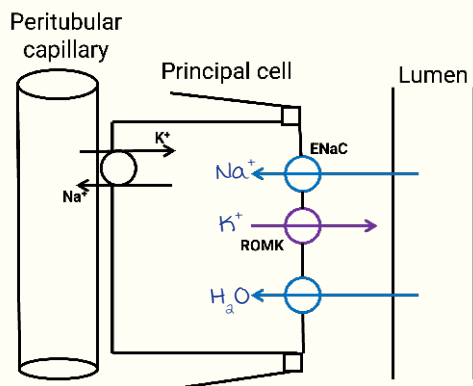
### Aldosterone :

Steroid hormone.

Actions :

Reabsorption :  $\text{Na}^+ \& \text{H}_2\text{O}$  } → At collecting duct { Principal cells.  
 Excretion :  $\text{K}^+ \& \text{H}^+$  } Intercalated cells.

Principal cell :



### • $\text{Na}^+$ reabsorption :

- Epithelial sodium channels (**ENaC**)
  - **Amiloride** : ENaC blocker →  $\text{Na}^+ \& \text{H}_2\text{O}$  excretion
  - **Liddle syndrome** : Overactive ENaC →  $\text{Na}^+ \& \text{H}_2\text{O}$  retention
- ↓
- Hypertension.

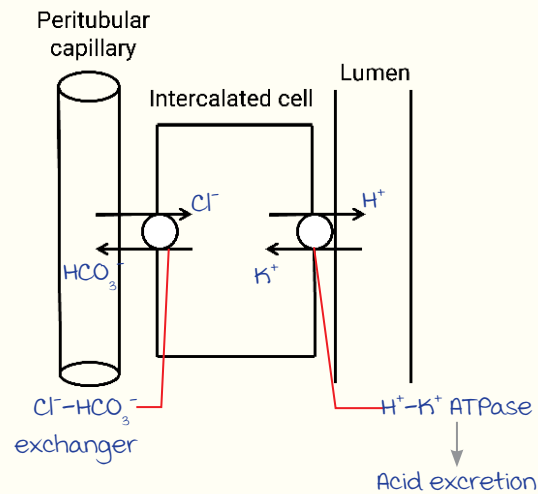
### • $\text{K}^+$ excretion :

- Renal outer medullary  $\text{K}^+$  channels (**ROMK**).
  - **Spironolactone** : Aldosterone receptor blocker
- ↓

$\text{Na}^+ \& \text{H}_2\text{O}$  excretion,  **$\text{K}^+$  sparing** diuretic.

Intercalated cell :

----- Active space -----



Cortisol :

Glucocorticoid.

Actions & clinical significance :

|                           | metabolic effect                                                                                                          | Cushing syndrome (↑)                   |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| Protein metabolism        | <ul style="list-style-type: none"> <li>muscle : <b>Proteolysis</b></li> <li>Liver : ↑ Plasma protein synthesis</li> </ul> | <ul style="list-style-type: none"> <li>muscle weakness</li> <li>Proximal myopathy</li> </ul>                              |
| Carbohydrate metabolism   | ↑ Blood glucose                                                                                                           | <ul style="list-style-type: none"> <li><b>Glucose intolerance</b></li> <li>Diabetes mellitus</li> </ul>                   |
| Lipid metabolism          | Lipolysis                                                                                                                 | <ul style="list-style-type: none"> <li>Round face (<b>moon facies</b>)</li> <li><b>Fat pad</b> on back of neck</li> </ul> |
| Immune system             | ↓ Inflammation                                                                                                            | Susceptibility to infections                                                                                              |
| Blood cells               | <ul style="list-style-type: none"> <li>↓ : Eosinophils, Lymphocytes</li> <li>↑ : RBC, neutrophils, platelets</li> </ul>   | <ul style="list-style-type: none"> <li>Eosinopenia</li> <li>Hypercoagulation state</li> </ul>                             |
| Nervous system            | <ul style="list-style-type: none"> <li>↑ Appetite</li> <li>Alters mood &amp; behaviour</li> </ul>                         | <ul style="list-style-type: none"> <li><b>Irritability</b></li> <li>Psychosis</li> </ul>                                  |
| Heart                     | <ul style="list-style-type: none"> <li>↑ Cardiac output</li> <li>↑ Vascular tone</li> </ul>                               | <ul style="list-style-type: none"> <li>Hypertension</li> <li>Atherosclerosis</li> </ul>                                   |
| Renal system              | <ul style="list-style-type: none"> <li>↑ GFR</li> <li>↑ <math>\text{Ca}^{2+}</math> excretion</li> </ul>                  | <ul style="list-style-type: none"> <li>Hypercalciuria</li> <li>Renal stones</li> </ul>                                    |
| GIT                       | ↑ Gastric acid secretion                                                                                                  | Peptic ulcer                                                                                                              |
| Bone & connective tissues | ⊖ fibroblast proliferation & collagen formation                                                                           | <b>Purple striae</b>                                                                                                      |
| Reproduction              | Negative regulator of reproduction (⊖ GnRH)                                                                               | <ul style="list-style-type: none"> <li>Amenorrhea</li> <li>Infertility</li> </ul>                                         |
| Permissive action         | Potentiates actions of catecholamines                                                                                     | <b>Hypertension</b>                                                                                                       |

----- Active space -----

**Adrenal Sex Steroids :**

- Dehydroepiandrosterone (DHEA) : major.
- Dehydroepiandrosterone sulphate (DHEAS).
- Androstenedione.
- Testosterone : Small amounts.

Actions :

- Axillary & pubic hair growth.
- Sexual desire.

Adrenogenital syndrome :

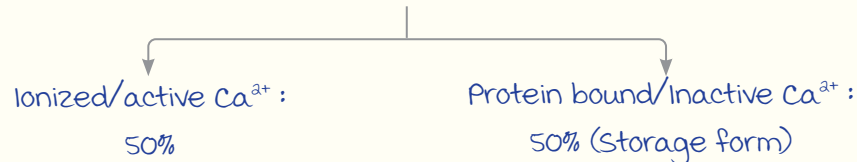
- Females : virilization (Baldness, hirsutism).
  - males : Precocious puberty.
- } Excess adrenal androgens.

**Calcium**

01:08:33

(N) : 9 - 11 mg/dL.

Distribution :

**Calcium Homeostasis :**

Three hormones :

- PTH
- Vitamin D
- Calcitonin

Three organs :

- Bone (major source)
- Intestine
- Kidney

**Parathormone, Vit. D & Calcitonin**

01:11:26

**PTH :**

Protein hormone (84 aa).

Source : Parathyroid gland (Chief cells).

Receptor : Calcium sensing receptor (CaSR).

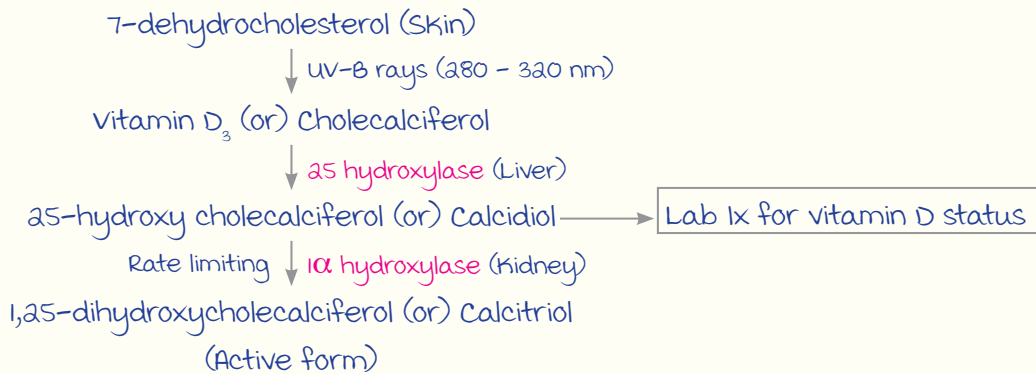
High  $\text{Ca}^{2+}$  → ⊖ PTH release.Low  $\text{Ca}^{2+}$  → ⊕ PTH release.

Drug form : Teriparatide.

Vitamin D :

----- Active space -----

Synthesis :

Calcitonin :

Source : Para follicular C cells.

Role in pregnancy &amp; lactation :

- D/t  $\uparrow$  Ca<sup>2+</sup> demand.
- Calcitonin  $\downarrow$  es bone loss.

Tumor marker : medullary carcinoma of thyroid.

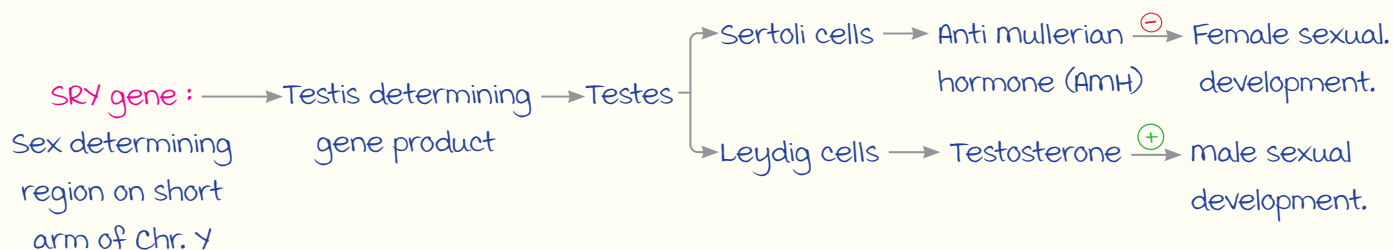
Drug form : **Salcatonin** →  $\ominus$  Osteoclast activity.Actions :

|           | PTH                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Vit. D                                                                                                                                                                              | Calcitonin                                                                                                        |
|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| Bone      | <ul style="list-style-type: none"> <li>• PTH <math>\xrightarrow{+}</math> PTH receptor on <b>osteoblasts</b></li> <li> <p style="text-align: center;">↓</p> <p style="text-align: center;">Osteoclasts <math>\xleftarrow{+}</math> RANKL</p> <p style="text-align: center;">↓</p> <p style="text-align: center;"><math>\uparrow</math> Bone resorption</p> </li> <li>• <b>Denosumab</b> :               <ul style="list-style-type: none"> <li>- RANKL blocker</li> <li>- Osteoporosis Rx</li> </ul> </li> </ul> | <ul style="list-style-type: none"> <li>• Bone mineralisation &amp; resorption depending on Ca<sup>2+</sup> levels</li> <li>• Vit. D receptors in <b>osteoblasts</b></li> </ul>      | <p style="text-align: center;"><math>\ominus</math> Bone resorption by <math>\ominus</math> <b>osteoclast</b></p> |
| Intestine | Indirectly $\uparrow$ Vit. D                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <p style="text-align: center;">Upregulates calbindin</p> <p style="text-align: center;">↓</p> <p style="text-align: center;"><math>\uparrow</math> Ca<sup>2+</sup> reabsorption</p> | $\downarrow$ Ca <sup>2+</sup> reabsorption                                                                        |
| Kidney    | <ul style="list-style-type: none"> <li>• <math>\uparrow</math> Phosphate excretion (By inhibiting <b>sodium phosphate cotransporter</b> in PCT)</li> <li>• DCT : <math>\uparrow</math> Ca<sup>2+</sup> reabsorption</li> </ul>                                                                                                                                                                                                                                                                                   | $\uparrow$ Ca <sup>2+</sup> & $\uparrow$ PO <sub>4</sub> <sup>3-</sup> reabsorption                                                                                                 | $\uparrow$ Ca <sup>2+</sup> excretion                                                                             |

# REPRODUCTIVE PHYSIOLOGY, EXERCISE PHYSIOLOGY & REGULATION OF BODY TEMPERATURE

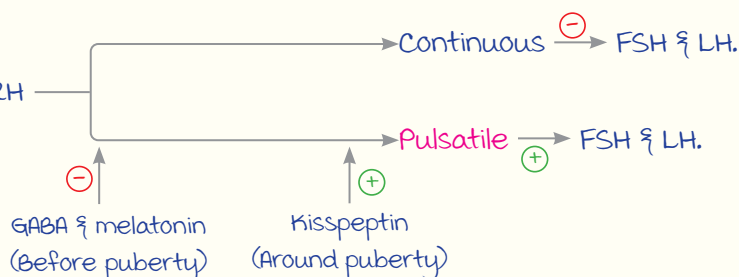
## Male Reproductive System

00:00:17



### Puberty :

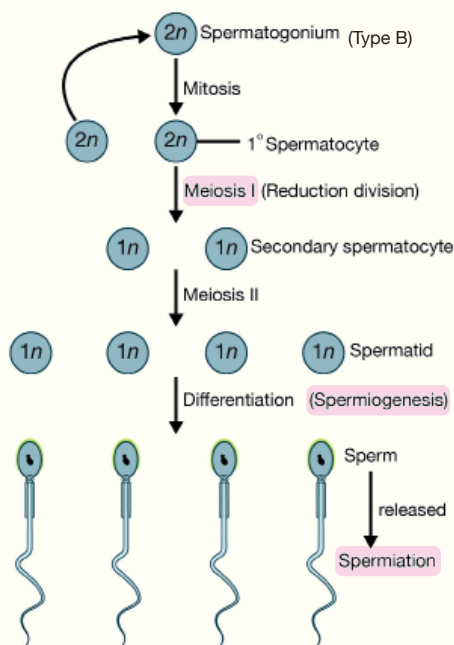
Secretion of GnRH



Permissive role : Leptin  $\oplus$  Adequate body fat → Puberty.

### Spermatogenesis :

In seminiferous tubules.



### Functions of Sertoli Cells :

----- Active space -----

- Secrete AMH.
- Form blood-testis barrier :  
Protects sperm.
- Nutrition to developing sperm  
(AKA nurse cells of testis).
- Produce androgen binding protein (ABP).
- Secrete **aromatase** enzyme :  
Testosterone → Estrogen.
- Secrete inhibin :  
Negative feedback regulation of FSH.
- Phagocytosis of dead sperm.

### Human Spermatozoon :

maturation :

- In epididymis.
- Activation of **CATSPER** ( $\text{Ca}^{2+}$  channel) → Forward progressive motility of sperm.

Capacitation :

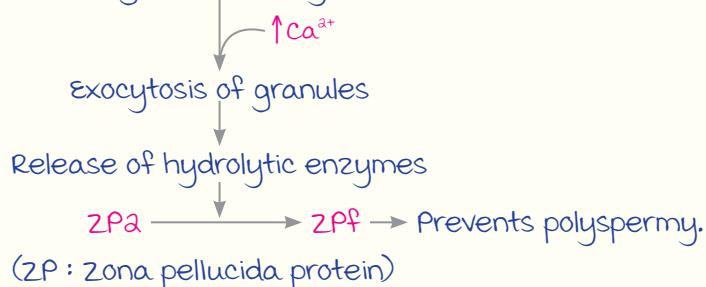
- Occurs in **female genital tract**.
- ↑ Forward motility of sperm.
- Facilitates acrosome reaction (Penetrate zona pellucida).

### Prevention of Polyspermy :

Reduction in membrane potential of ovum : Initial and transient.

Cortical reaction :

Cortical granules : Enzyme-filled vesicles



### Semen :

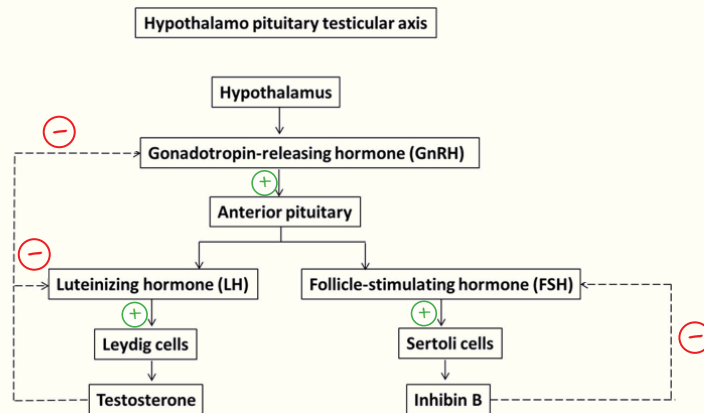
| Secretions from | % volume | Composition                                                                          |
|-----------------|----------|--------------------------------------------------------------------------------------|
| Seminal vesicle | 60%      | Fructose, phosphoryl choline ( <b>Florence test</b> ), ascorbic acid, prostaglandins |
| Prostate        | 30%      | Spermine ( <b>Barberio's test</b> ), fibrinolysin, Zn, acid phosphatase              |
| Vas deferens    | 10%      | -                                                                                    |

Buffers :  $\text{HCO}_3^-$  and  $\text{PO}_4^{3-}$ .

----- Active space -----

## Regulation of male Reproduction :

### Two cell-two gonadotropin model :



### Functions of testosterone :

- Wolffian system stimulation : Development of **internal** male genital organs.
- $\uparrow$  RBC.
- Anabolic effects :  $\uparrow$  muscle mass (**Youth** hormone).
- Bone growth stimulation.
- Kidney : Retention of  $\text{Na}^+$ ,  $\text{H}_2\text{O}$ ,  $\text{Ca}^{2+}$ .
- Inhibitory feedback to LH.
- male 2° sexual characters.
- $\uparrow$  Basal metabolic rate.

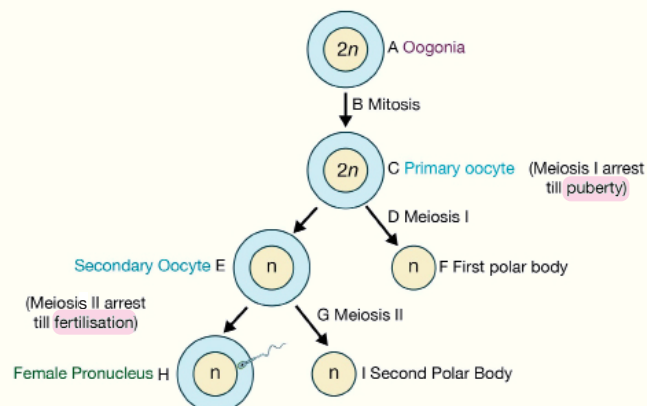
### Dihydrotestosterone :

- Testosterone  $\xrightarrow{\text{5-}\alpha\text{-reductase}}$  Dihydrotestosterone (more potent).
- Pubertal changes.
- masculinization of **external** male genital organs.
- Development of **prostate** :  $\text{5-}\alpha\text{-reductase}$  inhibitor (**Finasteride**) : Therapeutic use in BPH.
- 2° sexual characteristics : Facial hair, acne, temporal hair recession.

## Female Reproductive System

00:16:13

### Oogenesis :



menstrual Cycle :

----- Active space -----

Follicular phase :

FSH  $\xrightarrow{\text{stimulates}}$  Growth of 10 - 15 follicles  $\rightarrow$  1 dominant follicle  $\rightarrow$  Secretes estrogen.

LH surge :

- Due to positive feedback effect of  $\uparrow$  estrogen.
- Occurs :  
24 - 36 hrs before ovulation.

FSH surge (Smaller) :

Lysis of follicular cell

$\downarrow$

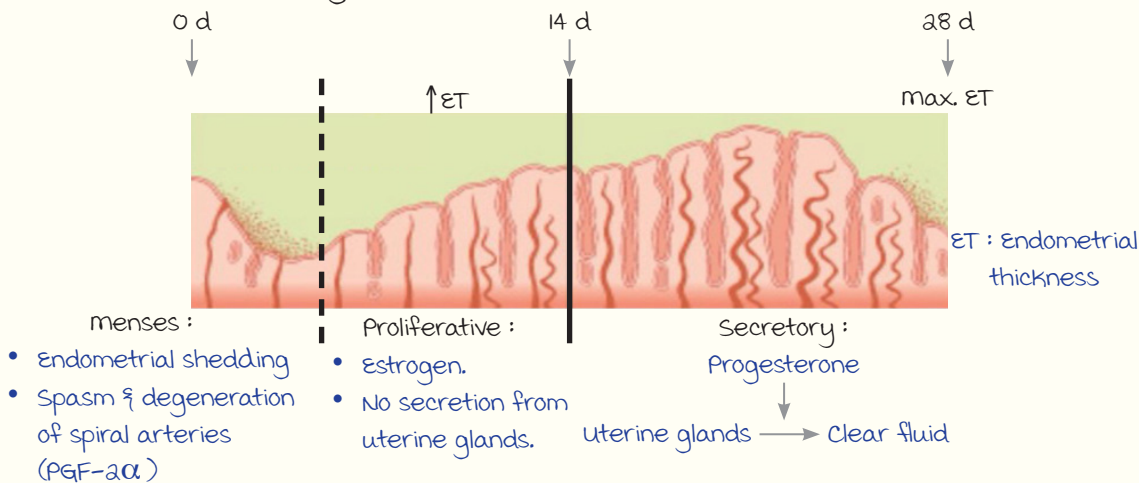
Release of ovum.

Luteal phase :

Lasts for a duration of 14 days after ovulation,

LH  $\xrightarrow{\text{stimulates}}$  Luteinization  $\rightarrow$  Formation of corpus luteum  
(Secretes progesterone)  
If no fertilisation  $\downarrow$   
Regresses on 24<sup>th</sup> day of cycle.

uterine endometrial changes :



Cervical changes :

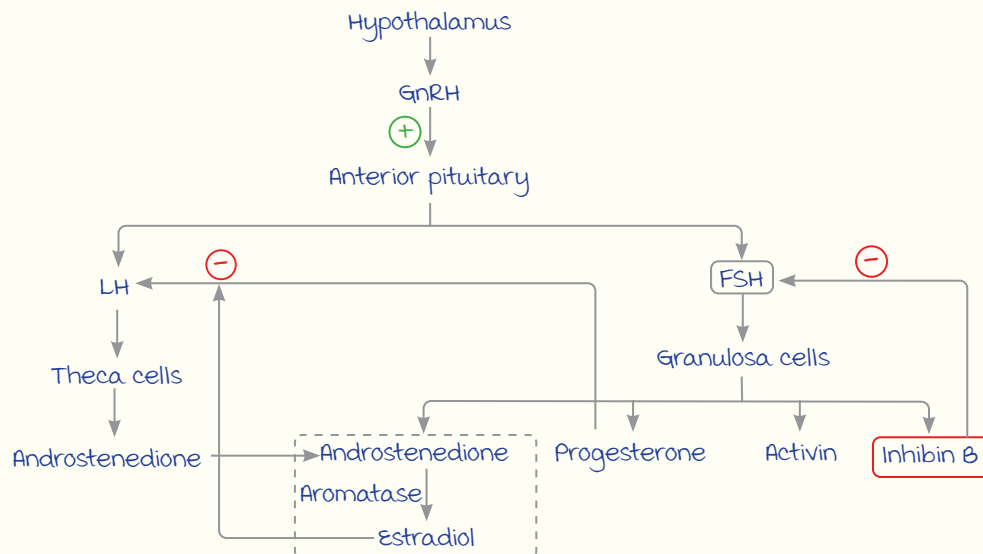
- Estrogen  $\rightarrow$  Cervical mucus :  $\rightarrow$  Facilitates sperm penetration.
  - Thin
  - $\uparrow$  stretchability (Spinnbarkeit effect)
  - $\uparrow$  NaCl (Fern pattern)
- Progesterone  $\rightarrow$  Thick cervical mucus  $\rightarrow$  Prevents sperm entry.

----- Active space ----- Vaginal changes :

| Cell seen    | Hormonal influence |
|--------------|--------------------|
| Parabasal    | Nil                |
| Intermediate | Progesterone       |
| Superficial  | Estrogen           |

### Regulation of Female Reproduction :

Two cell-two gonadotropin model :



Functions of estrogen and progesterone :

| Organs                    | Estrogen                                                                                           | Progesterone                                                           |
|---------------------------|----------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|
| uterus                    | Estrogen primed uterus :<br>• ↑ Contractility<br>• ↑ Excitability                                  | • ↓ Excitability.<br>• ↓ Contractility                                 |
| 2° sexual characteristics | Narrow shoulder, broad hips, converging thighs, diverging arms                                     | No effect                                                              |
| Breast                    | Stimulates ductal growth                                                                           | Lobulo-alveolar growth                                                 |
| Bone                      | • Induces apoptosis of osteoclasts → ⊖ Bone resorption<br>• ↓ Estrogen at menopause → Osteoporosis | No effect                                                              |
| Liver                     | ↑ Expression of LDL receptor → Plasma cholesterol lowering action                                  | No effect                                                              |
| CVS                       | • ↑ NO levels → Promotes vasodilation<br>• Inhibits platelet aggregation                           | No effect                                                              |
| CNS                       | Inhibits neuronal cell death → Neuroprotective action                                              | ↑ Basal body temperature<br>(Thermogenic action)                       |
| Kidney                    | Na <sup>+</sup> & H <sub>2</sub> O retention                                                       | ↑ Na <sup>+</sup> & H <sub>2</sub> O excretion                         |
| RS                        | No effect                                                                                          | Stimulates respiration<br>↓ Alveolar pCO <sub>a</sub> during ovulation |

Human chorionic gonadotropin (hCG) :

----- Active space -----

- AKA super LH : 80x activity of LH.
- Source : Syncytiotrophoblast.
- Sub-units :  $\alpha$  : Similar to  $\alpha$  subunit of LH, FSH, TSH.  
 $\beta$ .
- Functions :
  - Maintenance of corpus luteum till 6 weeks of pregnancy.
  - Stimulates Leydig cell  $\rightarrow$  Testosterone.
  - Immunosuppressive activity.
  - Secretion of relaxin from corpus luteum.

Lactation :

- Prolactin : milk production.
  - Estrogen : Ductal growth.
  - Progesterone : Lobuloalveolar growth.
  - Oxytocin : milk ejection.
    - Stimulus : Suckling (Nipple stimulation)
- $\downarrow$   
 Contraction of myoepithelial cells
- $\downarrow$   
 Galactokinesis.
- } Neuro endocrine reflex/  
 milk let down reflex.

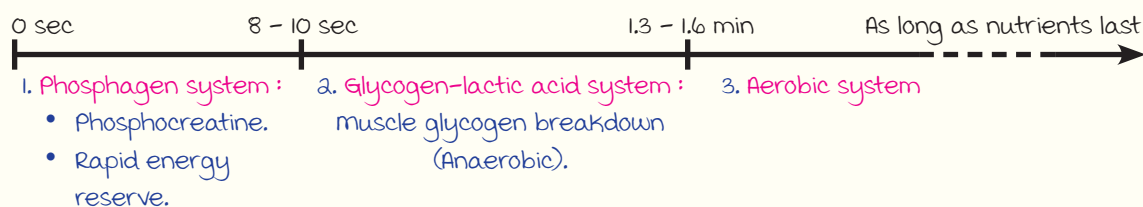
## Exercise Physiology

00:34:34

Types of exercise :

|                | Isotonic (Dynamic) | Isometric (Static)     |
|----------------|--------------------|------------------------|
| muscle tension | Constant           | Variable               |
| muscle length  | Variable           | Constant               |
| Example        | Treadmill running  | Pushing against a wall |

Energy source :



----- Active space -----

Redistribution of blood flow during exercise :

| Organ             | Effect                                      |
|-------------------|---------------------------------------------|
| Kidneys           | ↓                                           |
| GIT               | ↓                                           |
| Heart             | ↑                                           |
| Exercising muscle | ↑                                           |
| Brain             | No change (D/t autoregulation)              |
| Skin              | Initially ↓, later ↑ (For heat dissipation) |

Effects of Exercise :

CVS :

- HR : ↑ (Anticipatory tachycardia). m/l for exercise.
- CO : ↑.
- BP :

| Isotonic exercise                                                 | Isometric exercise                                                                |
|-------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| ↑ SBP                                                             | ↑ SBP                                                                             |
| ↓ DBP :<br>Accumulation of metabolites<br>(Vasodilators)<br>↓ TPR | ↑ DBP : Sustained muscle contraction<br>↓<br>Arterioles compression<br>↓<br>↑ TPR |

Respiratory system :

- $\text{MET} \propto \text{Resting } \text{O}_a \text{ consumption}$ ; MET = 250 mL/min.
- Peak  $\text{VO}_a$  : Highest  $\text{O}_a$  uptake at the end of test (Practical).
- $\text{VO}_a \text{ max}$  : Maximum possible  $\text{O}_a$  consumption (Theoretical value).

| Parameter                          | Effect                                             |
|------------------------------------|----------------------------------------------------|
| Respiratory rate                   | ↑ Even before exercise ;<br>anticipatory tachypnea |
| Pulmonary ventilation              | ↑                                                  |
| $\text{pO}_a$                      | No change                                          |
| $\text{pCO}_a$                     |                                                    |
| pH                                 |                                                    |
| Diffusion of gases                 | ↑                                                  |
| Resting $\text{O}_a$ consumption   | ↑                                                  |
| metabolic equivalents (MET)        | ↑                                                  |
| Release of $\text{O}_a$ to tissues | ↑ (R shift on ODC)                                 |

## Regulation of Body Temperature

00:45:00

----- Active space -----

Role of hypothalamus :

|          | Anterior hypothalamus                                                                                              | Posterior hypothalamus                                                                                                   |
|----------|--------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| Stimulus | Hot temperature                                                                                                    | Cold temperature                                                                                                         |
| effects  | <ul style="list-style-type: none"> <li>• Cutaneous vasodilation</li> <li>• Sweating</li> <li>• ↑ Thirst</li> </ul> | <ul style="list-style-type: none"> <li>• Vasoconstriction</li> <li>• Shivering</li> <li>• ↑ Urinary frequency</li> </ul> |
| Lesion   | Cannot lose heat                                                                                                   | Cannot gain heat                                                                                                         |

### Heat Disorders (mild to Severe) :

Heat cramps (mild) < Heat syncope < Heat exhaustion < Heat stroke, malignant hyperthermia (Severe).

Heat stroke :

- Due to loss of thermoregulatory mechanisms.
- **Sweating absent.**
- Forms :
  - Classic : Old age.
  - Exertional : Soldiers.
- **Triad** : Coma + hot & dry skin + rectal temperature  $> 106^{\circ}\text{F}$ .
- Rx : Cool place, cold water, cold clothes, fluid replacement.

malignant hyperthermia :

- Pathophysiology : ↑ Activity of ryanodine receptor (RYR) → ↑  $\text{Ca}^{2+}$  release  
 ↓  
 ↑ muscle contraction  
 ↓  
 Hyperthermia.
- Precipitating agents :
  - **Succinylcholine.**
  - Halothane.
- Rx : Dantrolene sodium (Blocks ryanodine receptor).

### Cold Disorders :

Hypothermia :

1. Therapeutic : Every  $1^{\circ}\text{C}$  ↓ in body temperature → 7% ↓ in cerebral blood flow (↓ Blood loss in neuro Sx).
2. Accidental.

